Supporting Materials for "Automatic Differentiation for Inference on Partially Observed Markov Process Models"

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S1 Proof of Theorem 1 (DMOP-α Functional Forms)

Theorem 1 follows immediately as a consequence of the following results, Lemmas S1 and S2. Here, an A superscript is used to denote ancestral trajectories, so that $x_{1:n,j}^{A,P,\theta}$ contains the history of the prediction particle $x_{n,j}^{P,\theta}$ at observation times $1:n$. Similarly, $g_{i,j}^{A,P,\theta}$ is the ancestral weight of $x_{n,j}^{P,\theta}$ at observation time i.

Lemma S1. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 1$,*

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{1}{J}\sum_{j=1}^J \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\big(y_{1:N}^*|x_{1:N,j}^{A,F,\theta};\theta\big), \tag{S1}$$

with the same undiscounted path-space score target as the bootstrap filter estimators of Poyiadjis et al. (2011) and Ścibior and Wood (2021).

Proof. Consider MOP-α with $\alpha = 1$. Hold ϕ, the simulator draws, and the reference-filter resampling indices fixed when differentiating, then evaluate at $\theta = \phi$. Write $k_{n,j}$ for k_j at time n. Algorithm 1 gives

$$w_{n,j}^{P,\theta} = w_{n-1,j}^{F,\theta}, \qquad w_{n,j}^{F,\theta} = w_{n,k_{n,j}}^{P,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = w_{n-1,k_{n,j}}^{F,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}}. \tag{S2}$$

The after-resampling likelihood estimate therefore telescopes:

$$\begin{aligned}\hat{\mathcal{L}}^1(\theta) &= \prod_{n=1}^N L_n^{A,\theta,1} = \prod_{n=1}^N \left(L_n^{\phi}\frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}\right)\\ &= \left(\prod_{n=1}^N L_n^{\phi}\right)\prod_{n=1}^N \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n-1,j}^{F,\theta}} = \left(\frac{1}{J}\sum_{j=1}^J w_{N,j}^{F,\theta}\right)\prod_{n=1}^N L_n^{\phi}.\end{aligned} \tag{S3}$$

Here each intermediate weight sum cancels, and the initial sum is J because $w_{0,j}^{F,\theta} = 1$. The log-derivative identity gives

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\nabla_\theta \hat{\mathcal{L}}^1(\theta)}{\hat{\mathcal{L}}^1(\theta)} = \frac{J^{-1}\sum_{j=1}^J \nabla_\theta w_{N,j}^{F,\theta}\prod_{n=1}^N L_n^{\phi}}{J^{-1}\sum_{j=1}^J w_{N,j}^{F,\theta}\prod_{n=1}^N L_n^{\phi}} = \frac{\sum_{j=1}^J \nabla_\theta w_{N,j}^{F,\theta}}{\sum_{j=1}^J w_{N,j}^{F,\theta}}. \tag{S4}$$

We next expand the filtered weight along its ancestry. Using the existing A convention, let $g_{i,j}^{A,F,\theta}$ denote the measurement density at time i along the filtered trajectory $x_{1:n,j}^{A,F,\theta}$; the endpoint n is implicit. At time n, this density is $g_{n,k_{n,j}}^{\theta}$. At earlier times, the trajectory agrees with that of the selected parent $x_{n-1,k_{n,j}}^{F,\theta}$. Starting from $w_{0,j}^{F,\theta} = 1$ and repeatedly substituting the filtered-weight recursion therefore gives

$$w_{n,j}^{F,\theta} = w_{n-1,k_{n,j}}^{F,\theta}\frac{g_{n,k_{n,j}}^{\theta}}{g_{n,k_{n,j}}^{\phi}} = \left(\prod_{i=1}^{n-1}\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right)\frac{g_{n,j}^{A,F,\theta}}{g_{n,j}^{A,F,\phi}} = \prod_{i=1}^{n}\frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}. \tag{S5}$$

The product through $n-1$ is empty when $n=1$; for $n>1$, it is the product already accumulated by the selected parent. The reference factors and ancestor indices are fixed under differentiation, so

$$\begin{aligned}\frac{\nabla_\theta w_{n,j}^{F,\theta}}{w_{n,j}^{F,\theta}} &= \nabla_\theta \log w_{n,j}^{F,\theta} = \nabla_\theta \log\left(\prod_{i=1}^{n} \frac{g_{i,j}^{A,F,\theta}}{g_{i,j}^{A,F,\phi}}\right) \\ &= \nabla_\theta \sum_{i=1}^{n}\left(\log g_{i,j}^{A,F,\theta} - \log g_{i,j}^{A,F,\phi}\right) = \sum_{i=1}^{n} \nabla_\theta \log g_{i,j}^{A,F,\theta}. \end{aligned} \tag{S6}$$

For the terminal filtered trajectory at time N, conditional independence of the observations gives

$$\begin{aligned}\sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} &= \nabla_\theta \log\left(\prod_{n=1}^{N} g_{n,j}^{A,F,\theta}\right) \\ &= \nabla_\theta \log\left(\prod_{n=1}^{N} f_{Y_n|X_n}\left(y_n^* \mid x_{n,j}^{A,F,\theta};\theta\right)\right) \\ &= \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right). \end{aligned} \tag{S7}$$

The derivative includes the dependence of the simulated particle locations on θ through the chain rule. Multiplying the log-weight derivative by $w_{N,j}^{F,\theta}$ yields

$$\nabla_\theta w_{N,j}^{F,\theta} = w_{N,j}^{F,\theta} \sum_{n=1}^{N} \nabla_\theta \log g_{n,j}^{A,F,\theta} = w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right). \tag{S8}$$

Finally, at $\theta=\phi$, every ratio in Equation (S5) equals one, so $w_{N,j}^{F,\theta}=1$ and $\sum_j w_{N,j}^{F,\theta}=J$. Substituting into Equation (S4) after differentiating gives

$$\begin{aligned}\nabla_\theta \hat{\ell}^1(\theta)\Big|_{\theta=\phi} &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta w_{N,j}^{F,\theta}\Big|_{\theta=\phi} \\ &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right)\Big|_{\theta=\phi}, \end{aligned} \tag{S9}$$

as claimed. This has the same undiscounted path-space score target as the estimators of Poyiadjis et al. (2011); Ścibior and Wood (2021) for the bootstrap filter. The reparameterization score contribution need not equal their complete-data likelihood-ratio score contribution particle by particle. □

For $\theta \neq \phi$, the same calculation instead gives the normalized weighted path score

$$\nabla_\theta \hat{\ell}^1(\theta) = \frac{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta} \nabla_\theta \log f_{Y_{1:N}|X_{1:N}}\left(y_{1:N}^* \mid x_{1:N,j}^{A,F,\theta};\theta\right)}{\displaystyle\sum_{j=1}^{J} w_{N,j}^{F,\theta}}. \tag{S10}$$

Growth of the unnormalized weight products alone does not establish a variance rate for this normalized expression. When $\theta = \phi$, a generic normalized path-functional inequality gives the coarse upper bound $O(N^4/J)$ after rescaling to the full unnormalized score. Section S7 gives an exact constant-weight calculation with order N^2/J at $\alpha = 1$ and explains why this example does not by itself provide a general finite-J upper bound.

Lemma S2. *Write $\nabla_\theta \hat{\ell}^\alpha(\theta)$ for the DMOP-α gradient estimate (or, equivalently, the gradient of MOP-α when $\theta = \phi$). Consider the case where we use the after-resampling conditional likelihood estimate so that $\hat{\mathcal{L}}(\theta) = \prod_{n=1}^N L_n^{A,\theta,\alpha}$. When $\alpha = 0$,*

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)\right), \tag{S11}$$

yielding the estimate of Naesseth et al. (2018) when applied to the bootstrap filter.

Proof. First, write

$$s_{n,j} = \frac{g^\theta_{n,k_{n,j}}}{g^\phi_{n,k_{n,j}}} \tag{S12}$$

as shorthand for the selected measurement density ratios, with the reference parameter and resampling indices held fixed. Observe that, when $\alpha = 0$, the likelihood estimate becomes

$$\begin{aligned}
\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,\alpha} &= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}} \right) && \text{(S13)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) && \text{(S14)} \\
&= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\theta}; \theta\right)}{f_{Y_n|X_n}\left(y_n^* | x_{n,k_{n,j}}^{P,\phi}; \phi\right)} \right). && \text{(S15)}
\end{aligned}$$

We lose the nice telescoping property observed in the MOP-1 case, but this expression still yields something useful. This is because its gradient when $\theta = \phi$ is therefore

$$\begin{aligned}
\nabla_\theta \hat{\ell}^0(\theta) &:= \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) && \text{(S16)} \\
&= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} && \text{(S17)} \\
&= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} && \text{(S18)} \\
&= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)}{f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\phi}; \phi)} && \text{(S19)}
\end{aligned}$$

$$= \frac{1}{J}\sum_{n=1}^{N}\sum_{j=1}^{J}\nabla_\theta \log\left(f_{Y_n|X_n}\left(y_n^*|x_{n,j}^{F,\theta};\theta\right)\right), \tag{S20}$$

where we use the log-derivative trick in the second equality, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$ in the fourth equality, and use the log-derivative trick again while noting that $\theta = \phi$ in the fifth equality. This yields the desired result. □

S2 Proof of Theorem 2 (Optimization Convergence Analysis)

All derivatives in this section are evaluated at the common filtering and evaluation parameter $\theta = \phi$. Let Z_n contain the complete ancestral path through time n, the parameter-independent simulator variables, and the first- and second-order gradients propagated by automatic differentiation. Set

$$G_n^\theta(Z_n) = f_{Y_n|X_n}(y_n^* \mid X_n^\theta;\theta), \qquad W_\theta(Z_N) = \prod_{n=1}^{N} G_n^\theta(Z_n), \tag{S21}$$

$$\psi_\theta(Z_N) = \nabla_\theta \log W_\theta(Z_N), \qquad \Xi_\theta(Z_N) = \nabla_\theta^2 \log W_\theta(Z_N). \tag{S22}$$

Differentiate through X_n^θ in $G_n^\theta(Z_n)$, holding the simulator draws and resampling indices fixed.

Write $\rho_N^J = J^{-1}\sum_{j=1}^{J}\delta_{Z_{N,j}}$ for the empirical measure of the after-resampling genealogical particles, and define the normalized Feynman–Kac path measure by

$$\rho_N(F) = \frac{\mathbb{E}\{W_\theta(Z_N)F(Z_N)\}}{\mathbb{E}\{W_\theta(Z_N)\}}. \tag{S23}$$

Assumptions (A3) and (A5) give $|\partial_{\theta_i} W_\theta| \le N\bar{G}^N G'(\theta)$, so dominated convergence and the log-derivative identity yield

$$\nabla_\theta \ell(\theta) = \rho_N(\psi_\theta), \qquad \nabla_\theta \hat{\ell}_J^{A,1}(\theta) = \rho_N^J(\psi_\theta). \tag{S24}$$

The second equality is Lemma S1; importantly, it is one terminal empirical average of the complete path score, not a sum of empirical filtering expectations at the individual observation times.

For completeness, multinomial resampling gives the required genetic Feynman–Kac particle system explicitly. For semigroup indexing, let Z_0 be the augmented prediction path at observation 1; for $q = 1,\dots,N-1$, let Z_q be the prediction path at observation $q+1$; and let Z_N be the terminal path after the final resampling. For $0 \le q \le N$, write $\rho_q^J = J^{-1}\sum_{j=1}^{J}\delta_{\mathsf{Z}_{q,j}}$, where the $\mathsf{Z}_{0,j}$ are independent draws from the unweighted prediction law of Z_0 and $\mathsf{Z}_{N,j} = Z_{N,j}$. Write $V_q^\theta(\mathsf{Z}_q) = G_{q+1}^\theta(\mathsf{Z}_q)$ for $q = 0,\dots,N-1$. Let the locally indexed augmented kernel $\mathcal{M}_{q+1}^\theta$ append the next independent simulator innovation and its gradients for $0 \le q \le N-2$, and put $\mathcal{M}_N^\theta = \mathrm{Id}$ for the final selection. The one-step population map is

$$\Phi_{q+1}^\theta(\mu)(dz') = \frac{\int \mu(dz)V_q^\theta(z)\mathcal{M}_{q+1}^\theta(z,dz')}{\mu(V_q^\theta)}, \qquad 0 \le q < N. \tag{S25}$$

Conditionally on the current cloud, the J children are independent with common law $\Phi_{q+1}^\theta(\rho_q^J)$. The final observation is handled by the $q = N-1$ transition, namely multinomial selection with potential $V_{N-1}^\theta = G_N^\theta$ followed by the identity mutation. Thus ρ_N^J is exactly the terminal empirical

measure of this augmented historical system. Systematic resampling does not have this conditional-independence property and is not covered by the calculation.

We next write out the particle concentration constant used below. For $0 \le q \le N$, let

$$Q^{\theta}_{q,N}F(z_q) = \mathbb{E}\left[F(\mathsf{Z}_N)\prod_{r=q}^{N-1} V^{\theta}_r(\mathsf{Z}_r)\,\middle|\,\mathsf{Z}_q = z_q\right], \tag{S26}$$

$$P^{\theta}_{q,N}F = \frac{Q^{\theta}_{q,N}F}{Q^{\theta}_{q,N}1}, \qquad g_{q,N} = \sup_{z,z'}\frac{Q^{\theta}_{q,N}1(z)}{Q^{\theta}_{q,N}1(z')}, \qquad b_{q,N} = \sup_{\mathrm{osc}(F)\le 1}\mathrm{osc}(P^{\theta}_{q,N}F), \tag{S27}$$

where $Q^{\theta}_{N,N}F = F$ and $\mathrm{osc}(F) = \sup_{z,z'}|F(z) - F(z')|$. Thus $0 \le b_{q,N} \le 1$. With

$$R_G = \frac{\bar{G}}{\underline{G}}, \qquad S_N = \sum_{q=0}^{N} g^3_{q,N}b_{q,N}, \qquad K_N = 32S_N, \qquad K^{\mathrm{pf}}_N = 32\sum_{r=0}^{N} R^{3r}_G, \tag{S28}$$

the potential bounds give, term by term,

$$\underline{G}^{N-q} \le Q^{\theta}_{q,N}1(z_q) \le \bar{G}^{N-q}, \qquad g_{q,N} \le R^{N-q}_G, \qquad K_N \le K^{\mathrm{pf}}_N. \tag{S29}$$

In particular, the constant used in the theorem is the closed-form quantity

$$K^{\mathrm{pf}}_N = \begin{cases} 32(N+1), & R_G = 1, \\ 32\dfrac{R^{3(N+1)}_G - 1}{R^3_G - 1}, & R_G > 1. \end{cases} \tag{S30}$$

Here is the constant calculation behind the tail bound used below. Let r^{DM}_N and β^{DM}_N denote the second- and first-order coefficients in Theorem 1.2 of Del Moral and Rio (2011). The Feynman–Kac bounds in their Appendix A.3 give

$$r^{\mathrm{DM}}_N \le 4S_N, \qquad (\beta^{\mathrm{DM}}_N)^2 \le 4\sum_{q=0}^{N} g^2_{q,N}b^2_{q,N} \le 4S_N, \tag{S31}$$

where the last inequality uses $g_{q,N} \ge 1$ and $0 \le b_{q,N} \le 1$. If $S_N = 0$, the particle error is identically zero. Hence suppose $S_N > 0$. For $u \in (0, 1/2]$, set $x = Ju^2/(32S_N)$. If $x \le \log 6$, then $6e^{-x} \ge 1$ and the desired probability bound is trivial. If $x > \log 6$, use $h^{-1}(x) \le 2x + 2\sqrt{x}$ for $h(t) = \{t - \log(1+t)\}/2$ and $1 + 2x + 2\sqrt{x} \le 8x$ to obtain

$$\frac{r^{\mathrm{DM}}_N}{J}\{1 + h^{-1}(x)\} \le \frac{u^2}{8x}\{1 + 2x + 2\sqrt{x}\} \le u^2 \le \frac{u}{2},$$
$$\beta^{\mathrm{DM}}_N\sqrt{\frac{2x}{J}} \le 2\sqrt{S_N}\sqrt{\frac{2x}{J}} = \frac{u}{2}.$$

Theorem 1.2 applied to both F and $-F$, after centering and scaling F to oscillation one, therefore gives the explicit Feynman–Kac corollary

$$\Pr\left(|(\rho^J_N - \rho_N)(F)| > \mathrm{osc}(F)u\right) \le 6\exp\left(-\frac{Ju^2}{K_N}\right) \le 6\exp\left(-\frac{Ju^2}{K^{\mathrm{pf}}_N}\right). \tag{S32}$$

for every F with $\mathrm{osc}(F) > 0$ and every $u \in (0, 1/2]$. If F is constant, $(\rho^J_N - \rho_N)(F) = 0$ identically. We use the constants in Equations S28– S32 in the concentration bounds below.

Lemma S3 (Explicit concentration of the DMOP-1 gradient)**.** *Assume multinomial resampling and let*

$$A_N(\theta) = \max_{1\le i\le p} \operatorname{osc}(\psi_{\theta,i}), \qquad u_g(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6p}{\delta}}. \tag{S33}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6p/\delta)$*, then, with probability at least* $1-\delta$*,*

$$\|\nabla_\theta \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta \ell(\theta)\|_2 \le A_N(\theta)\sqrt{p}\, u_g(J,\delta). \tag{S34}$$

Moreover, Assumption (A3) gives $A_N(\theta) \le 2NG'(\theta)$*. Consequently, the explicit condition*

$$J \ge K_N^{\mathrm{pf}} \log\frac{6p}{\delta}\max\left\{4, \frac{4pN^2G'(\theta)^2}{\epsilon^2}\right\} \tag{S35}$$

implies that the left side of equation S34 is at most ϵ *with probability at least* $1-\delta$*.*

Proof. The lower bound on J makes $u_g(J,\delta) \le 1/2$. Apply equation S32 to each normalized coordinate $\psi_{\theta,i}/A_N(\theta)$; a constant coordinate has zero error and needs no normalization. A union bound over the p coordinates gives

$$\max_{1\le i\le p} |(\rho_N^J - \rho_N)(\psi_{\theta,i})| \le A_N(\theta) u_g(J,\delta)$$

with probability at least $1-\delta$. Now use $\|v\|_2 \le \sqrt{p}\|v\|_\infty$ and equation S24. Finally, $|\psi_{\theta,i}| \le NG'(\theta)$, so $A_N(\theta) \le 2NG'(\theta)$; substituting this inequality and solving $2NG'(\theta)\sqrt{p}\, u_g(J,\delta) \le \epsilon$ gives equation S35. □

The next result bounds the particle Hessian, including the outer-product and empirical-mean terms produced by differentiating the logarithm of the terminal empirical average.

Lemma S4 (Explicit concentration of the DMOP-1 particle Hessian)**.** *Suppose in addition that Assumption (A3) gives the entrywise bound* $|\Xi_{\theta,ab}(Z_N)| \le NG''(\theta)$ *for* $1 \le a, b \le p$*. Define*

$$d_p = \frac{p(p+3)}{2}, \qquad H_N(\theta) = 2p\{NG''(\theta) + 3N^2G'(\theta)^2\}, \qquad u_H(J,\delta) = \sqrt{\frac{K_N^{\mathrm{pf}}}{J}\log\frac{6d_p}{\delta}}. \tag{S36}$$

If $J \ge 4K_N^{\mathrm{pf}}\log(6d_p/\delta)$*, then, with probability at least* $1-\delta$*,*

$$\|\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) - \nabla_\theta^2 \ell(\theta)\|_{\mathrm{op}} \le H_N(\theta) u_H(J,\delta). \tag{S37}$$

Fix $0 < \kappa < \gamma$*. If*

$$\gamma I_p \preceq -\nabla_\theta^2 \ell(\theta) \preceq \Gamma I_p \quad \textit{and} \quad J \ge K_N^{\mathrm{pf}} \log\frac{6d_p}{\delta}\max\left\{4, \frac{H_N(\theta)^2}{\kappa^2}\right\}, \tag{S38}$$

then, with probability at least $1-\delta$*,*

$$(\gamma-\kappa) I_p \preceq -\nabla_\theta^2 \hat{\ell}_J^{A,1}(\theta) \preceq (\Gamma+\kappa) I_p. \tag{S39}$$

In particular, the raw particle curvature is then positive definite and invertible.

Proof. For fixed reference-filter random numbers and resampling indices, the telescoping identity in the proof of Lemma S1 gives

$$\hat{\ell}_J^{A,1}(\theta) = C_\phi + \log\left\{\frac{1}{J}\sum_{j=1}^{J}\exp(a_j(\theta))\right\}, \qquad a_j(\theta) = \log w_{N,j}^{F,\theta}, \qquad a_j(\phi) = 0,$$

where C_ϕ is constant in θ. Twice differentiating this log-sum-exp at $\theta = \phi$ gives the exact finite-particle identity

$$\nabla_\theta^2 \hat{\ell}_J^{A,1} = \rho_N^J(\Xi_\theta + \psi_\theta \psi_\theta^T) - \rho_N^J(\psi_\theta)\rho_N^J(\psi_\theta)^T. \tag{S40}$$

Assumption (A3) also gives the entrywise dominating bound

$$|\partial^2_{\theta_a\theta_b} W_\theta| = W_\theta|\Xi_{\theta,ab} + \psi_{\theta,a}\psi_{\theta,b}| \le \bar{G}^N\{NG''(\theta) + N^2G'(\theta)^2\}.$$

Thus differentiation through second order may pass under the expectation, and differentiating $\log \mathbb{E}W_\theta$ gives the population identity

$$\nabla_\theta^2 \ell = \rho_N(\Xi_\theta + \psi_\theta\psi_\theta^T) - \rho_N(\psi_\theta)\rho_N(\psi_\theta)^T. \tag{S41}$$

Let $A_1 = 2NG'(\theta)$ and $A_2 = 2\{NG''(\theta) + N^2G'(\theta)^2\}$. The p scalar functions ψ_a have oscillation at most A_1, and the symmetric entries of $\Xi + \psi\psi^T$ have oscillation at most A_2. Apply equation S32 jointly to these functions, using $d_p = p + p(p+1)/2$ scalar events. On the resulting event,

$$\begin{aligned}\|(\rho_N^J - \rho_N)(\psi)\|_2 &\le A_1\sqrt{p}\, u_H(J,\delta),\\ \|(\rho_N^J - \rho_N)(\Xi + \psi\psi^T)\|_{\mathrm{op}} &\le pA_2 u_H(J,\delta).\end{aligned}$$

Both $\|\rho_N^J(\psi)\|_2$ and $\|\rho_N(\psi)\|_2$ are at most $\sqrt{p}\, NG'(\theta)$. Therefore

$$\begin{aligned}&\|\rho_N^J(\psi)\rho_N^J(\psi)^T - \rho_N(\psi)\rho_N(\psi)^T\|_{\mathrm{op}}\\ &\qquad \le 2\sqrt{p}\, NG'(\theta)\,\|(\rho_N^J - \rho_N)(\psi)\|_2 \le 4pN^2G'(\theta)^2 u_H(J,\delta).\end{aligned}$$

Combining the last three displays with equation S40 and equation S41 proves equation S37. Condition equation S38 makes its right side at most κ; Weyl's eigenvalue inequality then gives equation S39. □

For the optimization theorem, no stochastic Hessian claim is needed if the curvature is regularized deterministically. If $\widetilde{B} = Q\,\mathrm{diag}(\lambda_1,\dots,\lambda_p)Q^T$ is an orthogonal eigendecomposition of any symmetric raw estimate, define

$$B = Q\,\mathrm{diag}([\lambda_1]_c^C,\dots,[\lambda_p]_c^C)Q^T, \qquad [x]_c^C = \min\{C, \max\{c, x\}\}. \tag{S42}$$

For every $v \in \mathbb{R}^p$, direct expansion in the eigenbasis gives

$$c\|v\|_2^2 \le v^TBv \le C\|v\|_2^2, \qquad C^{-1}\|v\|_2^2 \le v^TB^{-1}v \le c^{-1}\|v\|_2^2. \tag{S43}$$

Thus $cI_p \preceq B \preceq CI_p$ and $C^{-1}I_p \preceq B^{-1} \preceq c^{-1}I_p$. Lemma S4 shows that the unclipped particle Hessian has these properties with $c = \gamma - \kappa$ and $C = \Gamma + \kappa$ under equation S38; equation S42 guarantees the chosen c, C bounds for every realization.

Proof of Theorem 2. Let $\mathcal{D}$ be the convex region in Theorem 2, and set

$$G'_{\mathcal{D}} = \sup_{\theta\in\mathcal{D}} G'(\theta), \qquad J_0 = \left\lceil K_N^{\mathrm{pf}} \log\!\left(\frac{6pM}{\delta}\right) \max\left\{4, \frac{4pN^2(G'_{\mathcal{D}})^2}{\epsilon^2}\right\}\right\rceil. \tag{S44}$$

Thus it is enough to take $J \geq J_0$. With $g_m = -\nabla_\theta \hat{\ell}_J^{A,1}(\theta_m)$, define

$$\tau = \min\left\{m \in \{0,\dots,M-1\} : \|g_m\|_2 \leq \sigma\epsilon\right\}, \qquad \min\varnothing = M, \tag{S45}$$

and

$$\mathcal{E}_M = \bigcap_{m=0}^{\min\{\tau,M-1\}} \left\{\|g_m - \nabla f(\theta_m)\|_2 \leq \epsilon\right\}. \tag{S46}$$

If $\mathcal{F}_m$ is the history before the mth particle-filter call, then θ_m and $\{m \leq \min(\tau, M-1)\}$ are $\mathcal{F}_m$-measurable. Lemma S3, with failure probability δ/M, gives

$$\begin{aligned}\Pr(\mathcal{E}_M^c \mid \theta_0) &\leq \sum_{m=0}^{M-1} \mathbb{E}\left[\mathbf{1}_{\{m\leq\min(\tau,M-1)\}} \Pr\left(\|g_m - \nabla f(\theta_m)\|_2 > \epsilon \mid \mathcal{F}_m\right) \middle| \theta_0\right] \\ &\leq \sum_{m=0}^{M-1} \frac{\delta}{M} = \delta.\end{aligned} \tag{S47}$$

This calculation is valid at the adaptive iterates because each call uses a fresh particle-randomness block.

Work on $\mathcal{E}_M$ and write $p_m = -B_m^{-1} g_m$. If $\|g_m\|_2 \leq \sigma\epsilon$, then

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon \leq (\sigma+1)\epsilon. \tag{S48}$$

Strong convexity implies $f(\theta) - f(\theta^*) \leq \|\nabla f(\theta)\|_2^2/(2\gamma)$; therefore

$$f(\theta_m) - f(\theta^*) \leq \frac{(\sigma+1)^2\epsilon^2}{2\gamma}. \tag{S49}$$

Suppose instead that $\|g_m\|_2 > \sigma\epsilon$. Since $g_m = -B_m p_m$ and $cI_p \preceq B_m \preceq CI_p$,

$$c\|p_m\|_2 \leq \|g_m\|_2 \leq C\|p_m\|_2, \qquad \epsilon\|p_m\|_2 < \frac{C}{\sigma}\|p_m\|_2^2. \tag{S50}$$

The smoothness inequality and the gradient-error event give

$$\begin{aligned}f(\theta_m + \eta p_m) - f(\theta_m) &\leq \eta \nabla f(\theta_m)^T p_m + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2 \\ &\leq -\eta p_m^T B_m p_m + \eta\epsilon\|p_m\|_2 + \frac{\Gamma\eta^2}{2}\|p_m\|_2^2.\end{aligned} \tag{S51}$$

The choices of σ and η imply

$$\frac{C}{\sigma}\|p_m\|_2^2 \leq \frac{c(1-\beta)}{2}\|p_m\|_2^2, \qquad \frac{\Gamma\eta}{2}\|p_m\|_2^2 \leq \frac{c(1-\beta)}{2}\|p_m\|_2^2.$$

Using $p_m^T B_m p_m \geq c\|p_m\|_2^2$ in equation S51 now gives

$$f(\theta_{m+1}) - f(\theta_m) \leq -\eta\beta p_m^T B_m p_m. \tag{S52}$$

On a nonstopping iteration,

$$\|\nabla f(\theta_m)\|_2 \leq \|g_m\|_2 + \epsilon < \frac{\sigma+1}{\sigma}\|g_m\|_2. \tag{S53}$$

Consequently, equation S43 and strong convexity give

$$\begin{aligned} p_m^T B_m p_m &= g_m^T B_m^{-1} g_m \\ &\geq \frac{1}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \|\nabla f(\theta_m)\|_2^2 \\ &\geq \frac{2\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2 \{f(\theta_m) - f(\theta^*)\}. \end{aligned} \tag{S54}$$

Combining equation S52 and equation S54 proves

$$f(\theta_{m+1}) - f(\theta^*) \leq q\{f(\theta_m) - f(\theta^*)\}, \qquad q = 1 - \frac{2\eta\beta\gamma}{C}\left(\frac{\sigma}{\sigma+1}\right)^2. \tag{S55}$$

The step-size bound also verifies the claimed range of the contraction:

$$0 < 1 - q < 2\beta(1-\beta)\frac{c\gamma}{C\Gamma} \leq \frac{1}{2}, \qquad \text{so} \qquad \frac{1}{2} < q < 1.$$

Iteration gives the q^m conclusion in the theorem, while equation S49 gives its stopping conclusion after substituting $f = -\ell$. □

The result covers DMOP-1. For $\alpha < 1$, the population limit is a discounted gradient, so extending the theorem requires a finite-sample bound on the discount bias relative to the exact score.

S3 Feynman-Kac Models and Monte Carlo Approximations

In this section, we introduce the Feynman-Kac convention of Del Moral (2004) that has since become commonplace (Karjalainen et al., 2023) for the analysis of the particle filter. The mathematical formalization and notation introduced here will be adopted in the remainder of the analysis, in order to prove Theorems 3, 4, and 5. Let $(\eta_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$ be the prediction and filtering probability measures on the current state space $\mathcal{X}$. The posterior measure ρ_n is on simulator histories through time n, including the base variables, as defined in Equation (S149). Its state-path marginal is $f_{X_{0:n}|Y_{1:n}}$. These are the measures we seek to approximate with the particle filter. For any measurable bounded functional h, we adopt the following functional-analytic notation, borrowed from Del Moral (2004); Chopin and Papaspiliopoulos (2020); Karjalainen et al. (2023). We choose our specific choice of notation and definitions to be in line with that of Karjalainen et al. (2023).

Markov kernels and the process model: A Markov kernel M with source $\mathcal{X}_1$ and target $\mathcal{X}_2$ is a map $M : \mathcal{X}_1 \times \mathcal{B}(\mathcal{X}_2) \to [0,1]$ such that for every set $A \in \mathcal{B}(\mathcal{X}_2)$ and every point $x \in \mathcal{X}_1$, the map $x \mapsto M(x, A)$ is a measurable function of x, and the map $A \mapsto M(x, A)$ is a probability measure on $\mathcal{X}_2$. The quantity $M(x, A)$ can be thought of as the probability of transitioning to the set A given that we are at the point x. If this yields a density, this then corresponds to the process density $f_{X_2|X_1}$ conditional on x and integrated over A.

Markov kernels and measures: For any measure η, any Markov kernel M on $\mathcal{X}$, any point $x \in \mathcal{X}$ and any measurable subset $A \subseteq \mathcal{X}$, let

$$\eta(h) = \int h \, d\eta = \int h(x)\eta(dx), \tag{S56}$$

$$(\eta M)(A) = \int \eta(dx)M(x, A), \tag{S57}$$

$$(Mh)(x) = \int M(x, dy)h(y). \tag{S58}$$

Compositions of Markov kernels: The composition of a Markov kernel M_1 with another Markov kernel M_2 is another Markov kernel, given by

$$(M_1 M_2)(x, A) = \int M_1(x, dy)M_2(y, A). \tag{S59}$$

Total variation distance: The total variation distance between two measures μ and ν on $\mathcal{X}$ is

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{\|h\|_\infty \leq 1/2} |\mu(h) - \nu(h)| = \sup_{\mathrm{osc}(h) \leq 1} |\mu(h) - \nu(h)|. \tag{S60}$$

Dobrushin contraction: The Dobrushin contraction coefficient β_{TV} of a Markov kernel M is given by

$$\beta_{\mathrm{TV}}(M) = \sup_{x,y \in \mathcal{X}} \big\|M(x, \cdot) - M(y, \cdot)\big\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}. \tag{S61}$$

Potential functions and the measurement model: A potential function $G : \mathcal{X} \to [0, \infty)$ is a non-negative function of an element of the state space $x \in \mathcal{X}$. In our case, this corresponds to the measurement model, and in our previous notation is written as $g_{n,j} = f_{Y_n|X_n}\big(y_n^*|x_{n,j}^P; \theta\big) = G_n(x_{n,j}^P)$, where we suppress the dependence on θ for notational simplicity. Note that $G_n(\cdot) = f_{Y_n|X_n}(y_n^*| \cdot)$ is the conditional density of the observed measurement at time n, evaluated at a state in $\mathcal{X}$. At the filtered particle, $G_n(x_{n,j}^F) = g_{n,k_{n,j}}$.

Feynman-Kac models: A Feynman-Kac model on $\mathcal{X}$ is a tuple $(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N)$ of an initial probability measure on the state space π_0, a sequence of transition kernels $(M_n)_{n=1}^N$, and a sequence of potential functions $(G_n)_{n=1}^N$. In the notation used in the main text, this corresponds to the starting distribution f_{X_0}, the sequence of transition densities $f_{X_n|X_{n-1}}$, and the measurement densities $f_{Y_n|X_n}$. This induces a set of mappings from the set of probability measures on $\mathcal{X}$ to itself, $\mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$, as follows:

- The update from the prediction to the filtering distributions is given by

$$\pi_n(dx) = \Psi_n(\eta_n)(dx) = \frac{G_n(x) \cdot \eta_n(dx)}{\eta_n(G_n)}. \tag{S62}$$

- The map from the prediction distribution at timestep n to timestep $n+1$ is given by

$$\Phi_{n+1}(\eta_n) = \Psi_n(\eta_n) M_{n+1}. \tag{S63}$$

- The composition of maps between prediction distributions yields the map from the prediction distribution at time k to the prediction distribution at time n where $k \leq n$,

$$\Phi_{k,n} = \Phi_n \circ ... \circ \Phi_{k+1}. \tag{S64}$$

The particle filter: The particle filter then yields a Monte Carlo approximation to the above Feynman-Kac model, via a sequence of mixture Dirac measures. When one resamples at every timestep, the prediction measure at timestep n is then given by

$$\eta_n^J = \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^P}, \tag{S65}$$

and the filtering measure at timestep n is given by

$$\pi_n^J = \frac{\sum_{j=1}^J g_{n,j} \delta_{x_{n,j}^P}}{\sum_{j=1}^J g_{n,j}} \approx \frac{1}{J} \sum_{j=1}^{J} \delta_{x_{n,j}^F}. \tag{S66}$$

In a slight abuse of notation, we will identify $x_{n,1:J}^P \equiv \eta_n^J$, and $x_{n,1:J}^F \equiv \pi_n^J$. As in Karjalainen et al. (2023), one can view this as an inhomogenous Markov process evolving on $\mathcal{X}^J$. For $\theta = \phi$ and multinomial resampling, the corresponding Markov transition kernel is

$$\mathbf{M}_n(x_{n-1,1:J}^P, \cdot) = \left(\Phi_n\left(\eta_{n-1}^J\right)\right)^{\otimes J} = \left(\Phi_n\left(\frac{1}{J}\sum_{j=1}^{J} \delta_{x_{n-1,j}^P}\right)\right)^{\otimes J}, \tag{S67}$$

and the composition of Markov kernels on particles from timestep n to timestep $n+k$ is written

$$\mathbf{M}_{n,n+k} = \mathbf{M}_{n+1} \cdots \mathbf{M}_{n+k}. \tag{S68}$$

The kernel acts on the full prediction cloud. Conditional on the preceding prediction cloud, its law is

$$\text{Law}\left(x_{n,1:J}^P \mid x_{n-1,1:J}^P\right) = \mathbf{M}_n(x_{n-1,1:J}^P, \cdot). \tag{S69}$$

Forgetting of the particle filter: The above formalization yields a result from Karjalainen et al. (2023) on the forgetting of the particle filter that we require for our analysis of the bias, variance, and error of MOP-α. That is, Karjalainen et al. (2023) show that

$$\beta_{\text{TV}}\left(\mathbf{M}_{n,n+k}\right) \leq (1-\epsilon)^{\lfloor k/(O(\log J))\rfloor}, \tag{S70}$$

for some ϵ dependent on $\bar{G}, \underline{\text{G}}, \bar{M}, \underline{\text{M}}$ in Assumptions (A3) and (A2). As a result, the mixing time of the particle filter is only on the order of $O(\log(J))$ timesteps.

S4 Proof of Theorem 3 via a Strong Law of Large Numbers for Triangular Arrays of Particles With Off-Parameter Resampling

Here, we prove Theorem 3 by deriving more general results, from which it will be clear that MOP-α targets the filtering distribution and is strongly consistent for the likelihood. We will prove a strong law of large numbers for triangular arrays of particles with off-parameter resampling, meaning that we resample the particles according to an arbitrary resampling rule that is not necessarily in proportion to the target distribution of interest. Using weights that encode the cumulative discrepancy between the resampling distribution and the target distribution (instead of resampling to equal weights, as in the basic particle filter) provides a sufficient correction to ensure almost sure convergence. We now introduce the precise definition of an off-parameter resampled particle filter.

Definition S1 (Off-Parameter Resampled Particle Filters)**.** *An off-parameter resampled particle filter is a Monte Carlo approximation to a Feynman-Kac model* $\left(\pi_0, (M_n)_{n=1}^N, (G_n)_{n=1}^N\right)$*, where we inductively define given some* $x_{n-1,j}^F$*,* $w_{n-1,j}^F$ *comprising the filtering measure approximation at time* $n-1$*:*

1. *Conditional on the filtered particles and weights at time* $n-1$*, draw independently* $x_{n,j}^P \sim M_n(x_{n-1,j}^F, \cdot)$ *using fresh simulator draws. Retain the corresponding parent weight:* $w_{n,j}^P = w_{n-1,j}^F$*.*
2. *The prediction measure at timestep* n *is given by*
$$\eta_n^J = \frac{\sum_{j=1}^J w_{n,j}^P \, \delta_{x_{n,j}^P}}{\sum_{j=1}^J w_{n,j}^P}. \tag{S71}$$
3. *The particles are resampled at every timestep* n*, yielding indices* k_j *according to some arbitrary probabilities* $(p_{n,j})_{j=1}^J$*.*
4. *The filtering measure at timestep* n *is given by*
$$\pi_n^J = \frac{\sum_{j=1}^J \delta_{x_{n,j}^P} w_{n,j}^P \, g_{n,j}}{\sum_{j=1}^J w_{n,j}^P \, g_{n,j}} \approx \frac{\sum_{j=1}^J \delta_{x_{n,k_j}^P} w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}}{\sum_{j=1}^J w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}} = \frac{\sum_{j=1}^J w_{n,j}^F \, \delta_{x_{n,j}^F}}{\sum_{j=1}^J w_{n,j}^F}. \tag{S72}$$
5. *The prediction weights at the next timestep are given by* $w_{n+1,j}^P = w_{n,j}^F = w_{n,k_j}^P \, g_{n,k_j}/p_{n,k_j}$*.*

To prove that the weight correction is sufficient for almost sure convergence, we first introduce what it means for a triangular array of particles to *target* a given target distribution.

Definition S2 (Targeting)**.** *A pair of random vectors* (X, W) *drawn from some measure* g ***targets*** *another measure* π *if, for any measurable and bounded functional* h*,*
$$E_g\big[h(X) \cdot W\big] = E_\pi\big[h(X)\big]. \tag{S73}$$
A particle approximation is a triangular array of pairs of random vectors $(X_j^J, W_j^J), j = 1, 2, \ldots, J$*. The particle approximation* ***targets*** π *if, for any measurable and bounded functional* h*,*
$$\frac{\sum_{j=1}^J h(X_j^J) \, W_j^J}{\sum_{j=1}^J W_j^J} \xrightarrow{a.s.} E_\pi\big[h(X)\big] \tag{S74}$$
as $J \to \infty$*.*

Chopin (2004) asserted without proof that common particle filter algorithms target the filtering distribution in this sense, while Chopin and Papaspiliopoulos (2020) proved a related result assuming bounded densities. We follow a similar approach to Chopin and Papaspiliopoulos (2020), based on showing strong laws of large numbers for triangular arrays, noting that triangular array strong laws do not generally hold without an additional regularity condition such as boundedness.

In order to prove the consistency of our variation on the particle filter, we now present three helper lemmas. The first follows from standard importance sampling arguments, the second from integrating out the marginal, and the third from Bayes' theorem. We state Lemma S5 assuming multinomial resampling, which is convenient for the proof though other resampling strategies may be preferable in practice.

Lemma S5 (Change of Weight Measure)**.** *Suppose that* $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Now, let* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *be a multinomial sample with indices* k_j *drawn from* $\{(\tilde{X}_j^J, U_j^J)\}$ *where* $(\tilde{X}_j^J, U_j^J)$ *is represented, on average, proportional to* $\pi_j^J J$ *times, with* $\pi_j^J > 0$*. Write*

$$(Y_j^J, V_j^J) = \big(\tilde{X}_{k_j}^J, U_{k_j}^J / \pi_{k_j}^J\big). \tag{S75}$$

If the importance sampling weights U_j/π_j *are bounded uniformly in* j, J *after normalizing* $\sum_m U_m^J = 1$*, then* $\{(Y_j^J, V_j^J), j = 1, \ldots, J\}$ *targets* f_X*.*

Proof. Normalize $\sum_j U_j^J = 1$, which leaves both weighted empirical measures unchanged. Note that as the Y_j^J are a subsample from X_j^J, h can be a function of Y as well as it is one for X. We then expand

$$\frac{\sum_j h(Y_j^J)\, V_j^J}{\sum_j V_j^J} = \frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}}. \tag{S76}$$

By hypothesis,

$$\frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}\big[h(X)\big]. \tag{S77}$$

We want to show

$$\frac{\sum_j h(\tilde{X}_{k_j}^J)\, \frac{U_{k_j}^J}{\pi_{k_j}^J}}{\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J}} - \frac{\sum_j h(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} 0. \tag{S78}$$

For this, it is sufficient to show that

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}\pi_j^J \overset{a.s.}{\to} 0 \tag{S79}$$

since an application of this result with $h(x) = 1$ provides almost sure convergence to 1 of J^{-1} times the denominator in (S78). Write $g_j^J = h(\tilde{X}_j^J)\frac{U_j^J}{\pi_j^J}$. We therefore need to show that

$$\frac{1}{J}\sum_j Z_j^J := \frac{1}{J}\sum_j \left(g_{k_j}^J - \sum_{m=1}^J g_m^J \pi_m^J \right) \overset{a.s.}{\to} 0. \tag{S80}$$

Because the functional h and importance sampling weights $U_{k_j}^J/\pi_{k_j}^J$ are bounded, we have that $\sup_{J,j} \mathbb{E}\big[(Z_j^J)^4\big] < \infty$. We can then follow the argument of Chopin and Papaspiliopoulos (2020) from this point on, where noting that the Z_j^J are conditionally independent, identically distributed, and centered given the $\tilde{X}_j^J$, U_j^J, and π_j^J,

$$\begin{aligned}
\mathbb{E}\left[\left(\sum_j Z_j^J\right)^4 \middle| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\right] &= J\,\mathbb{E}\left[(Z_1^J)^4 | (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\right] \\
&\quad + 3J(J-1)\left(\mathbb{E}\big[(Z_1^J)^2 \big| (\tilde{X}_j^J, U_j^J, \pi_j^J)_{j=1}^J\big]\right)^2 \quad \text{(S81)} \\
&\leq CJ^2, \quad \text{(S82)}
\end{aligned}$$

for some $C > 0$ independent of J. Taking expectations on both sides yields

$$\mathbb{E}\left[\left(\sum_j Z_j^J\right)^4\right] \leq CJ^2 \quad \text{(S83)}$$

by the tower property. Now by Markov's inequality,

$$\mathbb{P}\left(\left|\frac{1}{J}\sum_{j=1}^J Z_j^J\right| > \epsilon\right) \leq \frac{1}{\epsilon^4 J^4}\mathbb{E}\left[\left(\sum_{j=1}^J Z_j^J\right)^4\right] \leq \frac{C}{\epsilon^4 J^2}, \quad \text{(S84)}$$

and as these terms are summable we can apply Borel-Cantelli to conclude that these deviations happen only finitely often for every $\epsilon > 0$, giving us the almost-sure convergence for

$$\frac{1}{J}\sum_j h(\tilde{X}_{k_j}^J)\,\frac{U_{k_j}^J}{\pi_{k_j}^J} - \sum_j h(\tilde{X}_j^J)\,\frac{U_j^J}{\pi_j^J}\,\pi_j^J \overset{a.s.}{\to} 0. \quad \text{(S85)}$$

Similarly, we also have that

$$\sum_j \frac{U_j^J}{\pi_j^J}\pi_j^J - \frac{1}{J}\sum_j \frac{U_{k_j}^J}{\pi_{k_j}^J} \overset{a.s.}{\to} 0, \quad \text{(S86)}$$

and the result follows by taking the ratio, since $\sum_j U_j^J = 1$.

□

Remark: Note that Lemma S5 permits $\pi_{1:J}$ to depend on $\{(X_j, u_j)\}$ as long as the resampling is carried out independently of $\{(X_j, u_j)\}$, conditional on $\pi_{1:J}$.

Lemma S6 (Particle Marginals)**.** *Suppose that $\{(\tilde{X}_j^J, U_j^J), j = 1, \ldots, J\}$ targets f_X. Also suppose that conditional on $(\tilde{X}_m^J, U_m^J)_{m=1}^J$, the variables $\tilde{Z}_1^J, \ldots, \tilde{Z}_J^J$ are independent, with $\tilde{Z}_j^J \sim f_{Z|X}(\cdot \mid \tilde{X}_j^J)$, where $f_{Z|X}$ is a conditional probability density function corresponding to a joint density $f_{X,Z}$ with marginal densities f_X and f_Z. Then, if the U_j^J are uniformly bounded above and away from zero, $\{(\tilde{Z}_j^J, U_j^J)\}$ targets f_Z.*

Proof. We want to show that, for any measurable bounded h,

$$\frac{\sum_j h(\tilde{Z}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)] = \mathbb{E}_{f_X}\big[\mathbb{E}_{f_{Z|X}}[h(Z) \mid X]\big]. \tag{S87}$$

By assumption, for any measurable and bounded functional g with domain $\mathcal{X}$,

$$\frac{\sum_j g(\tilde{X}_j^J)\, U_j^J}{\sum_j U_j^J} \overset{a.s.}{\to} \mathbb{E}_{f_X}[g(X)]. \tag{S88}$$

Let $\bar{U}_j^J = JU_j^J / \sum_j U_j^J$. For constants $0 < c \le U_j^J \le C < \infty$, uniformly in j, J, we have $\bar{U}_j^J \le C/c$. Examine the numerator and denominator of the quantity

$$\frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} = \frac{J^{-1}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J}{J^{-1}\sum_j \bar{U}_j^J}. \tag{S89}$$

The denominator converges to 1 almost surely. The numerator, on the other hand, is

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J. \tag{S90}$$

Set

$$D_J = \frac{1}{J}\sum_j \bar{U}_j^J \big\{h(\tilde{Z}_j^J) - \mathbb{E}[h(Z) \mid X = \tilde{X}_j^J]\big\}.$$

Conditioning on $(\tilde{X}_j^J, U_j^J)_{j=1}^J$, we apply the fourth-moment argument of Lemma S5 to the centered, uniformly bounded summands, obtaining

$$\mathbb{E}[D_J^4] \le \frac{C_h}{J^2}, \qquad \mathbb{P}(|D_J| > \epsilon) \le \frac{C_h}{\epsilon^4 J^2},$$

for some $C_h < \infty$ independent of J and every $\epsilon > 0$. Thus $D_J \overset{a.s.}{\to} 0$ by Borel–Cantelli. For the expectation of (S90), the tower property gives

$$\lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J\right] = \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\big[h(\tilde{Z}_j^J)\, \bar{U}_j^J \mid \tilde{X}_j^J, \bar{U}_j^J\big]\right] \tag{S91}$$

$$= \lim_{J\to\infty} \mathbb{E}\left[\frac{1}{J}\sum_j \mathbb{E}\big[h(Z) \mid X = \tilde{X}_j^J\big]\, \bar{U}_j^J\right]. \tag{S92}$$

Applying (S88) with $g(x) = \mathbb{E}[h(Z) \mid X = x]$, the average on the right hand side converges almost surely to $\mathbb{E}\{\mathbb{E}[h(Z) \mid X]\} = \mathbb{E}[h(Z)]$. It remains to swap the limit and expectations. We can do so with the bounded convergence theorem. The almost-sure convergence of this average, together with $D_J \overset{a.s.}{\to} 0$, gives

$$\frac{1}{J}\sum_j h(\tilde{Z}_j^J)\, \bar{U}_j^J \overset{a.s.}{\to} \mathbb{E}_{f_Z}[h(Z)]. \tag{S93}$$

□

Remark. The conditional independence assumption in Lemma S6 holds for standard particle filters, and for our MOP filter it is guaranteed by step 1 of Definition S1.

Remark: The uniform boundedness condition in Lemma S6 can be guaranteed by construction in the context of Definition S1 for suitable choices of the resampling probabilities $\pi_{1:J}$ at an earlier iteration. In the context of DMOP-α, this is guaranteed if the measurement density is similarly bounded, which holds under Assumption (A2) for our theoretical analysis. It is trivially true for MOP-α at $\theta = \phi$, since in this case the weights at each propagation step are equal. The smoothness assumption (A5) therefore also guarantees this condition in a neighborhood of $\theta = \phi$.

Remark. For nonnegative weights with positive sum, the weight condition in Lemma S6 can be replaced by the weaker requirement

$$\max_{1\le j\le J} \frac{JU_j^J}{\sum_m U_m^J} \le B < \infty,$$

almost surely, for a constant B independent of J. The same proof applies.

Lemma S7 (Particle Posteriors)**.** *Suppose that* $\{(X_j^J, U_j^J), j = 1, \ldots, J\}$ *targets* f_X*. Also suppose that* $(X_j'^J, U_j'^J) = \big(X_j^J, U_j^J\, f_{Z|X}(z^*|X_j^J)\big)$*. Then, if* $x \mapsto f_{Z|X}(z^*|x)$ *is bounded and* $0 < f_Z(z^*) < \infty$*,* $\{(X_j'^J, U_j'^J)\}$ *targets* $f_{X|Z}(\cdot|z^*)$*.*

Proof. Again, we want to show that, for any measurable bounded h,

$$\frac{\sum_j h(X_j^J)\cdot U_j^J \cdot f_{Z|X}(z^*|X_j^J)}{\sum_j U_j^J \cdot f_{Z|X}(z^*|X_j^J)} \xrightarrow{a.s.} \mathbb{E}_{f_{X|Z}}\big[h(X)\big|z^*\big]. \tag{S94}$$

We already have that for any measurable bounded g,

$$\frac{\sum_j g(X_j^J)\, U_j^J}{\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[g(X)\big]. \tag{S95}$$

Consider the following:

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \times \left(\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J}\right)^{-1}. \tag{S96}$$

The functions $x \mapsto f_{Z|X}(z^*|x)$ and $x \mapsto h(x) f_{Z|X}(z^*|x)$ are bounded. We will apply Equation (S95) to the numerator and the denominator in the ratio above individually. The numerator converges to

$$\frac{J^{-1}\sum_j h(X_j^J)\, f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big], \tag{S97}$$

while the denominator converges to

$$\frac{J^{-1}\sum_j f_{Z|X}(z^*|X_j^J)\, U_j^J}{J^{-1}\sum_j U_j^J} \xrightarrow{a.s.} \mathbb{E}_{f_X}\big[f_{Z|X}(z^*|X)\big] = f_Z(z^*). \tag{S98}$$

Now we take advantage of the identities

$$\frac{\mathbb{E}_{f_X}\big[h(X)\, f_{Z|X}(z^*|X)\big]}{f_Z(z^*)} \;=\; \mathbb{E}_{f_X}\left[\frac{h(X)\, f_{Z|X}(z^*|X)}{f_Z(z^*)}\right] \tag{S99}$$

$$= \mathbb{E}_{f_X}\left[h(X)\frac{f_{X|Z}(X|z^*)}{f_X(X)}\right] = \mathbb{E}_{f_{X|Z}}\left[h(X)|z^*\right], \tag{S100}$$

to give the desired result.

□

Remark. In the MOP application, the boundedness and positivity conditions in Lemma S7 follow from Assumption (A3).

Theorem S1 (Off-Parameter Resampled Particle Filters Target the Filtering Distribution)**.** *The off-parameter resampled particle filter of Definition S1 targets the filtering distribution when the resampling probabilities $(p_{n,j})_{j=1}^J$ are chosen so that $g_{n,j}/(J\,p_{n,j})$ is uniformly bounded above and away from zero whenever $g_{n,j} > 0$.*

Proof. Recursively applying Lemmas S5, S6, and S7, we obtain that the off-parameter resampled particle filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X_{n-1,j}^F, w_{n-1,j}^F\right)\right\}$ targets π_{n-1}. Then, Lemma S6 tells us that $\left\{\left(X_{n,j}^P, w_{n,j}^P\right)\right\}$ targets η_n. Lemma S7 tells us that $\left\{\left(X_{n,j}^P, w_{n,j}^P g_{n,j}^\theta\right)\right\}$ therefore targets π_n. Lemma S5 guarantees that the resampling rule, given by

$$\left(X_{n,j}^F, w_{n,j}^F\right) = \left(X_{n,k_j}^P, w_{n,k_j}^P\, g_{n,k_j}/p_{n,k_j}\right), \tag{S101}$$

with resampling probabilities proportional to $p_{n,j}$, therefore also targets π_n. □

Proposition S1 (MOP-1 Targets the Filtering Distribution)**.** *When $\alpha = 1$ or $\phi = \theta$, MOP-α targets the filtering distribution.*

Proof. When $\theta = \phi$, regardless of the value of α, the ratio $\frac{g_{n,j}^\theta}{g_{n,j}^\phi} = 1$, and this reduces to the vanilla particle filter estimate.

When $\alpha = 1$, and $\theta \neq \phi$, the proof is identical to that of S1. Recursively applying Lemmas S5, S6, and S7, we obtain that the MOP-1 filter targets the posterior. Specifically, suppose inductively that $\left\{\left(X_{n-1,j}^{F,\theta}, w_{n-1,j}^{F,\theta}\right)\right\}$ is properly weighted for $f_{X_{n-1}|Y_{1:n-1}}(x_{n-1}|y_{1:n-1}^*;\theta)$. Then, Lemma S6 tells us that $\left\{\left(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\right)\right\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. Lemma S7 tells us that $\left\{\left(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta}\, g_{n,j}^\theta\right)\right\}$ therefore targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. Lemma S5 guarantees that the resampling rule, given by

$$\left(X_{n,j}^{F,\theta}, w_{n,j}^{F,\theta}\right) = \left(X_{n,k_j}^{P,\theta}, w_{n,k_j}^{P,\theta}\, g_{n,k_j}^\theta/g_{n,k_j}^\phi\right), \tag{S102}$$

with resampling weights proportional to $g_{n,j}^\phi$, therefore also targets $f_{X_n|Y_{1:n}}(x_n|y_{1:n}^*;\theta)$. □

This has addressed filtering, but not quite yet the likelihood evaluation. For this we use the following lemma.

Lemma S8 (Likelihood Proper Weighting)**.** *$f_{Y_n|Y_{1:n-1}}(y_n^*|y_{1_n-1}^*;\theta)$ is strongly consistently estimated by either the before-resampling estimate,*

$$L_n^{B,\theta} = \frac{\sum_{j=1}^J g_{n,j}^\theta\, w_{n,j}^{P,\theta}}{\sum_{j=1}^J w_{n,j}^{P,\theta}}, \tag{S103}$$

or by the after-resampling estimate,

$$L_n^{A,\theta} = L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}}. \tag{S104}$$

where L_n^{ϕ} is as defined in the various algorithms.

Here, (S103) is a direct consequence of our earlier result that $\{(X_{n,j}^{P,\theta}, w_{n,j}^{P,\theta})\}$ targets $f_{X_n|Y_{1:n-1}}(x_n|y_{1:n-1}^*;\theta)$. To see (S104), define the corrected weights before resampling and the reference resampling probabilities by

$$w_{n,j}^{FC,\theta} = w_{n,j}^{P,\theta} \frac{g_{n,j}^{\theta}}{g_{n,j}^{\phi}}, \qquad p_{n,j} = \frac{g_{n,j}^{\phi}}{JL_n^{\phi}}, \qquad \sum_{j=1}^{J} p_{n,j} = 1. \tag{S105}$$

The numerator of (S103) can then be written as

$$\sum_{j=1}^{J} w_{n,j}^{P,\theta} g_{n,j}^{\theta} = JL_n^{\phi} \sum_{j=1}^{J} p_{n,j} w_{n,j}^{FC,\theta}. \tag{S106}$$

Resampling selects $w_{n,j}^{F,\theta} = w_{n,k_j}^{FC,\theta}$. With the pre-resampling particles, weights, and reference densities fixed, the expectation over multinomial resampling satisfies

$$\begin{aligned} \mathbb{E}_{\text{res}}\left[L_n^{\phi} \sum_{j=1}^{J} w_{n,j}^{F,\theta} \right] &= JL_n^{\phi} \sum_{m=1}^{J} p_{n,m} w_{n,m}^{FC,\theta} \\ &= \sum_{m=1}^{J} w_{n,m}^{P,\theta} g_{n,m}^{\theta}. \end{aligned} \tag{S107}$$

The same conditional triangular-array argument as in Lemma S5 gives consistency of this resampling approximation under its boundedness conditions. Dividing by the unchanged prediction-weight sum gives (S104). Using Lemma S8, we obtain a likelihood estimate,

$$L^{A,\theta} = \prod_{n=1}^{N} \left(L_n^{\phi} \frac{\sum_{j=1}^{J} w_{n,j}^{F,\theta}}{\sum_{j=1}^{J} w_{n,j}^{P,\theta}} \right). \tag{S108}$$

Since $w_{n,j}^{F,\theta} = w_{n+1,j}^{P,\theta}$, this is a telescoping product. The remaining terms are $\sum_{j=1}^{J} w_{0,j}^{F,\theta} = J$ on the denominator and $\sum_{j=1}^{J} w_{N,j}^{F,\theta}$ on the numerator. This derives the MOP-1 likelihood estimates.

$L^{B,\theta}$ should generally be preferred in practice, since there is no reason to include the extra variability from resampling when calculating the conditional log likelihood, but it lacks the nice telescoping product that lets us derive exact expressions for the gradient in Theorem 1.

S5 Proof of Theorem 4 via Consistency of Off-Parameter Resampled Gradient Estimates

We now show that, under sufficient local regularity conditions, one can interchange differentiation and the particle limit. We apply Arzelà–Ascoli, working on a compact neighborhood of the

evaluation parameter. Consistency is used for the estimates themselves: differentiating a weighted particle estimate also differentiates its particles, weights, and normalizer, so boundedness alone does not make its derivative an empirical average to which Theorem S1 applies.

Theorem S2 (Off-Parameter Particle Filters Yield Strongly Consistent Estimates of Derivatives of Functionals)**.** *Let K be a closed ball of positive radius centered at ϕ and contained in the interior of $\Theta \subset \mathbb{R}^p$. Let $F_J(\theta)$ be scalar particle estimates, such as $\eta_n^J(h_\theta)$, defined on a common probability space. On one probability-one event, suppose every F_J has two continuous derivatives on a neighborhood of K, with*

$$\sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta F_J(\theta)\|_2 \leq G^*, \qquad \sup_{J\geq 1}\sup_{\theta\in K}\|\nabla_\theta^2 F_J(\theta)\|_{\mathrm{op}} \leq H^*$$

for finite constants G^*, H^*. Suppose $F_J(\theta) \to h_\theta^*$ almost surely for each fixed $\theta \in K$, where h^* is deterministic. Then h^* is continuously differentiable in the interior of K, and almost surely

$$\nabla_\theta F_J(\theta) \longrightarrow \nabla_\theta h_\theta^*$$

uniformly on every compact subset of that interior, in particular at $\theta = \phi$.

Proof. For each fixed pair $x, y \in K$, pointwise almost-sure convergence and the gradient bound give $|h_y^* - h_x^*| \leq G^*\|y - x\|_2$. Thus the deterministic function h^* is Lipschitz, and in particular continuous, on K.

Choose a countable dense subset D of K. Intersect the probability-one event of the derivative bounds with the events of convergence at each $\theta \in D$, and fix ω in this intersection. Suppress ω below. By the gradient bound, the functions F_J are uniformly Lipschitz on the convex set K. Convergence on D, compactness of K, and continuity of h^* therefore imply $F_J \to h^*$ uniformly on K: for any $\epsilon > 0$, choose a sufficiently fine finite net in D, use convergence at its finitely many points, and use the common Lipschitz bound and uniform continuity of h^* between those points.

The gradients are uniformly bounded and, by the Hessian bound, uniformly equicontinuous. Explicitly, for $\theta, \theta' \in K$,

$$\|\nabla_\theta F_J(\theta) - \nabla_\theta F_J(\theta')\|_2 \leq H^*\|\theta - \theta'\|_2. \tag{S109}$$

Thus, by Arzelà–Ascoli, every subsequence of gradients has a further subsequence converging uniformly on K to a continuous function g. We claim that every such limit is $\nabla_\theta h^*$ in the interior of K. For $x, y \in K$, the fundamental theorem of calculus gives

$$F_J(y) - F_J(x) = \int_0^1 \nabla_\theta F_J\big(x + t(y - x)\big)^{\mathsf{T}}(y - x)\, dt. \tag{S110}$$

Passing to the limit along the convergent subsequence yields

$$h_y^* - h_x^* = \int_0^1 g\big(x + t(y - x)\big)^{\mathsf{T}}(y - x)\, dt. \tag{S111}$$

Taking $y = x + te_i$ for each coordinate direction and letting $t \to 0$ shows that $\partial_{\theta_i} h_x^* = g_i(x)$ for every interior x. Since g is continuous, h^* is continuously differentiable there. Every subsequential gradient limit is therefore the same on the interior, and hence on K by continuity. Relative compactness and uniqueness of this limit imply convergence of the full gradient sequence. In particular, almost surely, for interior θ,

$$\lim_{J\to\infty} \nabla_\theta F_J(\theta) = \nabla_\theta \lim_{J\to\infty} F_J(\theta) = \nabla_\theta h_\theta^*, \tag{S112}$$

with convergence uniform on every compact subset of the interior, as required. □

For $F_J(\theta) = \eta_n^J(h_\theta)$, the conclusion is convergence to $\nabla_\theta \eta_n(h_\theta)$. No identity with $\eta_n(\nabla_\theta h_\theta)$ is asserted, since the target distribution can itself depend on θ.

Theorem 4 follows by taking $F_J(\theta) = \hat{\ell}_J^1(\theta)$ and $h_\theta^* = \ell(\theta)$. Keep the reference parameter ϕ fixed throughout K. For $\alpha = 1$, the off-parameter consistency result and Lemma S8 give $\hat{\mathcal{L}}_J^1(\theta) \to \mathcal{L}(\theta) > 0$ almost surely for each fixed $\theta \in K$, not only at $\theta = \phi$. The continuous mapping theorem gives the corresponding convergence of the log likelihoods. The neighborhood form of Assumption (A4) supplies their uniform gradient and Hessian bounds, and Assumption (A5) supplies their pathwise smoothness with the simulator draws and reference-filter resampling indices held fixed. Applying the theorem to these log likelihoods establishes score consistency at ϕ directly.

S6 Variance Analysis for DMOP-0

We first study DMOP-0 to introduce the particle variance and covariance estimates used in Section S7. Under (A6-J), the measurement gradient is a bounded function of the current joint state, allowing a direct particle-cloud mixing argument. Under (A6), its dependence on simulator history requires the additional decomposition developed in Section S7. We distinguish these two arguments below; the general discounting and MSE bounds are established in Section S7.

First, we relate cloud contraction to the strong-mixing coefficient used in the covariance bounds below.

Lemma S9. *Let $\mathcal{F}_m = \sigma(x_{0,1:J}^P, \ldots, x_{m,1:J}^P)$ and $\mathcal{G}_n = \sigma(x_{n,1:J}^P, \ldots, x_{N,1:J}^P)$ be the past and future sigma-fields of the Markov particle cloud. For $1 \le \ell \le N$, define*

$$a(\ell) = \sup_{0 \le m \le N-\ell} \sup_{A \in \mathcal{F}_m,\, B \in \mathcal{G}_{m+\ell}} \big| \Pr(A \cap B) - \Pr(A)\Pr(B) \big|. \tag{S113}$$

Then

$$a(\ell) \le \sup_{0 \le m \le N-\ell} \beta_{\mathrm{TV}}(\boldsymbol{M}_{m,m+\ell}). \tag{S114}$$

The same statement applies to the joint state–derivative cloud under (A6-J), using its transition kernel.

Proof. Fix $A \in \mathcal{F}_m$ and $B \in \mathcal{G}_n$, where $n = m + \ell$. By the Markov property, with $h(x) = \Pr(B \mid x_{n,1:J}^P = x) \in [0,1]$,

$$\Pr(B \mid \mathcal{F}_m) = (\mathbf{M}_{m,n} h)(x_{m,1:J}^P). \tag{S115}$$

The expectation of this conditional probability is $\Pr(B)$, so

$$\begin{aligned} \big| \Pr(A \cap B) - \Pr(A)\Pr(B) \big| &= \big| \mathbb{E}\left[\mathbf{1}_A \left\{ (\mathbf{M}_{m,n} h)(x_{m,1:J}^P) - \Pr(B) \right\} \right] \big| \\ &\le \operatorname{osc}(\mathbf{M}_{m,n} h) \le \beta_{\mathrm{TV}}(\mathbf{M}_{m,n}) \operatorname{osc}(h) \\ &\le \beta_{\mathrm{TV}}(\mathbf{M}_{m,n}). \end{aligned} \tag{S116}$$

Taking the suprema proves the claim. The coefficient is defined through sigma-fields, so the argument does not require joint Lebesgue densities. □

The same argument bounds dependence between past and future windows of simulation and resampling updates. A past window ending after resampling at time b is absorbed into the next

prediction cloud at $b+1$; condition there and at the prediction cloud starting the future window, and use the intervening cloud kernel. Fresh update variables are conditionally independent of earlier updates given the incoming cloud. Under (A6), apply this to the physical cloud and the derivative-age windows in Equation (S271); under (A6-J), use the joint cloud. The number of intervening transitions determines the lag in $a(\ell)$.

Assumptions (A2) and (A3) allow us to use the particle-filter forgetting results of Karjalainen et al. (2023). To control the measurement gradients, we additionally require either (A6) or (A6-J), stated in Section S7.

Theorem S3 (Variance of the DMOP-0 gradient estimate)**.** *Suppose the observations are fixed, $J \geq 2$, $\theta = \phi$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Then*

$$\mathrm{Var}\{\nabla_\theta \hat{\ell}^0(\theta)\} \lesssim \frac{NpG'(\theta)^2}{J} \begin{cases} (1+\log J)^2, & \textit{under (A6)}, \\ 1+\log J, & \textit{under (A6-J)}. \end{cases} \tag{S117}$$

Proof. First assume (A6-J). In this calculation, a current particle means the pair $(X_n, \partial_\theta X_n)$, and the particle-cloud transition is the one induced by its joint kernel. When $\alpha = 0$, we have

$$\hat{\mathcal{L}}^0(\theta) := \prod_{n=1}^N L_n^{A,\theta,0} = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{\sum_{j=1}^J w_{n,j}^{F,\theta,0}}{\sum_{j=1}^J w_{n,j}^{P,\theta,0}} \right) \tag{S118}$$

$$= \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) = \prod_{n=1}^N \left(L_n^\phi \cdot \frac{1}{J} \sum_{j=1}^J \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi} \right). \tag{S119}$$

Similarly to the proof of Lemma S2, we define

$$s_{n,j} = \frac{g_{n,k_{n,j}}^\theta}{g_{n,k_{n,j}}^\phi}, \tag{S120}$$

and from the exact same arguments conclude that its gradient when $\theta = \phi$ is therefore

$$\nabla_\theta \hat{\ell}^0(\theta) := \sum_{n=1}^N \nabla_\theta \log \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right) \tag{S121}$$

$$= \sum_{n=1}^N \frac{\nabla_\theta \left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)}{\left(L_n^\phi \frac{1}{J} \sum_{j=1}^J s_{n,j} \right)} \tag{S122}$$

$$= \sum_{n=1}^N \frac{\sum_{j=1}^J \nabla_\theta s_{n,j}}{\sum_{j=1}^J s_{n,j}} \tag{S123}$$

$$= \sum_{n=1}^N \frac{1}{J} \sum_{j=1}^J \frac{\nabla_\theta f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)}{f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\phi}; \phi\right)} \tag{S124}$$

$$= \frac{1}{J} \sum_{n=1}^N \sum_{j=1}^J \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right) \right), \tag{S125}$$

where we use the derivative of the logarithm, observe that $\sum_{j=1}^{J} s_{n,j} = J$ when $\theta = \phi$, and use the derivative of the logarithm where $\theta = \phi$ again. We do this to establish that the gradient of the log-likelihood estimate is given by the sum of terms over all N and J:

$$\nabla_\theta \hat{\ell}^0(\theta) = \frac{1}{J} \sum_{n=1}^{N} \sum_{j=1}^{J} \nabla_\theta \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)\right). \tag{S126}$$

Therefore, for a given θ_i in the parameter vector θ,

$$\begin{aligned}
\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}^0(\theta)\right) &= \frac{1}{J^2} \mathrm{Var}\left(\sum_{n=1}^{N} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}\left(y_n^* | x_{n,j}^{F,\theta}; \theta\right)\right)\right) && \text{(S127)} \\
&= \frac{1}{J^2} \sum_{n=1}^{N} \mathrm{Var}\left(\sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)\right)\right) \\
&\quad + 2 \sum_{m<n} \mathrm{Cov}\left(\frac{1}{J} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_m|X_m}(y_m^* | x_{m,j}^{F,\theta}; \theta)\right), \frac{1}{J} \sum_{j=1}^{J} \frac{\partial}{\partial_{\theta_i}} \log \left(f_{Y_n|X_n}(y_n^* | x_{n,j}^{F,\theta}; \theta)\right)\right) \\
&= \sum_{n=1}^{N} \mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta)\right) + 2 \sum_{m<n} \mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta)\right). && \text{(S128)}
\end{aligned}$$

Here (A6-J) and (A3) ensure strong mixing of the joint model. We know from Theorem 3.1 of Karjalainen et al. (2023) that when $\mathbf{M}_{n,n+k}$ is the k-step Markov operator from timestep n and $\beta_{\mathrm{TV}}(M) = \sup_{x,y \in E} \|M(x,\cdot) - M(y,\cdot)\|_{\mathrm{TV}} = \sup_{\mu,\nu \in \mathcal{P}, \mu \neq \nu} \frac{\|\mu M - \nu M\|_{\mathrm{TV}}}{\|\mu - \nu\|_{\mathrm{TV}}}$ is the Dobrushin contraction coefficient of a Markov operator,

$$\beta_{\mathrm{TV}}(\mathbf{M}_{n,n+k}) \leq (1-\epsilon)^{\lfloor k/(c \log(J))\rfloor}, \tag{S129}$$

i.e. the mixing time of this joint particle filter is $O(\log(J))$, where ϵ and c depend on $\bar{M}_D, \underline{\mathrm{M}}_D, \bar{G}, \underline{\mathrm{G}}$. The physical particle cloud satisfies a bound of the same form under (A2), with its own constants; that version is used under (A6) below.

Under (A6-J), apply Lemma 2.3 of Karjalainen et al. (2023) to the joint prediction cloud. Conditional resampling and the Bayes-ratio calculation give the centered L^4 bound below. Theorem 3.1 and Lemma S9 bound the strong-mixing coefficient $a(k)$ of the score sequence. The shift by one accounts for the endpoint resampling. At $\alpha = 0$ these results give

$$\begin{aligned}
\left\| \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta) - \mathbb{E}\left[\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta)\right]\right\|_{L^4} &\leq \frac{C G'(\theta)}{\sqrt{J}}, \\
a(k) &\leq (1-\epsilon)^{\lfloor (k-1)/[c\{1+\log J\}]\rfloor}. && \text{(S130)}
\end{aligned}$$

Davydov's inequality must be applied to the centered variables. Since covariance is unchanged by centering, Equation (S130) gives

$$\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}} \hat{\ell}_m^0(\theta), \frac{\partial}{\partial_{\theta_i}} \hat{\ell}_n^0(\theta)\right) \leq (1-\epsilon)^{\frac{1}{2}\lfloor \frac{|n-m|-1}{c\{1+\log(J)\}}\rfloor} \frac{G'(\theta)^2}{J}. \tag{S131}$$

The geometric sum is bounded by $C(1+\log J)$. Putting it all together, we see that

$$\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0(\theta)\right) = \sum_{n=1}^{N}\mathrm{Var}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) + 2\sum_{m<n}\mathrm{Cov}\left(\frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_m(\theta), \frac{\partial}{\partial_{\theta_i}}\hat{\ell}^0_n(\theta)\right) \tag{S132}$$

$$\leq \frac{NG'(\theta)^2}{J} + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\frac{G'(\theta)^2}{J} \tag{S133}$$

$$\leq \frac{G'(\theta)^2}{J}\left(N + 2\sum_{n=1}^{N-1}(N-n)(1-\epsilon)^{\frac{1}{2}\lfloor\frac{n-1}{c\{1+\log(J)\}}\rfloor}\right) \tag{S134}$$

$$\lesssim \frac{G'(\theta)^2 N(1+\log J)}{J}. \tag{S135}$$

It then follows that as $\theta \in \Theta \subseteq \mathbb{R}^p$,

$$\mathrm{Var}\big(\nabla_\theta\hat{\ell}^0(\theta)\big) \lesssim \frac{Np\,G'(\theta)^2(1+\log J)}{J}. \tag{S136}$$

Under (A6), apply the age decomposition and covariance calculation in Steps 1a–1d of Lemma S13 with $k=0$. In particular, Equation (S277) gives

$$\mathrm{Var}\{\nabla_\theta\hat{\ell}^0(\theta)\} = \mathrm{Var}\{\nabla_\theta\hat{s}^1_0(\theta)\} \lesssim NpG'(\theta)^2(1+\log J)^2/J.$$

That argument uses the physical-cloud mixing bound above and bounded finite-history inputs, not mixing of the full gradient cloud under (A6). □

The factor of N comes from using the unnormalized gradient. Dividing by $\sqrt{N}$ gives a variance bound independent of the horizon. Dividing by N gives a centered L^2 error bound of order $(1+\log J)/\sqrt{NJ}$.

The preceding calculation gives the order $NpG'(\theta)^2(1+\log J)^2/J$ for the DMOP-0 baseline. It is not asserted to be a lower bound for every model or every value of α.

S7 Proof of Theorem 5 (Bias-Variance Analysis)

Write $\nabla_\theta\ell^\alpha_n(\theta)$ for the nth conditional score of the population recursion. At $\theta=\phi$ and $\alpha=1$, it equals $\nabla_\theta\ell_n(\theta)$. Along each selected lineage,

$$\nabla_\theta \log w^{F,\theta}_{n,j}\Big|_{\theta=\phi} = \alpha\,\nabla_\theta \log w^{F,\theta}_{n-1,k_j}\Big|_{\theta=\phi} + \nabla_\theta \log g^{\theta}_{n,k_j}\Big|_{\theta=\phi}. \tag{S137}$$

The last term is the measurement gradient h_n defined in Equation (S151). This follows by differentiating

$$\begin{aligned}\log w^{P,\theta}_{n,j} &= \alpha \log w^{F,\theta}_{n-1,j},\\ \log w^{F,\theta}_{n,j} &= \alpha \log w^{F,\theta}_{n-1,k_j} + \log g^{\theta}_{n,k_j} - \log g^{\phi}_{n,k_j}.\end{aligned} \tag{S138}$$

at $\theta=\phi$. The observations are fixed throughout this section; expectations and variances are over the particle-filter randomness.

We next investigate conditions (A6) and (A6-J), repeating them here to assist the discussion.

(A6) Measurement-sensitivity decay. The simulator is C^1 in its incoming state. For a continuation $X_n^\theta(x_t, \omega_{t+1:n})$ from x_t, with $X_t^\theta(x_t) = x_t$,

$$\left\| D_{x_t} \log f_{Y_n|X_n}\big(y_n^* \mid X_n^\theta(x_t, \omega_{t+1:n}); \theta\big) \right\|_2 \le C\kappa^{n-t}, \qquad C < \infty, \quad 0 < \kappa < 1. \tag{S139}$$

The bound is uniform in $t \le n$, states, draws, and θ near ϕ. The parameter derivatives of the initial state, simulator, and log measurement density are bounded, holding incoming states fixed.

(A6-J) Joint-transition alternative. The log measurement density is C^1 in (x, θ). The joint process (X_n^θ, D_n^θ), $D_n^\theta = \partial_\theta X_n^\theta$, has transition densities satisfying

$$0 < \underline{\mathrm{M}}_D \le f_{(X_n, D_n)|(X_{n-1}, D_{n-1})}(x', v' \mid x, v; \theta) \le \bar{M}_D < \infty \tag{S140}$$

relative to a common finite measure on the invariant joint state space, uniformly in time, admissible pairs, and θ near ϕ.

The particle bounds assume either (A6) or (A6-J); the consistency results and population Lemmas S10 and S14 retain (A1)–(A5). Neither additional condition requires density evaluation by the algorithm. Their constants enter the particle bounds, while ϱ retains its physical-model definition.

Sufficient condition for (A6). Suppose the parameter derivatives in (A6) are bounded, the spatial log measurement derivatives are bounded by L_g, and the one-step simulator Jacobians are bounded by L. If, for fixed $b \ge 1$ and $q \in (0, 1)$, $\|\partial X_{t+b}^\theta / \partial X_t^\theta\|_{\mathrm{op}} \le q$ uniformly with the draws fixed, then writing $n - t = mb + v$, $0 \le v < b$, gives

$$\left\| D_{x_t} \log f_{Y_n|X_n}(y_n^* \mid X_n^\theta; \theta) \right\|_2 \le L_g L^v q^m \le C\kappa^{n-t}, \qquad \kappa = q^{1/b} \in (0, 1). \tag{S141}$$

Here $C = L_g \max_{0 \le v < b} (L/\kappa)^v$; a zero block Jacobian permits any $q \in (0, 1)$.

Example: physical mixing without (A6). Consider the annulus $\mathcal{X} = \{x \in \mathbb{R}^2 : 1 \le \|x\|_2 \le 2\}$. Let R_u denote planar rotation by u, and let independent simulator draws satisfy $U_n \sim \mathrm{Unif}[0, 2\pi]$ and $V_n \sim \mathrm{Unif}[1, 2]$. Set

$$X_0^\theta = R_\theta x_0, \qquad X_n^\theta = V_n R_{U_n} \frac{X_{n-1}^\theta}{\|X_{n-1}^\theta\|_2}, \qquad \Pr(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{x_1}{4\|x\|_2}. \tag{S142}$$

The physical transition forgets the incoming state in one step: its Lebesgue density on $\mathcal{X}$ is

$$f_{X_n|X_{n-1}}(x' \mid x; \theta) = \frac{1}{2\pi\|x'\|_2}, \qquad \frac{1}{4\pi} \le f_{X_n|X_{n-1}} \le \frac{1}{2\pi}. \tag{S143}$$

Both measurement probabilities lie in $[1/4, 3/4]$. For fixed draws, $X_n^\theta = R_\theta X_n^0$, so all parameter derivatives exist and are bounded; in particular the first two log measurement derivatives have magnitudes at most 1 and 2. The likelihood is smooth, and the MOP derivatives are bounded for each fixed horizon. Thus (A1)–(A5) hold.

Nevertheless, fix the observations $y_n^* = 1$. For $n > t$, the composition is

$$X_n^\theta(x, \omega_{t+1:n}) = V_n R_{U_{t+1} + \cdots + U_n} \frac{x}{\|x\|_2}. \tag{S144}$$

Its final angle is uniform, and a fixed-random-number change in the incoming angle is preserved. Differentiating the displayed measurement probability therefore gives

$$\mathbb{E}\left\|D_x \log f_{Y_n|X_n}\left(1 \mid X_n^\theta(x, \omega_{t+1:n}); \theta\right)\right\|_2^2 = \frac{1}{2\pi\|x\|_2^2} \int_0^{2\pi} \frac{\sin^2 u}{(2+\cos u)^2}\, du > 0. \tag{S145}$$

The right side is independent of $n - t$, so (A6) fails despite (A1)–(A5). Here $D_n^\theta = R_{\pi/2} X_n^\theta$, and the joint transition has the physical density relative to $dx'\, \delta_{R_{\pi/2}x'}(dv')$. Thus (A6-J) holds and the particle bounds still apply. In particular, (A6-J) does not imply (A6).

Differentiation and notation. By (A5), differentiation in θ holds the simulator draws and reference-filter resampling indices fixed and includes the dependence of X_n^θ on θ. Assumption (A3) gives, uniformly in n, j and θ near ϕ,

$$\left\|\nabla_\theta \log g_{n,j}^{A,F,\theta}\right\|_\infty \le G'(\theta). \tag{S146}$$

The product rule and Equation (S146) give

$$\begin{aligned}
&\left\|\nabla_\theta \prod_{s=1}^t G_s(X_s)\right\|_\infty \\
&\quad = \left\|\prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \nabla_\theta \log G_s(X_s)\right\|_\infty \\
&\le \prod_{s=1}^t G_s(X_s) \sum_{s=1}^t \left\|\nabla_\theta \log G_s(X_s)\right\|_\infty \\
&\le t\bar{G}^t G'(\theta).
\end{aligned} \tag{S147}$$

For fixed t, the mean-value theorem bounds the coordinatewise difference quotients by the integrable constant $t\bar{G}^t G'(\theta)$. Dominated convergence therefore gives

$$\nabla_\theta \mathbb{E}_U\left[\prod_{s=1}^t G_s(X_s)\right] = \mathbb{E}_U\left[\nabla_\theta \prod_{s=1}^t G_s(X_s)\right]. \tag{S148}$$

At $\theta = \phi$, let ρ_t be the posterior law of the simulator history through time t, including its base variables. Its $X_{0:t}$-marginal is the posterior path law, and

$$\rho_t(h) = \frac{\mathbb{E}_U\left[h \prod_{s=1}^t G_s(X_s)\right]}{\mathbb{E}_U\left[\prod_{s=1}^t G_s(X_s)\right]}. \tag{S149}$$

Here $\mathbb{E}_U$ integrates over the base simulator variables before measurement weighting. Expectations in particle-error calculations are instead over the particle-filter law, including resampling. Future simulator draws are independent of the past.

Iterating the weight recursion at $\theta = \phi$ gives

$$\nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} = \sum_{r=0}^{n-1} \alpha^r\ \nabla_\theta \log g_{n-r,j}^{A,F,\theta}\Big|_{\theta=\phi}. \tag{S150}$$

Write

$$h_k = \nabla_\theta \log g_k^{A,F,\theta}\Big|_{\theta=\phi} \tag{S151}$$

for the measurement gradient on an ancestral path. Following Del Moral and Jasra (2018), let

$$m(\xi_t) = \frac{1}{J}\sum_{j=1}^{J} \delta_{\xi_{t,j}}, \qquad m(\xi_t)(h_k) = \frac{1}{J}\sum_{j=1}^{J} h_k(\xi_{t,j}), \quad 1 \le k \le t, \tag{S152}$$

where $\xi_{t,j}$ is the simulator history along particle j's ancestry through time t, including the base variables needed to evaluate h_k and its pathwise derivative contributions. Thus $m(\xi_t)(h_k)$ averages the time-k gradients over these paths.

At $\theta = \phi$, all weights equal one. Differentiating the two sample averages in the definition of $L_n^{A,\theta,\alpha}$ gives

$$\begin{aligned}\nabla_\theta \hat{\ell}_n^{\alpha}(\theta) &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} - \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta}\Big|_{\theta=\phi} \\ &= \frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{F,\theta}\Big|_{\theta=\phi} - \frac{\alpha}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi}, \end{aligned} \tag{S153}$$

where the second equality uses $w_{n,j}^{P,\theta} = (w_{n-1,j}^{F,\theta})^\alpha$. In the MOP-$\alpha$ recursion, the discounted-weight accumulation and process propagation preserve the index j and occur before the time-n resampling step. The resampling index k_j enters only afterward, through

$$w_{n,j}^{F,\theta} = w_{n,k_j}^{P,\theta} \frac{g_{n,k_j}^{\theta}}{g_{n,k_j}^{\phi}}. \tag{S154}$$

Consequently, the second average in Equation (S153) is pathwise

$$\frac{1}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n,j}^{P,\theta}\Big|_{\theta=\phi} = \frac{\alpha}{J}\sum_{j=1}^{J} \nabla_\theta \log w_{n-1,j}^{F,\theta}\Big|_{\theta=\phi}, \tag{S155}$$

which is the empirical average over the time-$(n-1)$ filtered genealogy. Substituting Equation (S150) into Equation (S153) and changing the index in the second sum give

$$\begin{aligned}\nabla_\theta \hat{\ell}_n^{\alpha}(\theta) &= \sum_{r=0}^{n-1} \alpha^r m(\xi_n)(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r m(\xi_{n-1})(h_{n-r}) \\ &= m(\xi_n)(h_n) + \sum_{r=1}^{n-1} \alpha^r \left\{ m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}) \right\}. \end{aligned} \tag{S156}$$

Consequently, define

$$\begin{aligned} \Delta_{n,0}^J &= m(\xi_n)(h_n), & \Delta_{n,0} &= \rho_n(h_n), \\ \Delta_{n,r}^J &= m(\xi_n)(h_{n-r}) - m(\xi_{n-1})(h_{n-r}), & \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}), \quad 1 \le r < n. \end{aligned} \tag{S157}$$

For the population coefficients, Equations (S148) and (S151) give, at $\theta = \phi$,

$$\begin{aligned}
&\nabla_\theta \log \mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \right] \\
&\quad = \frac{\mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \sum_{s=1}^{n} h_s \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n} G_s(X_s) \right]} \\
&\quad = \sum_{s=1}^{n} \rho_n(h_s). \qquad \text{(S158)}
\end{aligned}$$

Subtracting the same identity at time $n-1$ gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) &= \sum_{s=1}^{n} \rho_n(h_s) - \sum_{s=1}^{n-1} \rho_{n-1}(h_s) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\
&= \sum_{r=0}^{n-1} \Delta_{n,r}. \qquad \text{(S159)}
\end{aligned}$$

Finally, differentiating the two population weighted averages in the deterministic MOP-α recursion and using Equation (S148) gives

$$\begin{aligned}
\nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1} \alpha^r \rho_n(h_{n-r}) - \sum_{r=1}^{n-1} \alpha^r \rho_{n-1}(h_{n-r}) \\
&= \rho_n(h_n) + \sum_{r=1}^{n-1} \alpha^r \{\rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r})\} \\
&= \sum_{r=0}^{n-1} \alpha^r \Delta_{n,r}. \qquad \text{(S160)}
\end{aligned}$$

Lemma S10 (Lagwise first-derivative forgetting)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), there are constants $C < \infty$ and $\varrho \in (0,1)$ such that, for $0 \le r < n$,*

$$\|\Delta_{n,r}\|_2 \le C\sqrt{p}\, G'(\theta) \varrho^r. \qquad \text{(S161)}$$

Proof. We use the posterior update and backward-kernel contraction.

Step 1: one-step posterior update. First suppose that $1 \le r < n$, and put $k = n - r$. Fix a coordinate $i \in \{1, \dots, p\}$, and let $h_{k,i}$ be the corresponding coordinate of h_k in Equation (S151). By Equation (S146),

$$|h_{k,i}| \le G'(\theta). \qquad \text{(S162)}$$

Because $h_{k,i}$ is measurable with respect to the canonical extended path through time $k \le n-1$, Equation (S149) and conditional expectation with respect to the new base simulator variable give

$$\rho_n(h_{k,i}) = \frac{\mathbb{E}_U \left[h_{k,i} \prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\} \right]}{\mathbb{E}_U \left[\prod_{s=1}^{n-1} G_s(X_s) \mathbb{E}_U\{G_n(X_n) \mid X_{n-1}\} \right]}$$

$$= \frac{\rho_{n-1}\{h_{k,i} M_n(G_n)\}}{\rho_{n-1} M_n(G_n)}, \tag{S163}$$

where the first equality uses the independence of the new base simulator variable from the path through time $n-1$. The second equality first uses the identity below and then divides its numerator and denominator by $\mathbb{E}_U\{\prod_{s=1}^{n-1} G_s(X_s)\}$. Using the manuscript's kernel notation, the required identity is

$$M_n(G_n)(x_{n-1}) = \int M_n(x_{n-1}, dx_n) G_n(x_n). \tag{S164}$$

The bounds on G_n and the fact that M_n is a Markov kernel imply for every x_{n-1} that

$$M_n(G_n)(x_{n-1}) \geq \underline{\mathrm{G}} \int M_n(x_{n-1}, dx_n) = \underline{\mathrm{G}}, \tag{S165}$$

$$M_n(G_n)(x_{n-1}) \leq \bar{G} \int M_n(x_{n-1}, dx_n) = \bar{G}. \tag{S166}$$

Step 2: backward-kernel contraction. For $s \geq 1$, the posterior backward kernel from X_s to X_{s-1} is

$$B_s(x_s, dx_{s-1}) = \frac{\pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)}. \tag{S167}$$

Assumption (A2) gives the following minorization for every measurable set A:

$$\begin{aligned} B_s(x_s, A) &= \frac{\int_A \pi_{s-1}(dx_{s-1}) f_{X_s|X_{s-1}}(x_s \mid x_{s-1}; \theta)}{\int \pi_{s-1}(du) f_{X_s|X_{s-1}}(x_s \mid u; \theta)} \\ &\geq \frac{\underline{\mathrm{M}}\, \pi_{s-1}(A)}{\bar{M} \int \pi_{s-1}(du)} \\ &= \frac{\underline{\mathrm{M}}}{\bar{M}} \pi_{s-1}(A). \end{aligned} \tag{S168}$$

Put $\epsilon = \underline{\mathrm{M}}/\bar{M}$. If $\epsilon < 1$, Equation (S168) permits the decomposition

$$B_s(x_s, \cdot) = \epsilon \pi_{s-1}(\cdot) + (1-\epsilon) R_s(x_s, \cdot), \tag{S169}$$

where

$$R_s(x_s, A) = \frac{B_s(x_s, A) - \epsilon \pi_{s-1}(A)}{1-\epsilon} \tag{S170}$$

is a Markov kernel. Hence, for every bounded real function f and any x_s, x_s',

$$\begin{aligned} B_s(f)(x_s) - B_s(f)(x_s') &= (1-\epsilon)\{R_s(f)(x_s) - R_s(f)(x_s')\}, \\ \left|B_s(f)(x_s) - B_s(f)(x_s')\right| &\leq (1-\epsilon)\operatorname{osc}(f). \end{aligned} \tag{S171}$$

If $\epsilon = 1$, Equation (S168) implies $B_s(x_s, \cdot) = \pi_{s-1}(\cdot)$, so the left-hand side is zero. Taking the supremum over x_s, x_s' gives

$$\operatorname{osc}\{B_s(f)\} \leq (1-\epsilon)\operatorname{osc}(f). \tag{S172}$$

Step 3: project and propagate the time-k contribution. For the historical gradient $h_{k,i}$, define

$$\bar{h}_{k,i}(X_k) = \mathbb{E}_{\rho_k}(h_{k,i} \mid X_k). \tag{S173}$$

To identify this conditional expectation under later posterior laws, for $t \geq k$ let

$$L_{k,t}(X_k) = \mathbb{E}_U \left\{ \prod_{s=k+1}^{t} G_s(X_s) \middle| X_k \right\}, \qquad L_{k,k} = 1. \tag{S174}$$

For every bounded function q of X_k, the Markov property and Equation (S149) give

$$\begin{aligned}
\rho_t\{h_{k,i} q(X_k)\} &= \frac{\mathbb{E}_U \left[h_{k,i} q(X_k) \prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[\prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k) \right]} \\
&= \frac{\mathbb{E}_U \left[\bar{h}_{k,i}(X_k) q(X_k) \prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k) \right]}{\mathbb{E}_U \left[\prod_{s=1}^{k} G_s(X_s) L_{k,t}(X_k) \right]} \\
&= \rho_t\{\bar{h}_{k,i}(X_k) q(X_k)\}.
\end{aligned} \tag{S175}$$

The second equality is the defining conditional-expectation identity in Equation (S173), multiplied by the X_k-measurable function $q(X_k)L_{k,t}(X_k)$. Hence

$$\mathbb{E}_{\rho_t}(h_{k,i} \mid X_k) = \bar{h}_{k,i}(X_k), \qquad t \geq k. \tag{S176}$$

Repeated conditioning through the physical backward kernels therefore gives

$$\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = x_{n-1}\} = (B_{n-1} B_{n-2} \cdots B_{k+1} \bar{h}_{k,i})(x_{n-1}). \tag{S177}$$

There are $n - 1 - k = r - 1$ backward kernels in this display. Moreover, Equation (S162) gives

$$\begin{aligned}
\|\bar{h}_{k,i}\|_\infty &\leq \|h_{k,i}\|_\infty \leq G'(\theta), \\
\operatorname{osc}(\bar{h}_{k,i}) &\leq 2\|\bar{h}_{k,i}\|_\infty \\
&\leq 2G'(\theta).
\end{aligned} \tag{S178}$$

If $r = 1$, Equation (S177) contains no backward kernel. If $r \geq 2$, applying Equation (S172) successively gives

$$\begin{aligned}
&\operatorname{osc}\big[\mathbb{E}_{\rho_{n-1}}\{h_{k,i} \mid X_{n-1} = \cdot\}\big] \\
&\qquad \leq 2G'(\theta) \begin{cases} 1, & r = 1, \\ (1-\epsilon)^{r-1}, & r \geq 2. \end{cases}
\end{aligned} \tag{S179}$$

Step 4: covariance identity and geometric bound. To finish, abbreviate the conditional expectation on the left of Equation (S179) by H, only for the next calculation. Since the X_{n-1}-marginal of ρ_{n-1} is π_{n-1}, the tower property gives

$$\rho_{n-1}(h_{k,i}) = \pi_{n-1}(H), \tag{S180}$$

$$\rho_{n-1}\{h_{k,i}M_n(G_n)\} = \pi_{n-1}\{HM_n(G_n)\}. \tag{S181}$$

Substituting Equations (S180)– (S181) into Equation (S163) and then subtracting $\rho_{n-1}(h_{k,i})$ gives the exact covariance identity

$$\begin{aligned}
\Delta_{n,r,i} &= \rho_n(h_{k,i}) - \rho_{n-1}(h_{k,i}) \\
&= \frac{\pi_{n-1}\{HM_n(G_n)\} - \pi_{n-1}(H)\pi_{n-1}M_n(G_n)}{\pi_{n-1}M_n(G_n)} \\
&= \frac{\pi_{n-1}[\{H - \pi_{n-1}(H)\}\{M_n(G_n) - \pi_{n-1}M_n(G_n)\}]}{\pi_{n-1}M_n(G_n)}.
\end{aligned} \tag{S182}$$

For any probability measure μ and bounded real f,

$$|f(x) - \mu(f)| \le \operatorname{osc}(f). \tag{S183}$$

The first positive lag also inherits one contraction from the forward transition. To see this directly, fix $x^\star$. Assumption (A2) gives, for every measurable set A,

$$\begin{aligned}
M_n(x, A) &= \int_A f_{X_n|X_{n-1}}(x_n \mid x; \theta)\, dx_n \\
&\ge \underline{\mathrm{M}} \int_A dx_n \\
&\ge \frac{\underline{\mathrm{M}}}{\bar{M}} \int_A f_{X_n|X_{n-1}}(x_n \mid x^\star; \theta)\, dx_n \\
&= \epsilon M_n(x^\star, A).
\end{aligned} \tag{S184}$$

If $\epsilon < 1$, Equation (S184) permits the decomposition

$$M_n(x, \cdot) = \epsilon M_n(x^\star, \cdot) + (1 - \epsilon)R_n(x, \cdot), \tag{S185}$$

where R_n is a Markov kernel. Therefore, for any x, x',

$$\begin{aligned}
M_n(G_n)(x) - M_n(G_n)(x') &= (1 - \epsilon)\{R_n(G_n)(x) - R_n(G_n)(x')\}, \\
\left|M_n(G_n)(x) - M_n(G_n)(x')\right| &\le (1 - \epsilon)\operatorname{osc}(G_n) \\
&\le (1 - \epsilon)(\bar{G} - \underline{\mathrm{G}}).
\end{aligned} \tag{S186}$$

Taking the supremum over x, x' gives

$$\operatorname{osc}\{M_n(G_n)\} \le (1 - \epsilon)(\bar{G} - \underline{\mathrm{G}}). \tag{S187}$$

If $\epsilon = 1$, Equation (S184) implies that $M_n(x, \cdot)$ does not depend on x, so the left side of Equation (S187) is zero and the same bound holds. Taking absolute values in Equation (S182), using Equations (S183)– (S187), and then using Equation (S179), yields

$$\begin{aligned}
|\Delta_{n,r,i}| &\le \frac{\operatorname{osc}(H)\operatorname{osc}\{M_n(G_n)\}}{\underline{\mathrm{G}}} \\
&\le \frac{2(\bar{G} - \underline{\mathrm{G}})}{\underline{\mathrm{G}}} G'(\theta) \begin{cases} (1 - \epsilon), & r = 1, \\ (1 - \epsilon)^r, & r \ge 2. \end{cases}
\end{aligned} \tag{S188}$$

For both cases, take

$$\varrho = 1 - \frac{\epsilon}{2} = 1 - \frac{\mathrm{M}}{2\bar{M}} \in (0,1). \tag{S189}$$

Since $1 - \epsilon \le \varrho$, the two cases in Equation (S188) are both bounded by ϱ^r. Summing the squared coordinate bounds and taking square roots give

$$\begin{aligned} \|\Delta_{n,r}\|_2 &= \left\{ \sum_{i=1}^{p} |\Delta_{n,r,i}|^2 \right\}^{1/2} \\ &\le C\sqrt{p}\, G'(\theta)\varrho^r, \qquad 1 \le r < n. \end{aligned} \tag{S190}$$

Step 5: the zero-lag case. Finally, when $r = 0$, Equation (S157) and Equation (S146) give

$$\begin{aligned} \|\Delta_{n,0}\|_2 &= \|\rho_n(h_n)\|_2 \\ &\le \rho_n(\|h_n\|_2) \\ &\le \sqrt{p}\, G'(\theta) \\ &\le C\sqrt{p}\, G'(\theta)\varrho^0. \end{aligned} \tag{S191}$$

Equations (S190) and (S191) prove Equation (S161). If $\epsilon = 1$, Equation (S182) and the zero oscillation in Equation (S187) show that $\Delta_{n,r} = 0$ for every $r \ge 1$, so any fixed $\varrho \in (0,1)$ proves the same result. The chosen ϱ depends only on the model minorization bounds, independently of N, J, α, k. The factor ϱ^r includes the first positive lag. This argument bounds linear posterior expectations; particle covariances are treated in Lemma S13.

□

Lemma S11 (Uniform weak bias of current-particle averages)**.** *Suppose $\theta = \phi$. Under Assumptions (A2), (A3), and (A5), with multinomial resampling, there is $C < \infty$ such that, for every bounded real function f of the current particle location and every $t \ge 1$,*

$$|\mathbb{E}m(\xi_t)(f) - \rho_t(f)| \le \frac{C}{J}\,\mathrm{osc}(f). \tag{S192}$$

The constant is independent of t and J. Under (A6-J), the same conclusion holds for a bounded function of the current pair $(X_t, \partial_\theta X_t)$, with constants depending on the joint-transition bounds. Under (A6), Lemma S13 handles h_t by decomposing its simulator history.

Proof. We telescope the bias over particle updates, using semigroup contraction.

Step 1: prediction-to-filtering reduction. First take f to be a function of the physical location, in the Feynman–Kac model defined in Section S3. The current-location marginal of ρ_t is the filtering distribution, and hence

$$\rho_t(f) = \pi_t(f) = \Psi_t(\eta_t)(f). \tag{S193}$$

Conditionally on the prediction cloud η_t^J, multinomial resampling draws the filtering particles independently from $\Psi_t(\eta_t^J)$. Therefore

$$\mathbb{E}\{m(\xi_t)(f) \mid \eta_t^J\} = \frac{1}{J}\sum_{j=1}^{J} \mathbb{E}\{f(x_{t,j}^F) \mid \eta_t^J\}$$

$$= \frac{1}{J}\sum_{j=1}^{J} \Psi_t(\eta_t^J)(f)$$
$$= \Psi_t(\eta_t^J)(f). \tag{S194}$$

Step 2: normalized-semigroup contraction. Take $\Phi_{t,t}$ to be the identity. For $1 \le p \le t$, define

$$\begin{aligned} Q_{t,t}(f)(x) &= G_t(x) f(x), \\ Q_{p,t}(f)(x) &= G_p(x) M_{p+1}\{Q_{p+1,t}(f)\}(x), \qquad p < t. \end{aligned} \tag{S195}$$

The definitions of $\Phi_{p,t}$ and Ψ_t give, for every probability measure μ,

$$\Psi_t\{\Phi_{p,t}(\mu)\}(f) = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}. \tag{S196}$$

For $p < t$,

$$Q_{p,t}(1)(x) = G_p(x) \int f_{X_{p+1}|X_p}(u \mid x; \theta) Q_{p+1,t}(1)(u)\, du. \tag{S197}$$

Thus, for every x, x',

$$\begin{aligned} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} &\le \frac{\bar{G}\bar{M} \int Q_{p+1,t}(1)(u)\, du}{\underline{\mathrm{GM}} \int Q_{p+1,t}(1)(u)\, du} \\ &= \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}}. \end{aligned} \tag{S198}$$

For $p = t$, the same ratio is at most $\bar{G}/\underline{\mathrm{G}}$. Put

$$R = \frac{\bar{G}\bar{M}}{\underline{\mathrm{GM}}} \ge 1. \tag{S199}$$

Then

$$\sup_{x,x'} \frac{Q_{p,t}(1)(x)}{Q_{p,t}(1)(x')} \le R, \qquad 1 \le p \le t. \tag{S200}$$

For $p \le s < t$, the normalized forward kernel is

$$P_{s,s+1}^{(t)}(x, A) = \frac{\displaystyle\int_A f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}{\displaystyle\int f_{X_{s+1}|X_s}(u \mid x; \theta) Q_{s+1,t}(1)(u)\, du}. \tag{S201}$$

Define, only within this proof,

$$\nu_{s,t}(A) = \frac{\displaystyle\int_A Q_{s+1,t}(1)(u)\, du}{\displaystyle\int Q_{s+1,t}(1)(u)\, du}. \tag{S202}$$

Assumption (A2) gives

$$P_{s,s+1}^{(t)}(x,A) \geq \frac{\underline{\mathrm{M}} \displaystyle\int_A Q_{s+1,t}(1)(u)\,du}{\bar{M} \displaystyle\int Q_{s+1,t}(1)(u)\,du} = \frac{\underline{\mathrm{M}}}{\bar{M}} \nu_{s,t}(A). \tag{S203}$$

Consequently, by Equation (S189), the Dobrushin coefficient of every kernel in Equation (S201) is at most ϱ. The factor $G_s(x)$ cancels from the normalized conditional expectation, and hence

$$\frac{Q_{p,t}(f)}{Q_{p,t}(1)} = P_{p,p+1}^{(t)} P_{p+1,p+2}^{(t)} \cdots P_{t-1,t}^{(t)} f. \tag{S204}$$

It follows that

$$\operatorname{osc}\left\{ \frac{Q_{p,t}(f)}{Q_{p,t}(1)} \right\} \leq \varrho^{t-p} \operatorname{osc}(f). \tag{S205}$$

Step 3: one complete particle update. For one selection–mutation update, if $p = 1$, put $\mu = \eta_1 = \pi_0 M_1$. For $p \geq 2$, condition on η_{p-1}^J and put

$$\mu = \Phi_p(\eta_{p-1}^J). \tag{S206}$$

In either case, the particles forming η_p^J are conditionally independent with common law μ. Define the normalizer q and centered fluctuation d by

$$q = \frac{Q_{p,t}(1)}{\mu Q_{p,t}(1)}, \qquad a = \frac{\mu Q_{p,t}(f)}{\mu Q_{p,t}(1)}, \qquad d = \frac{Q_{p,t}(f-a)}{\mu Q_{p,t}(1)}. \tag{S207}$$

Then

$$\mu(q) = 1, \qquad \mu(d) = 0, \tag{S208}$$

and Equation (S196) gives

$$\begin{aligned} &\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\mu)\}(f) \\ &\quad = \frac{\eta_p^J Q_{p,t}(f)}{\eta_p^J Q_{p,t}(1)} - a \\ &\quad = \frac{\eta_p^J Q_{p,t}(f-a)}{\eta_p^J Q_{p,t}(1)} \\ &\quad = \frac{\eta_p^J(d)}{\eta_p^J(q)}. \end{aligned} \tag{S209}$$

Equation (S200) implies

$$R^{-1} \leq q(x) \leq R, \qquad \eta_p^J(q) \geq R^{-1}, \qquad \|q - 1\|_\infty \leq R. \tag{S210}$$

Writing

$$F = \frac{Q_{p,t}(f)}{Q_{p,t}(1)}, \tag{S211}$$

we have $a = \mu(qF)$ and $\mu(q) = 1$. Thus a is an average of F under the probability measure $q\,d\mu$, so

$$\inf F \le a \le \sup F. \tag{S212}$$

Since $d = q(F - a)$, Equations (S210), (S212), and (S205) give

$$\begin{aligned} \|d\|_\infty &\le R\,\mathrm{osc}(F) \\ &\le R\varrho^{t-p}\,\mathrm{osc}(f). \end{aligned} \tag{S213}$$

Conditional independence and $\mu(d) = 0$ give

$$\begin{aligned} \mathbb{E}\{\eta_p^J(d) \mid \mu\} &= 0, \\ \mathbb{E}\{\eta_p^J(d)^2 \mid \mu\} &= \frac{1}{J}\mu(d^2), \\ \mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\} &= \frac{1}{J}\mu\{(q-1)^2\}. \end{aligned} \tag{S214}$$

Consequently,

$$\begin{aligned} &\mathbb{E}\left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \\ &\quad = -\mathbb{E}\left\{ \frac{\eta_p^J(d)\{\eta_p^J(q) - 1\}}{\eta_p^J(q)} \middle| \mu \right\}. \end{aligned} \tag{S215}$$

Conditional Cauchy–Schwarz and Equations (S210)– (S214) yield

$$\begin{aligned} &\left| \mathbb{E}\left\{ \frac{\eta_p^J(d)}{\eta_p^J(q)} \middle| \mu \right\} \right| \\ &\quad \le R\left[\mathbb{E}\{\eta_p^J(d)^2 \mid \mu\}\mathbb{E}\{(\eta_p^J(q) - 1)^2 \mid \mu\}\right]^{1/2} \\ &\quad = \frac{R}{J}\left[\mu(d^2)\mu\{(q-1)^2\}\right]^{1/2} \\ &\quad \le \frac{C}{J}\varrho^{t-p}\,\mathrm{osc}(f). \end{aligned} \tag{S216}$$

Step 4: telescope over update times. Since $\eta_t = \Phi_{1,t}(\eta_1)$,

$$\begin{aligned} &\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f) \\ &\quad = \mathbb{E}\left[\Psi_t\{\Phi_{1,t}(\eta_1^J)\}(f) - \Psi_t\{\Phi_{1,t}(\eta_1)\}(f)\right] \\ &\qquad + \sum_{p=2}^{t} \mathbb{E}\left[\Psi_t\{\Phi_{p,t}(\eta_p^J)\}(f) - \Psi_t\{\Phi_{p,t}(\Phi_p(\eta_{p-1}^J))\}(f)\right]. \end{aligned} \tag{S217}$$

Indeed, for $2 \le p \le t$,

$$\Phi_{p,t} \circ \Phi_p = \Phi_{p-1,t}, \tag{S218}$$

so consecutive terms cancel. The only terms left are $\Psi_t(\eta_t^J)(f)$ and $-\Psi_t(\eta_t)(f)$.

Apply Equation (S216) to every summand in Equation (S217). Then

$$\begin{aligned}\left|\mathbb{E}\Psi_t(\eta_t^J)(f) - \Psi_t(\eta_t)(f)\right| &\le \frac{C\,\mathrm{osc}(f)}{J}\sum_{p=1}^{t}\varrho^{t-p} \\ &\le \frac{C}{J(1-\varrho)}\,\mathrm{osc}(f). \end{aligned} \tag{S219}$$

Absorb $(1-\varrho)^{-1}$ into C and combine this display with Equations (S193) and (S194).

For (A6-J), apply the same calculation to the pair $(X_t, \partial_\theta X_t)$. In every display replace the physical transition density by that in Equation (S140), du by the common reference measure, $(\underline{\mathrm{M}}, \bar{M})$ by $(\underline{\mathrm{M}}_D, \bar{M}_D)$, and the contraction factor by $1-\underline{\mathrm{M}}_D/(2\bar{M}_D)$. The potential remains $G_t(X_t)$, so its bounds and all ratio and conditional-resampling identities are unchanged. For each coordinate, the current measurement gradient is

$$h_{t,i}(x,v) = \partial_{\theta_i}\log f_{Y_t|X_t}(y_t^* \mid x;\theta) + \sum_{a=1}^{d}\partial_{x_a}\log f_{Y_t|X_t}(y_t^* \mid x;\theta)v_{ai}, \qquad \theta = \phi. \tag{S220}$$

It is bounded by (A3) on the admissible joint state space. □

Lemma S12 (Full-score summation-by-parts identities)**.** *Suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then the particle and population scores satisfy Equations (S224) and (S225) below. Moreover, for every coordinate i and $t \le N$,*

$$\left\|\sum_{k=1}^{t}\alpha^{t-k}h_{k,i}\right\|_\infty \le G'(\theta)\sum_{k=1}^{t}\alpha^{t-k}. \tag{S221}$$

Proof. Step 1: particle summation by parts. Sum Equation (S156) over $n = 1,\dots,N$ and put $k = n - r$. Since the sums are finite,

$$\nabla_\theta\hat{\ell}^\alpha(\theta) = \sum_{k=1}^{N}\left[m(\xi_k)(h_k) + \sum_{t=k+1}^{N}\alpha^{t-k}\{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\}\right]. \tag{S222}$$

For each fixed k, collecting the coefficient of every $m(\xi_t)(h_k)$ gives

$$\begin{aligned} &m(\xi_k)(h_k) + \sum_{t=k+1}^{N}\alpha^{t-k}\{m(\xi_t)(h_k) - m(\xi_{t-1})(h_k)\} \\ &\quad = (1-\alpha)\sum_{t=k}^{N-1}\alpha^{t-k}m(\xi_t)(h_k) + \alpha^{N-k}m(\xi_N)(h_k). \end{aligned} \tag{S223}$$

Indeed, the coefficients are $1-\alpha$ at $t=k$, $\alpha^{t-k} - \alpha^{t+1-k} = (1-\alpha)\alpha^{t-k}$ for $k < t < N$, and α^{N-k} at $t = N$. Interchanging the finite sums once more yields

$$\nabla_\theta\hat{\ell}^\alpha(\theta) = (1-\alpha)\sum_{t=1}^{N-1}m(\xi_t)\left(\sum_{k=1}^{t}\alpha^{t-k}h_k\right) + m(\xi_N)\left(\sum_{k=1}^{N}\alpha^{N-k}h_k\right). \tag{S224}$$

Step 2: population identity and uniform bound. The same algebra applied to Equation (S160) gives

$$\nabla_\theta \ell^\alpha(\theta) = (1-\alpha)\sum_{t=1}^{N-1}\rho_t\left(\sum_{k=1}^{t}\alpha^{t-k}h_k\right) + \rho_N\left(\sum_{k=1}^{N}\alpha^{N-k}h_k\right). \tag{S225}$$

For every coordinate i, Equation (S146) gives

$$\left\|\sum_{k=1}^{t}\alpha^{t-k}h_{k,i}\right\|_\infty \le G'(\theta)\sum_{k=1}^{t}\alpha^{t-k}. \tag{S226}$$

Equation (S226) is identical to Equation (S221), completing the proof. □

Claim S1 (Conditional empirical moments)**.** *Let $\mathcal{F}$ be a sigma-field and let μ be an $\mathcal{F}$-measurable probability measure. Suppose that, conditionally on $\mathcal{F}$, $X_1,\dots,X_J$ are independent with common law μ. The following bounds hold for bounded $\mathcal{F}$-measurable real functions f, f_1, f_2:*

$$\mathbb{E}\left\{\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\,\middle|\,\mathcal{F}\right\} = 0, \tag{S227}$$

$$\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^2\middle|\,\mathcal{F}\right\} = \frac{\mu\{(f-\mu f)^2\}}{J} \le \frac{\operatorname{osc}(f)^2}{4J}, \tag{S228}$$

$$\begin{aligned}\mathbb{E}\left\{\left[\frac{1}{J}\sum_{j=1}^{J}\{f(X_j)-\mu(f)\}\right]^4\middle|\,\mathcal{F}\right\} &= \frac{\mu\{(f-\mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3}\mu\{(f-\mu f)^2\}^2 \\ &\le \frac{\mu\{(f-\mu f)^4\}}{J^3} + \frac{3}{J^2}\mu\{(f-\mu f)^2\}^2 \\ &\le \frac{3\operatorname{osc}(f)^4}{J^2}.\end{aligned} \tag{S229}$$

Consequently, for bounded f_1 and f_2,

$$\begin{aligned}&\left\|\left[\frac{1}{J}\sum_{j=1}^{J}\{f_1(X_j)-\mu(f_1)\}\right]\left[\frac{1}{J}\sum_{j=1}^{J}\{f_2(X_j)-\mu(f_2)\}\right]\right\|_{L^2|\mathcal{F}} \\ &\qquad\le \frac{\sqrt{3}}{J}\operatorname{osc}(f_1)\operatorname{osc}(f_2).\end{aligned} \tag{S230}$$

Proof. Put $Y_j = f(X_j)-\mu(f)$. Conditional independence and $\mathbb{E}(Y_j \mid \mathcal{F}) = 0$ give Equations (S227) and (S228). They also give

$$\begin{aligned}&\mathbb{E}\left\{\left(\frac{1}{J}\sum_{j=1}^{J}Y_j\right)^4\middle|\,\mathcal{F}\right\} \\ &\quad= \frac{1}{J^4}\left\{J\mu\{(f-\mu f)^4\} + 3J(J-1)\mu\{(f-\mu f)^2\}^2\right\}\end{aligned}$$

$$= \frac{\mu\{(f-\mu f)^4\}}{J^3} + \frac{3(J-1)}{J^3}\mu\{(f-\mu f)^2\}^2$$
$$\leq \frac{3}{J^2}\operatorname{osc}(f)^4, \tag{S231}$$

which is Equation (S229). Conditional Cauchy–Schwarz applied to the two fourth moments proves Equation (S230). □

Claim S2 (Normalized-ratio perturbations)**.** *Let μ and ν be probability measures. For $l \in \{0,1\}$, let $a_l > 0$ and b_l be bounded functions such that $\mu(a_l) = 1$, and put*

$$\mathcal{R}_l(\lambda) = \frac{\lambda(b_l)}{\lambda(a_l)}, \qquad D_\mu \mathcal{R}_l = b_l - \mathcal{R}_l(\mu)a_l. \tag{S232}$$

Then

$$\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) = (\nu-\mu)D_\mu\mathcal{R}_l - \frac{(\nu-\mu)(a_l)}{\nu(a_l)}(\nu-\mu)D_\mu\mathcal{R}_l, \tag{S233}$$

and hence

$$\begin{aligned}
&\{\mathcal{R}_1(\nu) - \mathcal{R}_0(\nu)\} - \{\mathcal{R}_1(\mu) - \mathcal{R}_0(\mu)\} \\
&\quad - (\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\} \\
&= -\frac{(\nu-\mu)(a_1-a_0)}{\nu(a_0)\nu(a_1)}(\nu-\mu)D_\mu\mathcal{R}_1 \\
&\quad - \frac{(\nu-\mu)(a_0)}{\nu(a_0)}(\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\}.
\end{aligned} \tag{S234}$$

If $\inf_{l\in\{0,1\}} \nu(a_l) \geq C^{-1}$, then the absolute value of the left side of Equation (S234) is at most

$$\begin{aligned}
&C^2\,|(\nu-\mu)(a_1-a_0)|\,|(\nu-\mu)D_\mu\mathcal{R}_1| \\
&\quad + C\,|(\nu-\mu)(a_0)|\,|(\nu-\mu)\{D_\mu\mathcal{R}_1 - D_\mu\mathcal{R}_0\}|\,.
\end{aligned} \tag{S235}$$

Proof. Since $\mu(a_l) = 1$,

$$\begin{aligned}
\mathcal{R}_l(\nu) - \mathcal{R}_l(\mu) &= \frac{\nu(b_l) - \mathcal{R}_l(\mu)\nu(a_l)}{\nu(a_l)} \\
&= \frac{(\nu-\mu)D_\mu\mathcal{R}_l}{\nu(a_l)} \\
&= (\nu-\mu)D_\mu\mathcal{R}_l - \frac{(\nu-\mu)(a_l)}{\nu(a_l)}(\nu-\mu)D_\mu\mathcal{R}_l,
\end{aligned} \tag{S236}$$

which proves Equation (S233). Moreover,

$$\begin{aligned}
&\frac{(\nu-\mu)(a_1)}{\nu(a_1)} - \frac{(\nu-\mu)(a_0)}{\nu(a_0)} \\
&\quad = \left(1 - \frac{1}{\nu(a_1)}\right) - \left(1 - \frac{1}{\nu(a_0)}\right)
\end{aligned}$$

$$= \frac{\nu(a_1) - \nu(a_0)}{\nu(a_0)\nu(a_1)}$$
$$= \frac{(\nu - \mu)(a_1 - a_0)}{\nu(a_0)\nu(a_1)}. \tag{S237}$$

Subtracting the cases $l = 0$ and $l = 1$ in Equation (S233) now proves Equation (S234). Taking absolute values there and using $\nu(a_0)^{-1} \le C$ and $\nu(a_1)^{-1} \le C$ proves Equation (S235). This exact identity is the paired specialization of the normalized Feynman–Kac perturbation calculation in Equations (3.1)–(3.4) and Appendix A.3 of Del Moral and Rio (2011). □

Lemma S13 (Particle error and variance of the DMOP-α score)**.** *Suppose $J \ge 2$, $\theta = \phi$, $\alpha \in [0, 1]$, Assumptions (A2), (A3), and (A5) hold, and multinomial resampling is used. Suppose also that either (A6) or (A6-J) holds. Then*

$$\operatorname{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} \lesssim \frac{NpG'(\theta)^2(1 + \log J)^2}{J} \left(\sum_{r=0}^{N-1} \alpha^r\right)^2, \tag{S238}$$

$$\left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2 \lesssim \frac{N\sqrt{p}\, G'(\theta)}{J} \sum_{r=0}^{N-1} \alpha^r, \tag{S239}$$

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 \lesssim pG'(\theta)^2 \left(\frac{N(1 + \log J)^2}{J} + \frac{N^2}{J^2}\right) \left(\sum_{r=0}^{N-1} \alpha^r\right)^2. \tag{S240}$$

The finite geometric sum is

$$\sum_{r=0}^{N-1} \alpha^r = \begin{cases} \dfrac{1 - \alpha^N}{1 - \alpha}, & 0 \le \alpha < 1, \\ N, & \alpha = 1. \end{cases} \tag{S241}$$

The constants suppressed by $\lesssim$ depend only on the fixed model and stability bounds and are independent of N, J, and α. As elsewhere, $\operatorname{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|_2^2$ for a random vector Z.

Proof. We bound lag-truncated scores uniformly in their starting time, then use their geometric mixture to recover DMOP-α. Under (A6), the chain rule first decomposes h_s by the age of its simulator contributions.

Step 1: auxiliary lag truncations. For $0 \le k < N$, define

$$\nabla_\theta \hat{s}_k^1(\theta) = \sum_{n=1}^{N} \sum_{r=0}^{\min(k, n-1)} \Delta_{n,r}^J, \qquad \nabla_\theta s_k^1(\theta) = \sum_{n=1}^{N} \sum_{r=0}^{\min(k, n-1)} \Delta_{n,r}. \tag{S242}$$

For a fixed creation time s, the terms in the first sum telescope:

$$m(\xi_s)(h_s) + \sum_{n=s+1}^{\min(N, s+k)} \{m(\xi_n)(h_s) - m(\xi_{n-1})(h_s)\}$$
$$= m(\xi_{\min(N, s+k)})(h_s). \tag{S243}$$

Summing Equation (S243) over $s = 1, \ldots, N$ gives

$$\nabla_\theta \hat{s}_k^1(\theta) = \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N)\left(\sum_{s=N-k}^{N} h_s\right), \tag{S244}$$

$$\nabla_\theta s_k^1(\theta) = \sum_{s=1}^{N-k-1} \rho_{s+k}(h_s) + \rho_N\left(\sum_{s=N-k}^{N} h_s\right). \tag{S245}$$

An empty sum in these displays is zero.

Step 1a: chain-rule decomposition. All derivatives in this step are evaluated at $\theta = \phi$, with reference resampling indices fixed. Write $X_u^\theta = \texttt{process}_u(X_{u-1}^\theta; \theta, \omega_u)$, and let $\partial_{\theta_i}\texttt{process}_u$ denote its partial derivative holding X_{u-1}^θ fixed. The chain rule, iterated from time zero, gives

$$\partial_{\theta_i} X_s^\theta = \frac{\partial X_s^\theta}{\partial X_0^\theta}\partial_{\theta_i} X_0^\theta + \sum_{u=1}^{s} \frac{\partial X_s^\theta}{\partial X_u^\theta}\partial_{\theta_i}\texttt{process}_u(X_{u-1}^\theta; \theta, \omega_u). \tag{S246}$$

Under (A6), decompose the coordinate-i measurement gradient at time s by the age of its chain-rule contributions. The superscript $[r]$ is an age index, not a power or derivative order: for $1 \le r < s$, $h_{s,i}^{[r]}$ is the contribution from the simulator's partial parameter derivative at time $s-r$, propagated to the measurement at time s. This age differs from the discount lag $n - s$, which gives the coefficient α^{n-s} when the time-s measurement gradient contributes at time n. Define

$$\begin{aligned}
h_{s,i}^{[0]} &= \partial_{\theta_i} \log f_{Y_s|X_s}(y_s^* \mid x; \theta)\big|_{x=X_s^\theta} + D_x \log f_{Y_s|X_s}(y_s^* \mid x; \theta)\big|_{x=X_s^\theta}\, \partial_{\theta_i}\texttt{process}_s(X_{s-1}^\theta; \theta, \omega_s),\\
h_{s,i}^{[r]} &= D_x \log f_{Y_s|X_s}\big(y_s^* \mid X_s^\theta(x, \omega_{s-r+1:s}); \theta\big)\Big|_{x=X_{s-r}^\theta} \partial_{\theta_i}\texttt{process}_{s-r}(X_{s-r-1}^\theta; \theta, \omega_{s-r}), \quad 1 \le r < s,\\
h_{s,i}^{[s]} &= D_x \log f_{Y_s|X_s}\big(y_s^* \mid X_s^\theta(x, \omega_{1:s}); \theta\big)\Big|_{x=X_0^\theta} \partial_{\theta_i} X_0^\theta.
\end{aligned} \tag{S247}$$

Here $h_{s,i}^{[0]}$ combines the direct parameter derivative of the log measurement density, holding x fixed, with the contribution from the time-s simulator's partial parameter derivative. The endpoint $h_{s,i}^{[s]}$ is the contribution from the initial-state derivative. Multiplying Equation (S246) by the spatial log measurement derivative gives

$$h_{s,i} = \sum_{r=0}^{s} h_{s,i}^{[r]}, \qquad \|h_{s,i}^{[r]}\|_\infty \le CG'(\theta)\kappa^r. \tag{S248}$$

For $r = 0$ use (A6) with $t = s$ and the bounded partial derivatives; for $1 \le r < s$ use it with $t = s - r$; for $r = s$ use $t = 0$ and the initial-gradient bound. For $r < s$, the displayed function depends only on X_{s-r-1} and the simulator variables from $s - r$ through s. For $r = s$ it also uses the initial simulator draw, whose particle replicates are independent. Set $h_{s,i}^{[r]} = 0$ for $r > s$.

Under (A6-J), take instead $h_{s,i}^{[0]} = h_{s,i}$, the bounded current function in Equation (S220), and $h_{s,i}^{[r]} = 0$ for $r \ge 1$. Use the joint pair as the state throughout the particle calculation. The sums over r below then contain only $r = 0$; κ is needed only under (A6).

Step 1b: finite-window particle error. Let f be a bounded real function of the state at time a and the subsequent simulator variables through time $b \ge a$, with no earlier history. Write $Q_{a,b}$ for the

path lift of the semigroup in Equation (S195). First suppose $a \geq 1$. Under (A6), the starting cloud is the physical cloud; under (A6-J), it is the joint cloud. Adding and subtracting the conditional population continuation gives

$$m(\xi_b)(f) - \rho_b(f) = \left\{ m(\xi_b)(f) - \frac{\eta_a^J Q_{a,b}(f)}{\eta_a^J Q_{a,b}(1)} \right\} + \left\{ \frac{\eta_a^J Q_{a,b}(f)}{\eta_a^J Q_{a,b}(1)} - \frac{\eta_a Q_{a,b}(f)}{\eta_a Q_{a,b}(1)} \right\}. \tag{S249}$$

For the first brace, condition on η_a^J. Let $\nu = m(\xi_b^P)$ and $\mu = \Phi_{a,b}(\eta_a^J)$ on the prediction window. The first prediction following selection from η_a^J has conditionally independent particles with common law $\Phi_{a,a+1}(\eta_a^J)$. To apply Theorems 3.1 and 3.3 of Del Moral and Jasra (2018), first note that future normalizers depend on the current physical state and satisfy Equation (S200). For a bounded function $\psi(X_b, \omega_b)$ of the last state and its innovation, its normalized continuation, for $b > a$, is

$$\begin{aligned} \frac{Q_{a,b}(\psi)}{Q_{a,b}(1)} &= P_{a,a+1}^{(b)} \cdots P_{b-2,b-1}^{(b)} \left[\frac{\mathbb{E}\{G_b(X_b)\psi(X_b, \omega_b) \mid X_{b-1} = \cdot\}}{M_b(G_b)} \right], \\ \operatorname{osc}\left\{ \frac{Q_{a,b}(\psi)}{Q_{a,b}(1)} \right\} &\leq \varrho^{b-a-1} \operatorname{osc}(\psi). \end{aligned} \tag{S250}$$

The product is empty when $b = a + 1$. The bracket is a conditional average of ψ, and the other kernels contract as in Equation (S205). Thus the innovation-augmented marginal has a summable forgetting bound, with a one-step delay. For prediction histories, omit the terminal G_b: the final bracket becomes $\mathbb{E}[\psi(X_b, \omega_b) \mid X_{b-1} = \cdot]$, and the same contraction argument gives the corresponding summable bound. Apply the path construction of Del Moral and Jasra (2018) to this marginal. Under (A6-J), use the joint marginal and its transition bounds instead, as in Lemma S11. The cited theorems now apply over at most $b - a$ updates, uniformly in a. For the second moment, put $\bar{f} = f - \mu(f)$ and choose c so that $0 \leq \bar{f} + c \leq \operatorname{osc}(f)$. Then $\mu(\bar{f} + c) = c$, and

$$\begin{aligned} \mathbb{E}\{|\nu(f) - \mu(f)|^2 \mid \eta_a^J\} &= \frac{1}{J^2} \sum_j \mathbb{E}\{\bar{f}(\xi_{b,j}^P)^2 \mid \eta_a^J\} + \frac{1}{J^2} \sum_{j \neq l} \mathbb{E}\{\bar{f}(\xi_{b,j}^P)\bar{f}(\xi_{b,l}^P) \mid \eta_a^J\}, \\ \mathbb{E}\{\bar{f}(\xi_{b,j}^P)\bar{f}(\xi_{b,l}^P) \mid \eta_a^J\} &= \mathbb{E}\{(\bar{f} + c)(\xi_{b,j}^P)(\bar{f} + c)(\xi_{b,l}^P) \mid \eta_a^J\} - c^2 \\ &\quad - c\{\mathbb{E}[(\bar{f} + c)(\xi_{b,j}^P) \mid \eta_a^J] - c\} \\ &\quad - c\{\mathbb{E}[(\bar{f} + c)(\xi_{b,l}^P) \mid \eta_a^J] - c\}. \end{aligned} \tag{S251}$$

The diagonal terms are at most $\operatorname{osc}(f)^2$. Theorem 3.3 bounds the first difference in the second identity by $C(b-a)\operatorname{osc}(f)^2/J$; Theorem 3.1 bounds each remaining difference by $C(b-a)\operatorname{osc}(f)/J$, and $|c| \leq \operatorname{osc}(f)$. Thus, initially for $J \geq c_0(b - a + 1)$,

$$\mathbb{E}\{|\nu(f) - \mu(f)|^2 \mid \eta_a^J\} \leq \frac{C(b - a + 1)}{J} \operatorname{osc}(f)^2, \tag{S252}$$

$$|\mathbb{E}\{\nu(f) - \mu(f) \mid \eta_a^J\}| \leq \frac{C(b - a + 1)}{J} \operatorname{osc}(f). \tag{S253}$$

When $a = b$, $\nu = \mu = \eta_a^J$, so both prediction errors are zero.

To pass to filtering, put $d = G_b\{f - \Psi_b(\mu)(f)\}$. Then $\mu(d) = 0$ and the exact ratio expansion is

$$\Psi_b(\nu)(f) - \Psi_b(\mu)(f) = \frac{(\nu - \mu)(d)}{\nu(G_b)}$$

$$= \frac{(\nu-\mu)(d)}{\mu(G_b)} - \frac{(\nu-\mu)(d)(\nu-\mu)(G_b)}{\nu(G_b)\mu(G_b)}. \tag{S254}$$

Both denominators are at least $\underline{G}$, and $\operatorname{osc}(d) \le 2\bar{G}\operatorname{osc}(f)$. Equations (S252) and (S253) consequently give

$$\begin{aligned}
\mathbb{E}\{|\Psi_b(\nu)(f) - \Psi_b(\mu)(f)|^2 \mid \eta_a^J\} &\le \frac{C(b-a+1)}{J}\operatorname{osc}(f)^2, \\
|\mathbb{E}\{\Psi_b(\nu)(f) - \Psi_b(\mu)(f) \mid \eta_a^J\}| &\le \frac{|\mathbb{E}\{(\nu-\mu)(d) \mid \eta_a^J\}|}{\underline{G}} + \frac{\|(\nu-\mu)(d)\|_{L^2(\cdot|\eta_a^J)}\|(\nu-\mu)(G_b)\|_{L^2(\cdot|\eta_a^J)}}{\underline{G}^2} \\
&\le \frac{C(b-a+1)}{J}\operatorname{osc}(f).
\end{aligned} \tag{S255}$$

Finally, conditional multinomial resampling gives

$$\begin{aligned}
\mathbb{E}\{m(\xi_b)(f) - \Psi_b(\nu)(f) \mid \xi_b^P\} &= 0, \\
\mathbb{E}\{|m(\xi_b)(f) - \Psi_b(\nu)(f)|^2 \mid \xi_b^P\} &\le \operatorname{osc}(f)^2/J.
\end{aligned} \tag{S256}$$

Adding this centered term proves the same orders for the first brace of Equation (S249).

For the second brace, put

$$q = \frac{Q_{a,b}(1)}{\eta_a Q_{a,b}(1)}, \qquad d = \frac{Q_{a,b}\{f - \rho_b(f)\}}{\eta_a Q_{a,b}(1)}. \tag{S257}$$

These functions of the starting state satisfy

$$\eta_a(q) = 1, \quad \eta_a(d) = 0, \quad R^{-1} \le q \le R, \quad \|d\|_\infty \le R\operatorname{osc}(f). \tag{S258}$$

Karjalainen et al.'s Lemma 2.3 applies to these bounded current prediction functions. The prediction weak-bias bound follows from Lemma S11, since for $a \ge 2$

$$\begin{aligned}
\mathbb{E}\eta_a^J(u) - \eta_a(u) &= \mathbb{E}m(\xi_{a-1})(M_a u) - \rho_{a-1}(M_a u), \\
|\mathbb{E}\eta_a^J(u) - \eta_a(u)| &\le C\operatorname{osc}(u)/J.
\end{aligned} \tag{S259}$$

At initialization the corresponding empirical averages are independent and unbiased. Applying these bounds to d and q yields

$$\mathbb{E}\left|\frac{\eta_a^J(d)}{\eta_a^J(q)}\right|^2 \le R^2\mathbb{E}|\eta_a^J(d)|^2 \le C\operatorname{osc}(f)^2/J, \tag{S260}$$

$$\begin{aligned}
\left|\mathbb{E}\frac{\eta_a^J(d)}{\eta_a^J(q)}\right| &= \left|\mathbb{E}\eta_a^J(d) - \mathbb{E}\frac{\eta_a^J(d)\{\eta_a^J(q)-1\}}{\eta_a^J(q)}\right| \\
&\le |\mathbb{E}\eta_a^J(d)| + R\{\mathbb{E}|\eta_a^J(d)|^2\,\mathbb{E}|\eta_a^J(q)-1|^2\}^{1/2} \le C\operatorname{osc}(f)/J.
\end{aligned} \tag{S261}$$

Combine the two braces using $|u+v|^2 \le 2|u|^2 + 2|v|^2$. For $a = 0$ and $b \ge 1$, the time-one prediction histories, including the independent initial draw and its gradient when present, are unconditionally independent with their common population prediction law. Apply the same diagonal–off-diagonal calculation to these histories over the remaining $b-1$ updates, followed by the displayed Bayes-ratio and final resampling steps. The normalizer and marginal-forgetting bounds above still apply,

because future transitions depend on these histories only through X_1 under (A6), or the joint pair under (A6-J). This gives bounds $Cb\,\mathrm{osc}(f)^2/J$ and $Cb\,\mathrm{osc}(f)/J$. For $a = b = 0$, the empirical average is unbiased and its variance is at most $\mathrm{osc}(f)^2/J$. For $J < c_0(b-a+1)$ the pathwise bound $2\|f\|_\infty$ gives the same orders after a constant shift of f and enlargement of C. For every $J \geq 2$ we have therefore proved

$$\begin{aligned} \|m(\xi_b)(f) - \rho_b(f)\|_{L^2} &\leq C\,\mathrm{osc}(f)\sqrt{(b-a+1)/J}, \\ |\mathbb{E}m(\xi_b)(f) - \rho_b(f)| &\leq C\,\mathrm{osc}(f)(b-a+1)/J. \end{aligned} \tag{S262}$$

Step 1c: sum over gradient ages. Apply Equation (S262) to $f = h_{s,i}^{[r]}$ at $b = s+q$. Under (A6) start at $a = \max(0, s-r-1)$; when $r = s$ include the independent initial draw. The window contains at most $q+r+2$ times. Under (A6-J), use only $r = 0$ and start at s. Thus

$$\begin{aligned} \|m(\xi_{s+q})(h_{s,i}^{[r]}) - \rho_{s+q}(h_{s,i}^{[r]})\|_{L^2} &\leq \frac{CG'(\theta)\kappa^r}{\sqrt{J}}\sqrt{q+r+2}, \\ |\mathbb{E}m(\xi_{s+q})(h_{s,i}^{[r]}) - \rho_{s+q}(h_{s,i}^{[r]})| &\leq \frac{CG'(\theta)\kappa^r}{J}(q+r+2). \end{aligned} \tag{S263}$$

Under (A6-J), only the $r = 0$ inequalities are used. Since $q+r+2 \leq (q+1)(r+2)$,

$$\begin{aligned} \sum_{r=0}^{s} \kappa^r\sqrt{q+r+2} &\leq \sqrt{q+1}\sum_{r=0}^{\infty}\kappa^r\sqrt{r+2} \leq C_\kappa\sqrt{q+1}, \\ \sum_{r=0}^{s}\kappa^r(q+r+2) &\leq \frac{q+2}{1-\kappa} + \frac{\kappa}{(1-\kappa)^2} \leq C_\kappa(q+1). \end{aligned} \tag{S264}$$

Minkowski's inequality, the triangle inequality for expectations, and Equation (S248) give

$$\mathbb{E}|m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i})|^2 \leq C(q+1)G'(\theta)^2/J, \tag{S265}$$

$$|\mathbb{E}m(\xi_{s+q})(h_{s,i}) - \rho_{s+q}(h_{s,i})| \leq C(q+1)G'(\theta)/J. \tag{S266}$$

Taking $q = 0$ proves the uniform J^{-1} weak-bias bound for h_s.

For the terminal block, put $s = N-q$ and apply Minkowski:

$$\begin{aligned} &\left\| m(\xi_N)\left(\sum_{s=N-k}^{N} h_{s,i}\right) - \rho_N\left(\sum_{s=N-k}^{N} h_{s,i}\right)\right\|_{L^2} \\ &\quad\leq \sum_{q=0}^{k}\|m(\xi_N)(h_{N-q,i}) - \rho_N(h_{N-q,i})\|_{L^2} \leq \frac{CG'(\theta)}{\sqrt{J}}\sum_{q=0}^{k}\sqrt{q+1} \leq \frac{CG'(\theta)(k+1)^{3/2}}{\sqrt{J}}. \end{aligned} \tag{S267}$$

Consequently,

$$\mathbb{E}\left| m(\xi_N)\left(\sum_{s=N-k}^{N} h_{s,i}\right) - \rho_N\left(\sum_{s=N-k}^{N} h_{s,i}\right)\right|^2 \leq C(k+1)^3G'(\theta)^2/J, \tag{S268}$$

$$\left|\mathbb{E}m(\xi_N)\left(\sum_{s=N-k}^{N} h_{s,i}\right) - \rho_N\left(\sum_{s=N-k}^{N} h_{s,i}\right)\right| \leq \frac{CG'(\theta)}{J}\sum_{q=0}^{k}(q+1) \leq C(k+1)^2G'(\theta)/J. \tag{S269}$$

Step 1d: covariance summation. Fix r and consider the interior terms $m(\xi_{s+k})(h_{s,i}^{[r]})$, $1 \le s < N-k$. Under (A6) these depend on the physical cloud at $\max(0, s-r-1)$ and subsequent variables through $s+k$. Under (A6-J) the $r=0$ terms depend on the joint cloud at s and the next k updates. Write $a(u)$ for the mixing coefficient bound in Equation (S130), using the physical cloud in the first case and the joint cloud in the second. Equation (S263) and Cauchy–Schwarz give

$$|\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \le \frac{CG'(\theta)^2\kappa^{2r}(k+r+2)}{J}. \tag{S270}$$

For $t-s > k+r+2$, the prediction clouds separating the windows are at least $t-s-k-r-2$ transitions apart, allowing one transition to absorb the past endpoint resampling. The centered variables have L^4 norms at most $2CG'(\theta)\kappa^r$ by their pathwise bounds, so Davydov's inequality gives

$$|\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \le CG'(\theta)^2\kappa^{2r}a(t-s-k-r-2)^{1/2}. \tag{S271}$$

Under (A6-J), the larger gap $t-s-k-1$ is available; the weaker displayed bound with $r=0$ suffices. Combining these inequalities gives

$$\begin{aligned}&\sum_{t=s+1}^{N-k-1} |\mathrm{Cov}\{m(\xi_{s+k})(h_{s,i}^{[r]}), m(\xi_{t+k})(h_{t,i}^{[r]})\}| \\ &\quad \le CG'(\theta)^2\kappa^{2r}\left\{\frac{(k+r+2)^2}{J} + \sum_{u=1}^{\infty}\min\left(\frac{k+r+2}{J}, a(u)^{1/2}\right)\right\}.\end{aligned} \tag{S272}$$

Choose $u_0 \le C(1+\log J)^2$ so that the geometric bound for $a(u_0)^{1/2}$ is at most J^{-1}. The geometric tail has sum at most $C(1+\log J)/J$, and hence

$$\sum_{u=1}^{\infty}\min\left(\frac{k+r+2}{J}, a(u)^{1/2}\right) \le \frac{u_0(k+r+2)}{J} + \frac{C(1+\log J)}{J} \le \frac{C(k+r+2)(1+\log J)^2}{J}. \tag{S273}$$

Retaining this logarithmic factor, expanding the variance as the diagonal sum plus twice the covariance sum gives

$$\mathrm{Var}\left\{\sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_{s,i}^{[r]})\right\} \lesssim \frac{NG'(\theta)^2\kappa^{2r}(k+r+2)^2(1+\log J)^2}{J}. \tag{S274}$$

Center each age sum and apply Minkowski:

$$\begin{aligned}&\left\|\sum_{s=1}^{N-k-1}\{m(\xi_{s+k})(h_{s,i}) - \mathbb{E}m(\xi_{s+k})(h_{s,i})\}\right\|_{L^2} \\ &\quad \le \sum_{r=0}^{N}\left\|\sum_{s=1}^{N-k-1}\{m(\xi_{s+k})(h_{s,i}^{[r]}) - \mathbb{E}m(\xi_{s+k})(h_{s,i}^{[r]})\}\right\|_{L^2} \\ &\quad \lesssim G'(\theta)(1+\log J)\sqrt{N/J}\sum_{r=0}^{N}\kappa^r(k+r+2) \le C_\kappa G'(\theta)(1+\log J)\sqrt{N/J}(k+1).\end{aligned} \tag{S275}$$

Again the joint alternative uses only $r = 0$. The terminal variance is bounded by its second moment in Equation (S268); since $k + 1 \le N$,

$$\mathrm{Var}\left\{ m(\xi_N) \left(\sum_{s=N-k}^{N} h_{s,i} \right) \right\} \le C(k+1)^3 G'(\theta)^2 / J \le CN(k+1)^2 G'(\theta)^2 / J. \tag{S276}$$

Use $\mathrm{Var}(A + B) \le 2\mathrm{Var}(A) + 2\mathrm{Var}(B)$ in Equation (S244) and sum coordinates:

$$\mathrm{Var}\{\nabla_\theta \hat{s}_k^1(\theta)\} \lesssim NpG'(\theta)^2 (k+1)^2 (1 + \log J)^2 / J. \tag{S277}$$

Finally, the one-input and terminal bias bounds yield

$$\begin{aligned} \|\mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) - \nabla_\theta s_k^1(\theta)\|_2 &\le \frac{C\sqrt{p}\, G'(\theta)}{J} \{(N-k-1)(k+1) + (k+1)^2\} \\ &= \frac{CN\sqrt{p}\, G'(\theta)(k+1)}{J}. \end{aligned} \tag{S278}$$

Step 2: geometric mixture. For $0 \le k < N - 1$, put

$$\beta_k = (1-\alpha)\alpha^k, \qquad \beta_{N-1} = \alpha^{N-1}. \tag{S279}$$

These nonnegative numbers sum to one. Moreover, for every $0 \le r < N$,

$$\begin{aligned} \sum_{k=r}^{N-1} \beta_k &= \sum_{k=r}^{N-2} (1-\alpha)\alpha^k + \alpha^{N-1} \\ &= \alpha^r (1 - \alpha^{N-1-r}) + \alpha^{N-1} \\ &= \alpha^r. \end{aligned} \tag{S280}$$

Equations (S156), (S160), and (S242) therefore give the pathwise identities

$$\nabla_\theta \hat{\ell}^\alpha(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_\theta \hat{s}_k^1(\theta), \qquad \nabla_\theta \ell^\alpha(\theta) = \sum_{k=0}^{N-1} \beta_k \nabla_\theta s_k^1(\theta). \tag{S281}$$

The coefficient of each $\Delta_{n,r}^J$ or $\Delta_{n,r}$ is the tail sum in Equation (S280). For $N \ge 2$, the mixture weight on $\nabla_\theta \hat{s}_0^1(\theta)$ is $\beta_0 = 1 - \alpha$. Every truncated score contains $\Delta_{n,0}^J$, so its coefficient is

$$\sum_{k=0}^{N-1} \beta_k = 1. \tag{S282}$$

In particular, at $\alpha = 0$,

$$\nabla_\theta \hat{\ell}_n^0(\theta) = \Delta_{n,0}^J, \qquad \nabla_\theta \ell_n^0(\theta) = \Delta_{n,0}. \tag{S283}$$

For $N = 1$, the only mixture weight is $\beta_0 = 1$ for every α.

The mean lag in this censored geometric mixture satisfies

$$\sum_{k=0}^{N-1} \beta_k (k+1) = \sum_{r=0}^{N-1} \sum_{k=r}^{N-1} \beta_k$$

$$= \sum_{r=0}^{N-1} \alpha^r. \tag{S284}$$

Center the first identity in Equation (S281). Minkowski's inequality, Equation (S277), and Equation (S284) give

$$\begin{aligned}
\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta)\right\|_{L^2} &\leq \sum_{k=0}^{N-1} \beta_k \left\|\nabla_\theta \hat{s}_k^1(\theta) - \mathbb{E}\nabla_\theta \hat{s}_k^1(\theta)\right\|_{L^2} \\
&\lesssim C(1+\log J)\sqrt{\frac{NpG'(\theta)^2}{J}} \sum_{k=0}^{N-1} \beta_k(k+1) \\
&\lesssim C(1+\log J)\sqrt{\frac{NpG'(\theta)^2}{J}} \sum_{r=0}^{N-1} \alpha^r.
\end{aligned} \tag{S285}$$

Squaring proves Equation (S238). Taking expectations in Equation (S281), applying the triangle inequality and Equation (S278), and then using Equation (S284) give

$$\begin{aligned}
\left\|\mathbb{E}\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2 &\leq \sum_{k=0}^{N-1} \beta_k \left\|\mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 \\
&\leq \frac{CN\sqrt{p}\,G'(\theta)}{J} \sum_{k=0}^{N-1} \beta_k(k+1) \\
&= \frac{CN\sqrt{p}\,G'(\theta)}{J} \sum_{r=0}^{N-1} \alpha^r,
\end{aligned} \tag{S286}$$

which proves Equation (S239). Finally, for a random vector Z and deterministic z,

$$\mathbb{E}\|Z - z\|_2^2 = \mathrm{Var}(Z) + \|\mathbb{E}Z - z\|_2^2. \tag{S287}$$

Apply this identity with $Z = \nabla_\theta \hat{\ell}^\alpha(\theta)$ and $z = \nabla_\theta \ell^\alpha(\theta)$, and substitute Equations (S238) and (S239). This proves Equation (S240). □

Corollary S1 (Particle bias of the DMOP-α score)**.** *Under the conditions of Lemma S13, Equation (S239) holds.*

Proof. This is the second conclusion of Lemma S13. □

Lemma S14 (Population discounting bias)**.** *Under the conditions of Lemma S10, suppose $\theta = \phi$ and $\alpha \in [0,1]$. Then*

$$\begin{aligned}
\|\nabla_\theta \ell_n^\alpha(\theta) - \nabla_\theta \ell_n(\theta)\|_2 &\leq C\sqrt{p}\,G'(\theta) \sum_{r=1}^{n-1} (1-\alpha^r)\varrho^r \\
&\leq C\sqrt{p}\,G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}.
\end{aligned} \tag{S288}$$

For the full score,

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad \le C\sqrt{p}\, G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q \\
&\quad = C\sqrt{p}\, G'(\theta)\varrho(1-\alpha)\frac{(N-1)-N\alpha\varrho+(\alpha\varrho)^N}{(1-\alpha\varrho)^2} \\
&\quad \le C(N-1)\sqrt{p}\, G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}.
\end{aligned} \tag{S289}$$

For $N = 1$, the finite sum is empty, the displayed fraction is zero, and the full-score difference is zero. The dependence on the transition bounds can be sharpened further. With the existing $\varrho = 1 - \underline{M}/(2\bar{M})$ from Equation (S189), for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad \le 2\sqrt{p}\, G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)\left\{\alpha\frac{2\varrho-1}{3-2\varrho}\right\}^q \\
&\quad \le 2(N-1)\sqrt{p}\, G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.
\end{aligned} \tag{S290}$$

If every transition is independent of its starting location, in particular if $\varrho = \frac{1}{2}$, the population discounting bias is zero for every N and α.

Proof. Step 1: exact lagwise difference. Equation (S160) at α and at $\alpha = 1$ gives

$$\begin{aligned}
\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta) &= \sum_{r=0}^{n-1}(1-\alpha^r)\Delta_{n,r} \\
&= \sum_{r=1}^{n-1}(1-\alpha^r)\Delta_{n,r},
\end{aligned} \tag{S291}$$

The coefficient of $\Delta_{n,0}$ is one in both scores, so its difference is zero. In particular, at the left endpoint Equation (S283) gives

$$\nabla_\theta \ell_n^0(\theta) = \Delta_{n,0}, \qquad \nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^0(\theta) = \sum_{r=1}^{n-1}\Delta_{n,r}. \tag{S292}$$

Thus DMOP-0 retains the zero-lag contribution with coefficient one and retains no positive-lag contribution.

Step 2: apply forgetting and sum. Taking norms, applying the triangle inequality and Lemma S10, and then extending the nonnegative scalar sum to infinity give

$$\|\nabla_\theta \ell_n(\theta) - \nabla_\theta \ell_n^\alpha(\theta)\|_2 \le \sum_{r=1}^{n-1}(1-\alpha^r)\|\Delta_{n,r}\|_2$$

$$
\begin{aligned}
&\le C\sqrt{p}\, G'(\theta) \sum_{r=1}^{n-1} (1-\alpha^r)\varrho^r \\
&\le C\sqrt{p}\, G'(\theta) \sum_{r=1}^{\infty} (1-\alpha^r)\varrho^r \\
&= C\sqrt{p}\, G'(\theta) \left\{ \sum_{r=1}^{\infty} \varrho^r - \sum_{r=1}^{\infty} (\alpha\varrho)^r \right\} \\
&= C\sqrt{p}\, G'(\theta) \left\{ \frac{\varrho}{1-\varrho} - \frac{\alpha\varrho}{1-\alpha\varrho} \right\} \\
&= C\sqrt{p}\, G'(\theta) \frac{\varrho(1-\alpha)}{(1-\varrho)(1-\alpha\varrho)}. \qquad \text{(S293)}
\end{aligned}
$$

Step 3: exact summation by parts for the full score. Reindexing Equation (S160) by $s = n - r$ shows that the contribution of h_s to the full DMOP-α population score is

$$
\begin{aligned}
&\rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&\quad = \rho_s(h_s) + \sum_{q=1}^{N-s} \alpha^q \rho_{s+q}(h_s) - \sum_{q=0}^{N-s-1} \alpha^{q+1} \rho_{s+q}(h_s) \\
&\quad = (1-\alpha) \sum_{q=0}^{N-s-1} \alpha^q \rho_{s+q}(h_s) + \alpha^{N-s} \rho_N(h_s). \qquad \text{(S294)}
\end{aligned}
$$

The contribution of h_s to the exact score in Equation (S158) is $\rho_N(h_s)$. Subtracting Equation (S294) from this contribution gives

$$
\begin{aligned}
&\rho_N(h_s) - \rho_s(h_s) - \sum_{q=1}^{N-s} \alpha^q \{\rho_{s+q}(h_s) - \rho_{s+q-1}(h_s)\} \\
&\quad = (1-\alpha) \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \qquad \text{(S295)}
\end{aligned}
$$

Summing Equation (S295) over $s = 1, \ldots, N-1$ yields the exact identity

$$
\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) = (1-\alpha) \sum_{s=1}^{N-1} \sum_{q=0}^{N-s-1} \alpha^q \{\rho_N(h_s) - \rho_{s+q}(h_s)\}. \qquad \text{(S296)}
$$

Step 4: contract each partial-smoother comparison. Fix $s \le N-1$, $0 \le q \le N-s-1$, and a coordinate $i \in \{1, \ldots, p\}$. The Markov property implies that observations after time $s+q$ are conditionally independent of the path through time s given X_{s+q}. Hence the following two conditional expectations are the same function of x:

$$
\mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\} = \mathbb{E}_{\rho_{s+q}}\{h_{s,i} \mid X_{s+q} = x\}
$$

$$= \begin{cases} \bar{h}_{s,i}(x), & q = 0, \\ (B_{s+q} B_{s+q-1} \cdots B_{s+1} \bar{h}_{s,i})(x), & q \geq 1. \end{cases} \tag{S297}$$

Here $\bar{h}_{s,i}$ and B_t are already defined in Equations (S173) and (S167). We first sharpen the bound on the change of a probability expectation under a positive likelihood.

For a probability measure μ, a positive function L with $0 < a \leq L \leq b$, and a bounded real function f, suppose first that $a < b$. Convexity of the absolute value gives, pointwise,

$$|L - \mu(L)| \leq \frac{b - L}{b - a}\{\mu(L) - a\} + \frac{L - a}{b - a}\{b - \mu(L)\}. \tag{S298}$$

Subtract the midpoint of the range of f inside $\mu[f\{L - \mu(L)\}]$, and then integrate Equation (S298). Since $\mu\{L - \mu(L)\} = 0$, this gives

$$\begin{aligned} \left| \frac{\mu(fL)}{\mu(L)} - \mu(f) \right| &\leq \frac{\operatorname{osc}(f)}{2\mu(L)} \mu|L - \mu(L)| \\ &\leq \operatorname{osc}(f) \frac{\{b - \mu(L)\}\{\mu(L) - a\}}{\mu(L)(b - a)} \\ &= \operatorname{osc}(f) \frac{a + b - \mu(L) - ab/\mu(L)}{b - a} \\ &\leq \operatorname{osc}(f) \frac{\sqrt{b} - \sqrt{a}}{\sqrt{b} + \sqrt{a}}. \end{aligned} \tag{S299}$$

The last inequality is $\mu(L) + ab/\mu(L) \geq 2\sqrt{ab}$. When $a = b$, L is constant and the difference is zero.

Equation (S189) gives $\underline{M}/\bar{M} = 2(1 - \varrho)$ and $\frac{1}{2} \leq \varrho < 1$. For $\frac{1}{2} < \varrho < 1$, apply Equation (S299) to the two backward kernels at x and x'. In this application set

$$L(u) = \frac{f_{X_t|X_{t-1}}(x \mid u; \theta)}{f_{X_t|X_{t-1}}(x' \mid u; \theta)}. \tag{S300}$$

Equation (S167) and (A2) give

$$2(1 - \varrho) \leq L(u) \leq \{2(1 - \varrho)\}^{-1}, \qquad B_t(f)(x) = \frac{B_t(fL)(x')}{B_t(L)(x')}. \tag{S301}$$

Taking $\mu = B_t(x', \cdot)$, $a = 2(1 - \varrho)$, and $b = \{2(1 - \varrho)\}^{-1}$ in Equation (S299) therefore gives

$$\operatorname{osc}\{B_t(f)\} \leq \frac{2\varrho - 1}{3 - 2\varrho} \operatorname{osc}(f). \tag{S302}$$

Applying this inequality to the q kernels in Equation (S297), starting from $\operatorname{osc}(\bar{h}_{s,i}) \leq 2G'(\theta)$, yields

$$\operatorname{osc}[x \longmapsto \mathbb{E}_{\rho_N}\{h_{s,i} \mid X_{s+q} = x\}] \leq 2G'(\theta) \left(\frac{2\varrho - 1}{3 - 2\varrho} \right)^q \leq 2G'(\theta)\varrho^q. \tag{S303}$$

This also covers $q = 0$, when the product contains no backward kernel.

Next, the future likelihood already defined in Equation (S174) satisfies, since $s+q<N$,

$$L_{s+q,N}(x)=\int M_{s+q+1}(x,du)G_{s+q+1}(u)L_{s+q+1,N}(u),$$
$$L_{s+q,N}(x)\geq 2(1-\varrho)L_{s+q,N}(x') \qquad \text{for every } x,x'. \tag{S304}$$

The second line follows by applying $M_{s+q+1}(x,\cdot)\geq 2(1-\varrho)M_{s+q+1}(x',\cdot)$ to the nonnegative integrand. Thus its supremum divided by its infimum is at most $\{2(1-\varrho)\}^{-1}$. Apply Equation (S299) with $\mu=\pi_{s+q}$ and $L=L_{s+q,N}$ to obtain

$$\left|\frac{\pi_{s+q}(fL_{s+q,N})}{\pi_{s+q}(L_{s+q,N})}-\pi_{s+q}(f)\right|\leq\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\operatorname{osc}(f). \tag{S305}$$

The tower property and the common conditional expectation in Equation (S297) identify the two quantities being compared:

$$\rho_N(h_{s,i})=\frac{\rho_{s+q}\big[\mathbb{E}_{\rho_{s+q}}\{h_{s,i}\mid X_{s+q}\}L_{s+q,N}(X_{s+q})\big]}{\rho_{s+q}\{L_{s+q,N}(X_{s+q})\}},$$
$$\rho_{s+q}(h_{s,i})=\rho_{s+q}\big[\mathbb{E}_{\rho_{s+q}}\{h_{s,i}\mid X_{s+q}\}\big]. \tag{S306}$$

Consequently, Equations (S303)– (S306) give

$$\begin{aligned}|\rho_N(h_{s,i})-\rho_{s+q}(h_{s,i})|&\leq 2G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\left(\frac{2\varrho-1}{3-2\varrho}\right)^q\\&\leq 2G'(\theta)\varrho^{q+1}.\end{aligned} \tag{S307}$$

For the second inequality, each of $(1-\sqrt{2(1-\varrho)})/(1+\sqrt{2(1-\varrho)})$ and $(2\varrho-1)/(3-2\varrho)$ is at most $2\varrho-1\leq\varrho$. Summing the squared coordinate bounds and taking square roots gives

$$\begin{aligned}\|\rho_N(h_s)-\rho_{s+q}(h_s)\|_2&\leq 2\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\left(\frac{2\varrho-1}{3-2\varrho}\right)^q\\&\leq 2\sqrt{p}\,G'(\theta)\varrho^{q+1}.\end{aligned} \tag{S308}$$

If every transition is independent of its starting location, as (A2) implies when $\varrho=\frac{1}{2}$, then $L_{s+q,N}$ is constant, Equation (S306) has equal right-hand sides, and the full bias is zero.

Step 5: sum the boundary comparisons. Insert Equation (S308) into Equation (S296), apply the triangle inequality, and reverse the two finite sums. This gives

$$\begin{aligned}&\|\nabla_\theta\ell(\theta)-\nabla_\theta\ell^\alpha(\theta)\|_2\\&\quad\leq 2\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}(\alpha\varrho)^q\\&\quad=2\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q\\&\quad\leq 2(N-1)\sqrt{p}\,G'(\theta)\varrho(1-\alpha)\sum_{q=0}^{\infty}(\alpha\varrho)^q\end{aligned}$$

$$= 2(N-1)\sqrt{p}\,G'(\theta)\frac{\varrho(1-\alpha)}{1-\alpha\varrho}. \tag{S309}$$

Finally, multiplying the finite sum by $1-\alpha\varrho$ gives

$$\begin{aligned}
&(1-\alpha\varrho)\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q \\
&\quad= (N-1)-\sum_{q=1}^{N-1}(\alpha\varrho)^q \\
&\quad= (N-1)-\frac{\alpha\varrho\{1-(\alpha\varrho)^{N-1}\}}{1-\alpha\varrho}.
\end{aligned} \tag{S310}$$

Multiplying Equation (S310) once more by $1-\alpha\varrho$ and collecting terms therefore gives

$$\sum_{q=0}^{N-2}(N-1-q)(\alpha\varrho)^q = \frac{(N-1)-N\alpha\varrho+(\alpha\varrho)^N}{(1-\alpha\varrho)^2}. \tag{S311}$$

Equations (S309) and (S311) prove Equation (S289). To obtain the sharper bound, retain the first line of Equation (S308) in the same finite sum. For $\frac{1}{2}<\varrho<1$,

$$\begin{aligned}
&\|\nabla_\theta\ell(\theta)-\nabla_\theta\ell^\alpha(\theta)\|_2 \\
&\quad\le 2\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)\left\{\alpha\frac{2\varrho-1}{3-2\varrho}\right\}^q \\
&\quad\le 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{\infty}\left\{\alpha\frac{2\varrho-1}{3-2\varrho}\right\}^q \\
&\quad= 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)}.
\end{aligned} \tag{S312}$$

The ratio in the geometric series is strictly less than one even at $\alpha=1$. This proves Equation (S290). □

Remark 1 (Signed cancellation in the population bias)**.** *Retaining signs gives a first-order expansion of the population bias. The score is a finite polynomial in α, so*

$$\begin{aligned}
\frac{\partial}{\partial\alpha}\nabla_\theta\ell^\alpha(\theta)\Big|_{\alpha=1} &= \sum_{n=1}^{N}\sum_{r=1}^{n-1} r\Delta_{n,r} \\
&= \sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}\{\rho_N(h_s)-\rho_{s+q}(h_s)\}.
\end{aligned} \tag{S313}$$

The first equality follows by differentiating Equation (S160) and summing over n. For the second, divide Equation (S296) by $1-\alpha$ and let $\alpha\uparrow 1$; all sums are finite. Subtracting $(1-\alpha)$ times Equation (S313) from Equation (S296) gives exactly

$$\nabla_\theta\ell(\theta)-\nabla_\theta\ell^\alpha(\theta)-(1-\alpha)\,\frac{\partial}{\partial\alpha}\nabla_\theta\ell^\alpha(\theta)\Big|_{\alpha=1}$$

$$= -(1-\alpha)\sum_{s=1}^{N-1}\sum_{q=0}^{N-s-1}(1-\alpha^q)\{\rho_N(h_s)-\rho_{s+q}(h_s)\}. \tag{S314}$$

Thus Equation (S308), the triangle inequality, and reversal of the finite sums give, for $\frac{1}{2} < \varrho < 1$,

$$\begin{aligned}
&\left\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta) - (1-\alpha)\,\frac{\partial}{\partial\alpha}\nabla_\theta \ell^\alpha(\theta)\Big|_{\alpha=1}\right\|_2 \\
&\quad\le 2\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\sum_{q=0}^{N-2}(N-1-q)(1-\alpha^q)\left(\frac{2\varrho-1}{3-2\varrho}\right)^q \\
&\quad\le 2(N-1)\sqrt{p}\,G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}(1-\alpha)\left\{\frac{3-2\varrho}{4(1-\varrho)}-\frac{3-2\varrho}{3-2\varrho-\alpha(2\varrho-1)}\right\} \\
&\quad= (N-1)\sqrt{p}\,G'(\theta)\frac{(1-\sqrt{2(1-\varrho)})^2(3-2\varrho)(1-\alpha)^2}{2(1-\varrho)\{3-2\varrho-\alpha(2\varrho-1)\}}.
\end{aligned} \tag{S315}$$

The last equality uses $2\varrho - 1 = (1-\sqrt{2(1-\varrho)})(1+\sqrt{2(1-\varrho)})$. *For* $\varrho = \frac{1}{2}$, *the score difference and its signed derivative are both zero, so the same bound holds. In particular, the population bias is bounded by*

$$\begin{aligned}
&\|\nabla_\theta \ell(\theta) - \nabla_\theta \ell^\alpha(\theta)\|_2 \\
&\quad\le (1-\alpha)\left\|\sum_{n=1}^{N}\sum_{r=1}^{n-1} r\Delta_{n,r}\right\|_2 + (N-1)\sqrt{p}\,G'(\theta)\frac{(1-\sqrt{2(1-\varrho)})^2(3-2\varrho)(1-\alpha)^2}{2(1-\varrho)\{3-2\varrho-\alpha(2\varrho-1)\}}.
\end{aligned} \tag{S316}$$

For every fixed observation sequence, if the signed sum vanishes, this gives $O\{N(1-\alpha)^2\}$ *bias and* $O\{N^2(1-\alpha)^4\}$ *squared bias, with fixed model constants. The next example shows that the assumptions do not ensure this cancellation.*

Remark 2 (Sharpness of the fixed-observation population order)**.** *The powers of* N *and* $1-\alpha$ *in Equation (S289) cannot be improved uniformly under the conditions of Lemma S14. To see this, take state space* $\{0,1\}$ *with counting measure,*

$$X_0 \sim \text{Bernoulli}(3/4), \qquad \Pr(X_n = X_{n-1} \mid X_{n-1}) = \frac{3}{4}, \tag{S317}$$

fix $y_n^* = 1$ *for every* n, *and, for* $|\theta| \le 1/4$, *set*

$$g_\theta(1 \mid 0) = \frac{1}{4}, \qquad g_\theta(1 \mid 1) = \frac{9}{20}(1+\theta), \tag{S318}$$

with the probabilities at observation zero defined by complementation. At $\theta = \phi = 0$,

$$h_n(x) = x. \tag{S319}$$

If the filtering probability of state one at time $n-1$ *is* $3/4$, *its one-step prediction and update are*

$$\Pr(X_n = 1 \mid y_{1:n-1}^*) = \frac{3}{4}\frac{3}{4} + \frac{1}{4}\frac{1}{4} = \frac{5}{8},$$

$$\Pr(X_n = 1 \mid y^*_{1:n}) = \frac{(5/8)(9/20)}{(5/8)(9/20) + (3/8)(1/4)} = \frac{3}{4}. \tag{S320}$$

The law of X_0 initializes this induction, so the filtered probability is $3/4$ at every time. The resulting time-homogeneous posterior backward kernel satisfies

$$B(1 \mid 1) = \frac{(3/4)(3/4)}{5/8} = \frac{9}{10}, \qquad B(1 \mid 0) = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}. \tag{S321}$$

Moreover, the next-observation likelihoods conditional on X_{n-1} are

$$M_n(G_n)(1) = \frac{3}{4}\frac{9}{20} + \frac{1}{4}\frac{1}{4} = \frac{2}{5}, \qquad M_n(G_n)(0) = \frac{1}{4}\frac{9}{20} + \frac{3}{4}\frac{1}{4} = \frac{3}{10}. \tag{S322}$$

Consequently,

$$\begin{aligned} \Pr_{\rho_n}(X_{n-1} = 1) &= \frac{(3/4)(2/5)}{(3/4)(2/5) + (1/4)(3/10)} = \frac{4}{5}, \\ \Pr_{\rho_{n-1}}(X_{n-1} = 1) &= \frac{3}{4}. \end{aligned} \tag{S323}$$

Equation (S321) maps every affine function of the state to an affine function whose slope is multiplied by

$$B(1 \mid 1) - B(1 \mid 0) = \frac{2}{5}. \tag{S324}$$

Applying the $r-1$ backward kernels between X_{n-r} and X_{n-1} and then using Equation (S323) therefore gives, for $1 \le r < n$,

$$\begin{aligned} \Delta_{n,r} &= \rho_n(h_{n-r}) - \rho_{n-1}(h_{n-r}) \\ &= \left(\frac{4}{5} - \frac{3}{4}\right)\left(\frac{2}{5}\right)^{r-1} = \frac{1}{20}\left(\frac{2}{5}\right)^{r-1}. \end{aligned} \tag{S325}$$

All these quantities are nonnegative. Equation (S291), summed over n, now implies

$$\begin{aligned} \nabla_\theta \ell(0) - \nabla_\theta \ell^\alpha(0) &= \frac{1}{20}\sum_{r=1}^{N-1}(N-r)(1-\alpha^r)\left(\frac{2}{5}\right)^{r-1} \\ &\ge \frac{(N-1)(1-\alpha)}{20}. \end{aligned} \tag{S326}$$

Thus the squared population bias is at least $(N-1)^2(1-\alpha)^2/400$. This model satisfies the uniform transition and measurement bounds on $|\theta| \le 1/4$, its simulator is independent of θ, and h_n is a bounded current function. A uniform improvement therefore requires additional assumptions, such as signed cancellation.

Theorem S4 (MSE bounds for DMOP-α)**.** *Under the conditions of Lemmas S13 and S14, take the same* $\varrho = 1 - \underline{M}/(2\bar{M}) \in [\frac{1}{2}, 1)$*. For* $\alpha \in [0,1]$*,*

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \Bigg[\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right) \left(\sum_{r=0}^{N-1} \alpha^r\right)^2 \\
&\qquad + \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2} \left\{ N - 1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho}\right)^q \right\}^2 \Bigg].
\end{aligned} \tag{S327}$$

For $0 \le \alpha < 1$*, this implies*

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \Bigg[\frac{N(1+\log J)^2}{J(1-\alpha)^2} + \frac{N^2}{J^2(1-\alpha)^2} \\
&\qquad + N^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 \Bigg].
\end{aligned} \tag{S328}$$

The finite sum beginning at $q = 1$ *is empty for* $N \le 2$*. At* $\varrho = \frac{1}{2}$*, the displayed population terms are zero. In particular, the two endpoints satisfy*

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^0(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \left[\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2} + (N-1)^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \right\}^2 \right],
\end{aligned} \tag{S329}$$

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2 \min\left\{N^2, \frac{N^3}{J}\right\}. \tag{S330}$$

At $\alpha = 0$*, the variance is at most order* $NpG'(\theta)^2(1+\log J)^2/J$*; at* $\alpha = 1$*, the population discounting bias is exactly zero. Remark 1 gives a complementary bound retaining signed cancellation.*

Proof. Add and subtract $\nabla_\theta \ell^\alpha(\theta)$ and use $\|a+b\|_2^2 \le 2\|a\|_2^2 + 2\|b\|_2^2$. This gives

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \le 2\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell^\alpha(\theta)\right\|_2^2 + 2\left\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2.
\end{aligned} \tag{S331}$$

Lemma S13 bounds the first term by

$$CpG'(\theta)^2 \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right) \left(\sum_{r=0}^{N-1} \alpha^r\right)^2. \tag{S332}$$

The sharper bound in Equation (S290) of Lemma S14 gives

$$\begin{aligned}&\|\nabla_\theta \ell^\alpha(\theta) - \nabla_\theta \ell(\theta)\|_2^2 \\ &\quad \le 4pG'(\theta)^2 \, \frac{(1-\sqrt{2(1-\varrho)})^2(1-\alpha)^2}{(1+\sqrt{2(1-\varrho)})^2} \left\{ N - 1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho} \right)^q \right\}^2 . \end{aligned} \tag{S333}$$

Here the $q = 0$ summand has been written separately as $N - 1$. For $\varrho = \frac{1}{2}$, both sides of this last inequality are zero by Lemma S14. Substitution into Equation (S331), absorbing fixed numerical factors into $\lesssim$, proves the first bound. For $0 \le \alpha < 1$,

$$\sum_{r=0}^{N-1} \alpha^r \le \frac{1}{1-\alpha}. \tag{S334}$$

Also,

$$\begin{aligned} &N - 1 + \sum_{q=1}^{N-2} (N-1-q) \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho} \right)^q \\ &\quad \le (N-1) \left\{ 1 + \sum_{q=1}^{\infty} \left(\alpha \frac{2\varrho - 1}{3 - 2\varrho} \right)^q \right\} \\ &\quad = \frac{(N-1)(3-2\varrho)}{3 - 2\varrho - \alpha(2\varrho - 1)}. \end{aligned} \tag{S335}$$

The geometric ratio is in $[0, 1)$, including when $\varrho = \frac{1}{2}$. These inequalities and $N - 1 \le N$ prove Equation (S328). Alternatively, Equation (S316) may be substituted in Equation (S331), taking the smaller population bound without changing the particle contribution.

For $\alpha = 0$, the geometric sum is one and every summand with $q \ge 1$ is zero. Substitution in Equation (S327) gives Equation (S329); Equation (S238) gives the stated variance. For $\alpha = 1$, use the terminal-block estimate directly, without separately bounding and squaring the particle bias. Equation (S281) and the two telescopes in Equations (S244)– (S245), with $k = N - 1$, give

$$\nabla_\theta \hat{\ell}^1(\theta) = m(\xi_N) \left(\sum_{s=1}^{N} h_s \right), \qquad \nabla_\theta \ell^1(\theta) = \rho_N \left(\sum_{s=1}^{N} h_s \right) = \nabla_\theta \ell(\theta). \tag{S336}$$

Consequently, Equation (S268) implies

$$\begin{aligned} &\mathbb{E} \left\| \nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta) \right\|_2^2 \\ &\quad = \sum_{i=1}^{p} \mathbb{E} \left| m(\xi_N) \left(\sum_{s=1}^{N} h_{s,i} \right) - \rho_N \left(\sum_{s=1}^{N} h_{s,i} \right) \right|^2 \le \frac{CN^3 pG'(\theta)^2}{J}. \end{aligned} \tag{S337}$$

Since $\|h_s\|_2 \le \sqrt{p}\, G'(\theta)$, the same difference satisfies, pathwise,

$$\left\| \nabla_\theta \hat{\ell}^1(\theta) - \nabla_\theta \ell(\theta) \right\|_2 \le m(\xi_N) \left(\left\| \sum_{s=1}^{N} h_s \right\|_2 \right) + \rho_N \left(\left\| \sum_{s=1}^{N} h_s \right\|_2 \right)$$

$$\leq 2N\sqrt{p}\,G'(\theta). \tag{S338}$$

Squaring and combining with Equation (S337) proves Equation (S330). □

Remark 3 (Neutral-model check)**.** *To check the rates in Lemma S13, take iid* Uniform$[0,1]$ *states, constant weights at* $\theta = \phi = 0$*, and*

$$\Pr_\theta(Y_n = 1 \mid X_n = x) = \frac{1}{2} + \frac{\theta}{4}(2x-1), \qquad y_n^* = 1. \tag{S339}$$

Then $h_n = (2X_n - 1)/2$ *has variance* $\sigma^2 = 1/12$*, and direct conditional multinomial calculation gives*

$$\operatorname{Var}\left\{\nabla_\theta \hat{\ell}^\alpha(\theta)\right\} = \frac{\sigma^2}{J}\left[N(2 - J^{-1}) + \sum_{r=1}^{N-1}(N-r)\alpha^{2r}(1-J^{-1})^{r+1}\right]. \tag{S340}$$

For fixed $\alpha < 1$*, the variance is of order* N/J*. Its genealogical contribution is of order* $N/\{J(1-\alpha^2)\}$ *before horizon saturation, so the general factor* $(1-\alpha)^{-2}$ *is loose in this model. At* $\alpha = 1$ *and* $J/N \to \infty$*, the same exact formula gives*

$$\operatorname{Var}\left\{\nabla_\theta \hat{\ell}^1(\theta)\right\} \sim \frac{\sigma^2}{J}\left\{2N + \sum_{r=1}^{N-1}(N-r)\right\} = \frac{\sigma^2 N(N+3)}{2J}. \tag{S341}$$

Thus the general N^3/J *endpoint upper bound is also conservative in this model.*

Remark 4 (Rates and normalization)**.** *The finite particle term in Lemma S13 is*

$$pG'(\theta)^2\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right)\left(\sum_{r=0}^{N-1}\alpha^r\right)^2. \tag{S342}$$

At $\alpha = 0$*, the particle-MSE upper bound is* $pG'(\theta)^2\{N(1+\log J)^2/J + N^2/J^2\}$*. Its variance bound is* $NpG'(\theta)^2(1+\log J)^2/J$*, whereas its MSE relative to the full score also contains the population term displayed in Equation (S329). Every fixed* $\alpha < 1$ *has an* $N(1+\log J)^2/J$ *particle-variance upper bound and, when* $J \geq N$*, the same leading particle-MSE order, with constants depending on* α*. The particle bound increases with the effective memory* $\sum_{r=0}^{N-1}\alpha^r$*.*

The population bound in Theorem S4 decreases to zero as $\alpha \uparrow 1$*. At* $\alpha = 1$*, the terminal* L^2 *estimate gives Equation (S330) directly. Substitution into Equation (S342) would instead give* $N^3(1+\log J)^2/J + N^4/J^2$*. The neutral-model example shows that the* $(1-\alpha)^{-2}$ *dependence and the* N^3/J *endpoint bound can be loose.*

A particle MSE bound for $\operatorname{osc}(f) \leq 1$ *is multiplied by* $\operatorname{osc}(f)^2$ *when it is applied to an unnormalized functional. Thus a generic path-space bound of order* N^2/J *for an oscillation-one functional becomes* N^4/J *if it is applied once to a full score whose oscillation is of order* N*. Lemma S13 bounds each* h_s *before summing lags, so Theorem S4 needs no further rescaling. The resulting* N^4/J *generic upper bound is separate from the quadratic asymptotic-variance lower bound of Poyiadjis et al. (2011).*

Remark 5 (Choosing α in the upper bound)**.** *We tune the upper bound. For* $0 \leq \alpha < 1$*,*

$$\sum_{r=0}^{N-1}\alpha^r \leq \min\{N, (1-\alpha)^{-1}\} \tag{S343}$$

and Theorem S4 give

$$\begin{aligned}
&\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \\
&\quad \lesssim pG'(\theta)^2 \Bigg[\left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right) \min\{N^2, (1-\alpha)^{-2}\} \\
&\qquad\qquad + N^2 \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{(1-\alpha)(3-2\varrho)}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 \Bigg].
\end{aligned} \tag{S344}$$

For fixed $\frac{1}{2} < \varrho < 1$, before the effective memory reaches the horizon, balancing these two terms is equivalent to

$$\frac{(1+\log J)^2}{NJ} + \frac{1}{J^2} \asymp \left\{ \frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}} \frac{3-2\varrho}{3-2\varrho-\alpha(2\varrho-1)} \right\}^2 (1-\alpha)^4. \tag{S345}$$

Suppressing these fixed model constants gives the following rate-attaining choices for $N \geq 2$:

regime	$1-\alpha$	*MSE upper bound*/$\{pG'(\theta)^2\}$
$J(1+\log J)^2 \lesssim N$	$\asymp J^{-1/2}$	N^2/J
$N \lesssim J(1+\log J)^2 \lesssim N^3$	$\asymp \{(1+\log J)^2/(NJ)\}^{1/4}$	$N^{3/2}(1+\log J)/J^{1/2}$
$J(1+\log J)^2 \gtrsim N^3$	0	$N^3/J.$

(S346)

The first boundary is where the two particle coefficients have the same order. In the middle regime, compare the balanced bound with the independent terminal bound:

$$\min\left\{ \frac{N^{3/2}(1+\log J)}{\sqrt{J}}, \frac{N^3}{J} \right\}.$$

They agree when $J(1+\log J)^2$ has order N^3. The choice $\alpha = 1$ attains the last rate by Equation (S330). To check minimization of the finite bound in Equation (S327), for $N \geq 2$,

$$\frac{\partial}{\partial \alpha}\left\{ \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right) \left(\sum_{r=0}^{N-1} \alpha^r\right)^2 \right\}\Bigg|_{\alpha=1} = \left(\frac{N(1+\log J)^2}{J} + \frac{N^2}{J^2}\right) N^2(N-1) > 0, \tag{S347}$$

The squared population term has derivative zero there, so decreasing α slightly from one reduces this finite upper bound. This need not change the rate or reduce the actual MSE. The first-row choice $J^{-1/2}$ is a rate for $1-\alpha$; the corresponding MSE upper bound is N^2/J.

For fixed $\varrho \in (\frac{1}{2}, 1)$, its factors affect the tuning constants. At $\varrho = \frac{1}{2}$, the population term is zero and $\alpha = 0$ minimizes the finite particle upper bound. For $N = 1$ all choices of α give the same estimator. Finally, if the signed sum in Equation (S313) vanishes, Remark 1 gives, before horizon saturation and with fixed model constants,

$$1-\alpha \asymp \left(\frac{(1+\log J)^2}{NJ} + \frac{1}{J^2}\right)^{1/6},$$

$$\mathbb{E}\left\|\nabla_\theta \hat{\ell}^\alpha(\theta) - \nabla_\theta \ell(\theta)\right\|_2^2 \lesssim pG'(\theta)^2 N^2 \left(\frac{(1+\log J)^2}{NJ} + \frac{1}{J^2}\right)^{2/3}. \tag{S348}$$

Indeed, the particle contribution is of order $\{N(1+\log J)^2/J + N^2/J^2\}(1-\alpha)^{-2}$ *and the squared population contribution is then at most order* $N^2(1-\alpha)^4$*. This cancellation is an additional condition for the last rate.*

Remark 6 (Geometric discounting and lag truncation)**.** *For* $0 \le k < N$*, the truncations defined in Lemma S13 satisfy*

$$\begin{aligned}\nabla_\theta \hat{s}_k^1(\theta) &= \sum_{s=1}^{N-k-1} m(\xi_{s+k})(h_s) + m(\xi_N)\left(\sum_{s=N-k}^{N} h_s\right),\\ \nabla_\theta \ell(\theta) - \nabla_\theta s_k^1(\theta) &= \sum_{s=1}^{N-k-1}\sum_{u=s+k+1}^{N} \Delta_{u,u-s}.\end{aligned} \tag{S349}$$

Equation (S308) with $q = k$ *gives, for* $\frac{1}{2} < \varrho < 1$*,*

$$\left\|\nabla_\theta \ell(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 \le 2(N-k-1)\sqrt{p}\, G'(\theta)\frac{1-\sqrt{2(1-\varrho)}}{1+\sqrt{2(1-\varrho)}}\left(\frac{2\varrho-1}{3-2\varrho}\right)^k. \tag{S350}$$

At $\varrho = \frac{1}{2}$ *the population error is zero for every* k*; at* $k = N-1$ *it is zero for every* ϱ*. The finite-window particle calculation in Equations (S277) and (S278) gives*

$$\begin{aligned}\mathrm{Var}\{\nabla_\theta \hat{s}_k^1(\theta)\} &\lesssim \frac{NpG'(\theta)^2(k+1)^2(1+\log J)^2}{J},\\ \left\|\mathbb{E}\nabla_\theta \hat{s}_k^1(\theta) - \nabla_\theta s_k^1(\theta)\right\|_2 &\lesssim \frac{N\sqrt{p}\, G'(\theta)(k+1)}{J}.\end{aligned} \tag{S351}$$

Equation (S281) writes DMOP-α *as their geometric average. Its effective memory is*

$$\sum_{k=0}^{N-1} \beta_k(k+1) = \sum_{r=0}^{N-1} \alpha^r \tag{S352}$$

Thus every fixed $\alpha < 1$ *has an* $N(1+\log J)^2/J$ *particle-variance upper bound, including the DMOP-*0 *case.*

S8 Implementation and Figures for Bayesian Inference

S8.1 A common sampling target

For a comparison with a common posterior, we implemented NUTS and particle Metropolis–Hastings (PMMH) using the Dhaka model in Pypomp. The process simulator uses Gaussian noise with 20 Euler steps per monthly observation. We fix the initial state proportions at the library values, with $\rho = 0$, $c = 1$, $\alpha = 1$ and $\delta = 0.02$. Here α is a model parameter, distinct from the DMOP discount parameter. The observation distribution is Gaussian, with mean equal to the simulated monthly deaths and standard deviation equal to τ times that mean. We omit the

additive measurement-density floor used in the library's optimization code. Paths that leave the nonnegative state space are assigned zero likelihood for the observed series.

Let x denote the 18 estimation coordinates. These comprise the logarithms of the five positive parameters $\gamma, m, \epsilon, \sigma, \tau$, the trend coordinate $100\beta_{\text{trend}}$, and the six coefficients in each seasonal function. The unused constant environmental-transmission parameter in the earlier implementation is excluded. We use independent Gaussian priors for these coordinates, common to all chains and both methods. For each coordinate, the mean is the midpoint of the interval in Table S-1, and the standard deviation is half its width. These intervals come from the parameter search ranges; they do not truncate the Gaussian distributions. Thus, writing a_i and b_i for the endpoints,

$$z_i = \frac{x_i - (a_i + b_i)/2}{(b_i - a_i)/2}, \qquad p(z) = \prod_{i=1}^{18} \phi(z_i; 0, 1).$$

This specifies a reference prior without fitting a density to IF2 particles. The prior density is evaluated in z, so the sampling calculation requires no further Jacobian factor. On the natural parameter scale, the priors for the five positive parameters are lognormal.

Estimation coordinate	a_i	b_i
$\log \gamma$	$\log 10$	$\log 40$
$\log m$	$\log 0.03$	$\log 0.6$
$\log \epsilon$	$\log 0.2$	$\log 30$
$100\beta_{\text{trend}}$	-1	0
$\log \sigma$	$\log 1$	$\log 5$
$\log \tau$	$\log 0.1$	$\log 0.5$
Seasonal transmission coefficients 1 and 3	-4	4
Seasonal transmission coefficients 2, 4, 5 and 6	0	8
Each seasonal environmental-transmission coefficient	-10	0

Table S-1: Intervals specifying the means and standard deviations of the common Gaussian prior on the estimation scale. A sensitivity calculation doubles all prior standard deviations, keeping the means fixed.

Let u denote the particle-filter random numbers, with law m, and let $\widehat{L}_J(z, u)$ be the bootstrap particle-filter likelihood estimate. The augmented target is

$$\widetilde{\pi}(z, u) \;\propto\; \widehat{L}_J(z, u)\, p(z)\, m(u). \tag{S353}$$

Unbiasedness of the likelihood estimate gives the posterior for the specified Euler model as the marginal distribution of z (Andrieu et al., 2010). The log likelihood and log prior enter without division by the number of observations. We use $J = 10,000$ particles and double precision for this likelihood calculation in both methods.

PMMH jointly proposes $z' = z + hB\xi$, where $\xi \sim N(0, I)$, and independent random numbers $u' \sim m$. Its acceptance probability is

$$1 \wedge \frac{\widehat{L}_J(z', u')p(z')}{\widehat{L}_J(z, u)p(z)}.$$

	NUTS				PMMH			
Coordinate	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE
$\log \gamma$	**1.009**	**243**	**645**	**0.00749**	1.206	15	58	0.028
$\log m$	**1.019**	**257**	**439**	**0.0118**	1.121	25	122	0.0358
$\log \epsilon$	**1.025**	**226**	**267**	**0.0305**	1.032	105	259	0.0385
$100\beta_{\text{trend}}$	**1.014**	**333**	**745**	**0.00593**	1.025	73	165	0.0124
$\log \sigma$	**1.015**	**303**	**551**	**0.00287**	1.170	18	69	0.0116
$\log \tau$	**1.012**	**328**	**537**	**0.00562**	1.177	16	94	0.0274
β_1	**1.006**	**215**	**529**	**0.0313**	1.300	11	26	0.125
β_2	**1.009**	**197**	**460**	**0.0247**	1.266	12	43	0.0813
β_3	**1.011**	**196**	**284**	**0.0917**	1.312	10	30	0.319
β_4	**1.008**	**282**	**549**	**0.0101**	1.178	19	62	0.0368
β_5	**1.004**	**295**	**591**	**0.00877**	1.303	11	149	0.0436
β_6	**1.007**	**341**	**640**	**0.00578**	1.114	34	200	0.0181
ω_1	**1.059**	**60**	**214**	**0.135**	1.114	34	61	0.188
ω_2	**1.027**	**297**	**416**	**0.0634**	1.079	39	129	0.157
ω_3	**1.027**	**220**	**381**	**0.0463**	1.155	17	49	0.153
ω_4	**1.015**	**278**	**542**	**0.0405**	1.177	15	37	0.166
ω_5	**1.010**	**293**	**335**	**0.118**	1.168	18	68	0.495
ω_6	**1.056**	**109**	**116**	**0.396**	1.111	42	59	0.528

Table S-2: Diagnostics for four chains of each method, using 2,000 retained iterations per chain after 500 warmup iterations. Rank-normalized split $\widehat{R}$, bulk and tail ESS, and the Monte Carlo standard error of the mean (MCSE) are calculated across chains on the displayed estimation coordinates. All chains have the common prior and target specified above. Bold entries indicate $\widehat{R}$ closer to 1, larger ESS, or smaller MCSE.

On rejection, both z and u are retained, together with the cached likelihood estimate. The matrix BB^T is estimated from pilot PMMH draws and held fixed; the scalar h is adapted during 500 warmup iterations. Preparation of the starting points and proposal scales includes numerical optimization with derivatives, although the PMMH transition itself requires only process simulation and measurement-density evaluation.

NUTS first proposes independent random numbers $u' \sim m$, accepting with probability $1 \wedge \widehat{L}_J(z, u')/\widehat{L}_J(z, u)$, and then updates z conditional on the retained u. We use BlackJAX's NUTS trajectory construction, with maximum tree depth 5. DMOP with discount parameter 0.95 and 1,000 particles supplies the proposal force, evaluated in single precision with deterministic GPU operations. Every trajectory weight uses the double-precision likelihood in Equation S353 and the common prior. The force and likelihood are therefore separate numerical calculations. Step sizes are adapted during 500 warmup iterations and then held fixed. The implementation, prior specification and checks of the sampling kernels are available in `code/bayesian`.

S8.2 Sampling settings and diagnostics

We ran four chains of each method from the same four starting points, obtained by optimization followed by a pilot PMMH run. Figure 6 compares the first 2,000 retained iterations of each chain, after 500 warmup iterations. Figure S-1 shows the marginal sample distributions of the individual chains over this window.

Both methods use fixed covariance matrices estimated from preliminary PMMH runs. The PMMH covariance uses 1% diagonal shrinkage and a $10^{-6}I$ ridge. For NUTS, we combine the pooled covariance from the first 5,000 retained PMMH iterations per chain with this matrix, with weights 0.75 and 0.25, and rescale to the trace of the PMMH proposal covariance. These matrices affect the proposals but not the prior. Only the scalar step size is adapted during warmup, with target acceptance probabilities 0.65 for NUTS and 0.234 for PMMH. Initial step sizes are 0.07 and 0.5, respectively. The final NUTS step sizes range from 0.048 to 0.060.

Table S-2 gives diagnostics for all 18 coordinates in the 2,000-iteration comparison. For the trend coordinate, NUTS has $\widehat{R} = 1.014$ and bulk ESS 333, compared with 1.025 and 73 for PMMH. The largest NUTS $\widehat{R}$ is 1.059, and the smallest bulk ESS is 60, both for the first seasonal environmental-transmission coefficient. There are two divergent NUTS transitions among 8,000 retained iterations. Almost all trajectories reach the depth limit of 5, corresponding to 31 integration steps. Thus the traces show improved exploration without establishing convergence of all coordinates.

The longer PMMH runs give broadly similar marginal distributions ~~(Figure S-1)~~; their largest $\widehat{R}$ is 1.014 and smallest bulk ESS is 202. Doubling all prior standard deviations has little effect on the sampled trend standard deviation (0.112 versus 0.115), but gives wider lower tails for some seasonal environmental-transmission coefficients. These tails also mix slowly, so the comparison does not establish prior insensitivity. The prior overlays in Figure S-3 make this issue visible.

S8.3 Additional sampling figures

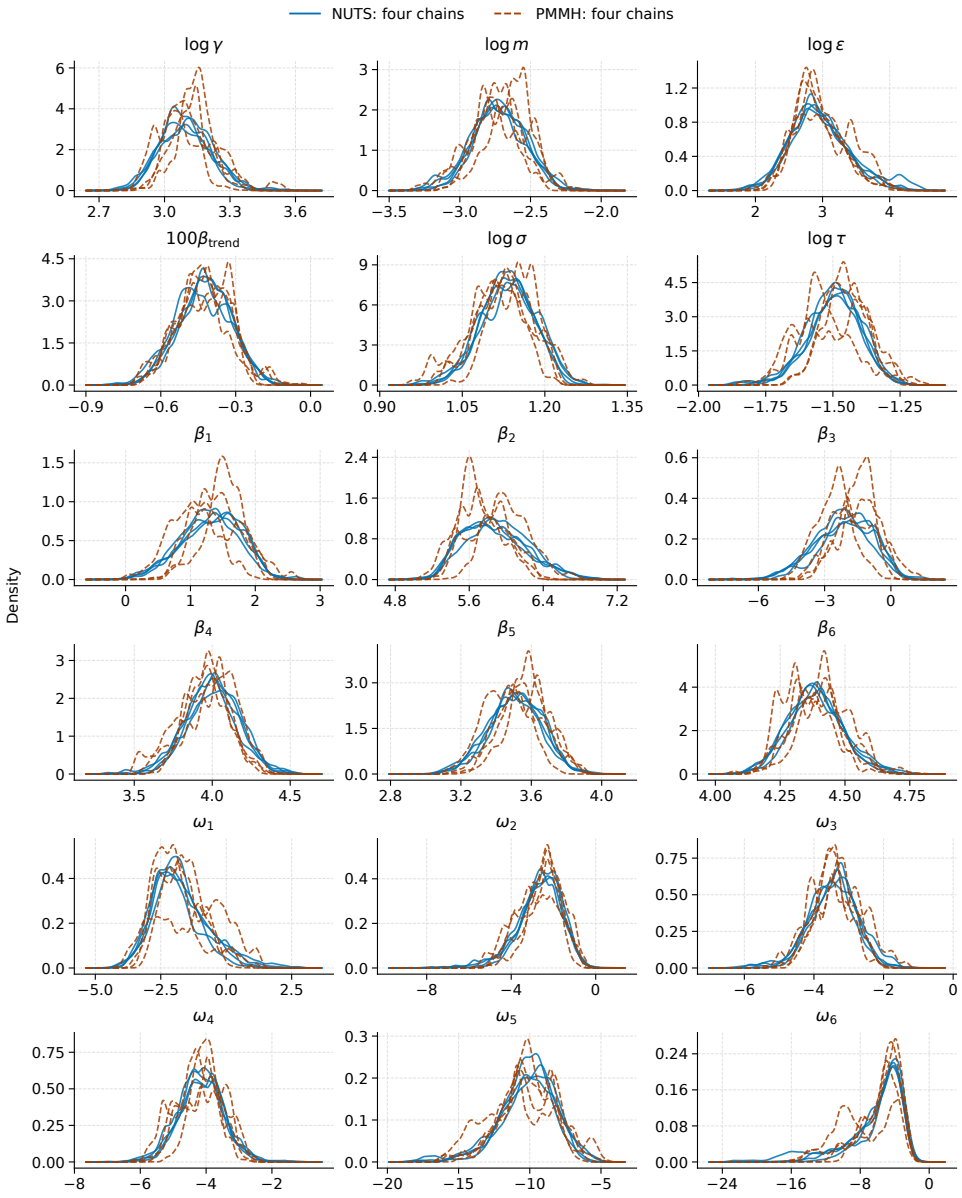


Figure S-1: Marginal sample distributions for the four NUTS chains (blue solid) and four PMMH chains (orange dashed), using all 2,000 retained iterations per chain after 500 warmup iterations. Each curve represents one chain. For each coordinate, all eight density estimates use the same Gaussian-kernel bandwidth, selected by Scott's rule from the pooled draws of both methods. The prior, target and iteration window are those of Table S-2.

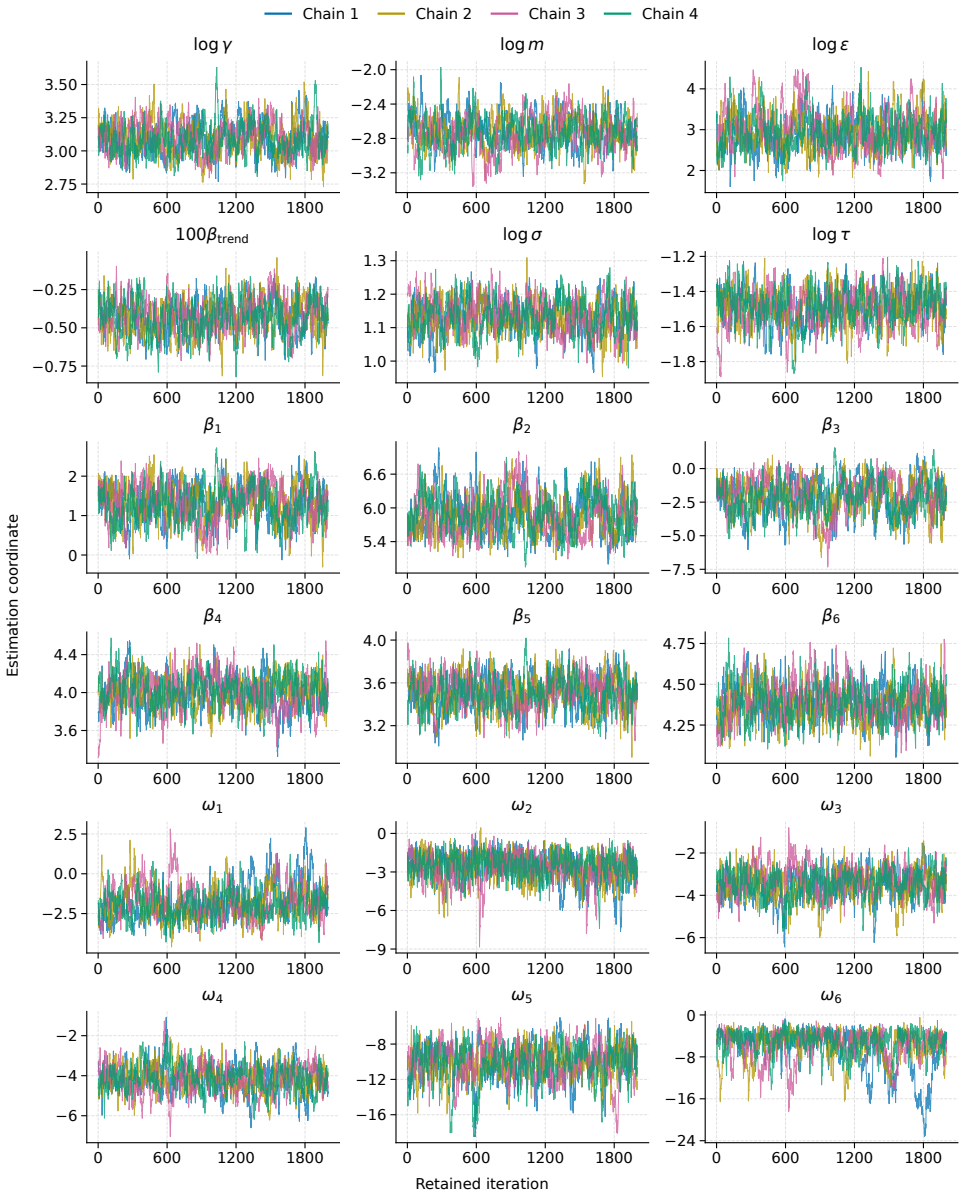


Figure S-2: NUTS traces for all 18 estimation coordinates. Colors identify the four chains, with 2,000 retained iterations per chain after 500 warmup iterations. The long excursions of some seasonal environmental-transmission coefficients are less apparent in the trend diagnostic alone.

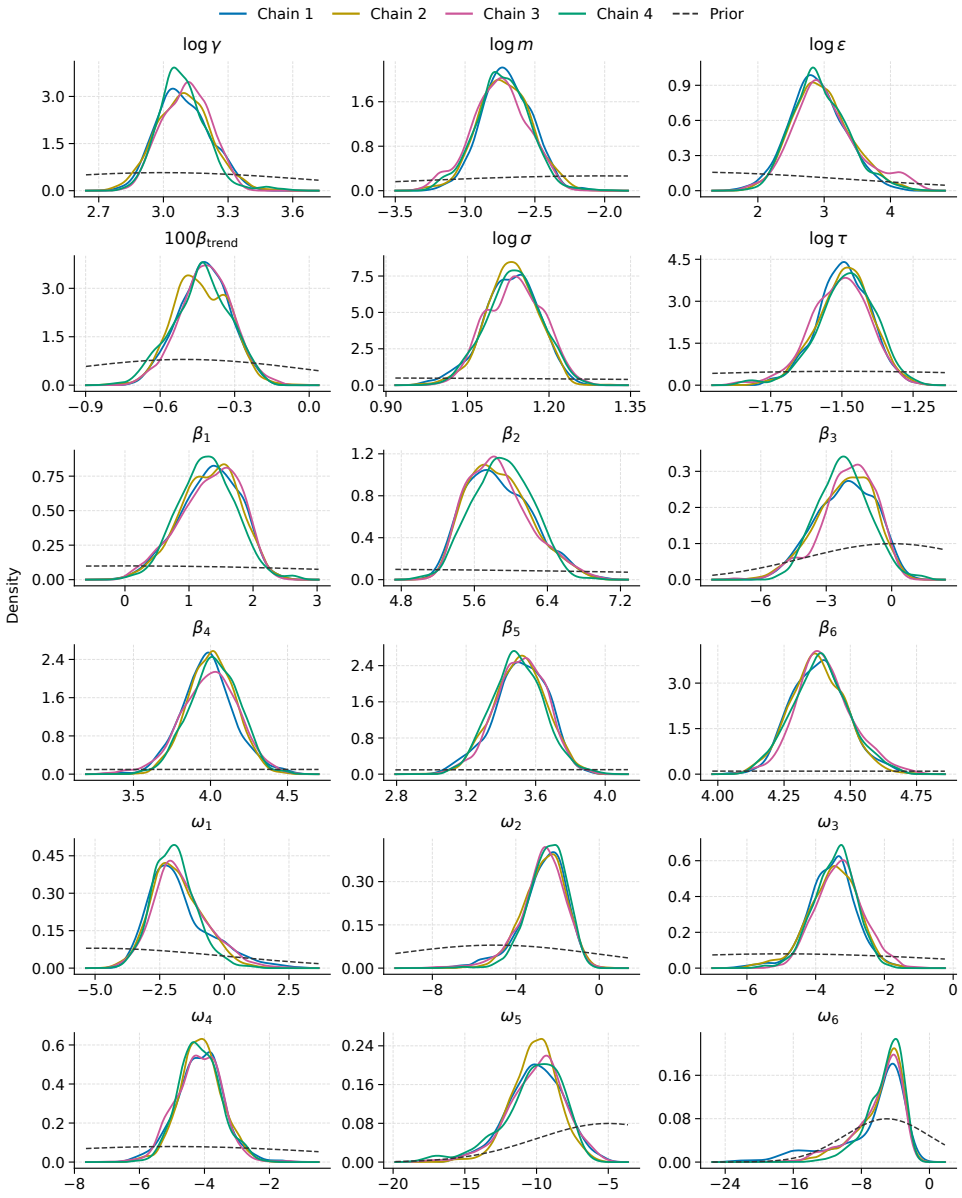


Figure S-3: Marginal sample distributions from the four NUTS chains for all 18 estimation coordinates, using 2,000 retained iterations per chain. Dashed curves show the common Gaussian prior. Density estimates use Gaussian kernels with Scott bandwidths.

S8.4 Sampling without covariance tuning

To assess the effect of covariance tuning, we also ran NUTS with an identity metric on the estimation coordinates x defined in Section S8.1, and PMMH with a fixed diagonal proposal. The model, prior, likelihood particle count and four starting points are the same as in the preceding comparison. For PMMH, the proposal standard deviation is one hundredth of the search-interval width in each coordinate, equivalently $z' = z + 0.02\xi$ with $\xi \sim N(0, I)$. This proposal is held fixed throughout the run. For NUTS, the metric is held fixed but the step size is adapted during 500 warmup iterations, starting at 0.002. The final step sizes are 0.000811, 0.001238, 0.001121 and 0.000963. The maximum tree depth remains 5, and the DMOP force uses 1,000 particles with discount parameter 0.95.

Each chain has 1,500 iterations. We compare iterations 501–1,500 for both methods, discarding the first 500 PMMH iterations for this comparison although PMMH has no adaptive warmup. Figure S-4 shows greater movement of the trend coordinate under NUTS, with $\widehat{R} = 1.380$ and bulk ESS 8.9, compared with 2.395 and 4.9 for PMMH. However, substantial differences between chains remain for both methods (Figures S-5 and S-6). The largest $\widehat{R}$ is 3.491 for NUTS and 2.580 for PMMH, and the smallest bulk ESS is 4.4 and 4.8, respectively (Table S-3). These runs do not provide reliable posterior summaries.

There are no divergent NUTS transitions, but every retained trajectory reaches the depth limit. Thus removing divergences by adapting the step size does not ensure adequate exploration. PMMH accepts between 9.8% and 13.9% of proposals during the displayed iterations.

	Identity-metric NUTS				Diagonal PMMH			
Coordinate	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE	$\widehat{R}$	Bulk ESS	Tail ESS	MCSE
$\log \gamma$	**1.560**	**7.1**	30.4	**0.0204**	2.524	4.8	**31.5**	0.0489
$\log m$	2.475	4.8	11.2	0.117	**2.246**	**5.1**	**18.2**	**0.101**
$\log \epsilon$	2.827	4.6	13.1	0.204	**1.895**	**5.7**	**13.9**	**0.131**
$100\beta_{\text{trend}}$	**1.380**	**8.9**	**73.7**	**0.0399**	2.395	4.9	11.6	0.049
$\log \sigma$	**1.239**	**12.5**	**164.3**	**0.0127**	1.776	6.3	15.7	0.0183
$\log \tau$	**1.437**	**8.3**	11.4	0.0416	2.407	5.0	**14.1**	**0.0359**
β_1	**1.870**	**5.8**	**33.6**	**0.119**	2.129	5.3	22.5	0.192
β_2	2.025	5.4	19.0	**0.109**	**1.813**	**6.0**	**39.0**	0.154
β_3	2.741	4.7	**14.2**	**0.302**	**2.572**	**4.8**	12.4	0.583
β_4	1.373	9.0	22.4	**0.0445**	**1.314**	**10.1**	**74.5**	0.0546
β_5	1.688	6.4	16.7	**0.0431**	**1.528**	**7.3**	**32.6**	0.0508
β_6	**1.383**	8.8	25.8	0.034	1.386	**9.3**	**37.8**	**0.0333**
ω_1	3.074	4.5	13.6	0.374	**1.811**	**6.0**	**31.8**	**0.232**
ω_2	**2.179**	**5.2**	13.3	**0.21**	2.580	4.8	**17.3**	0.504
ω_3	2.324	5.0	**37.4**	0.209	**1.492**	**7.6**	18.8	**0.119**
ω_4	2.424	4.9	14.5	**0.145**	**2.196**	**5.2**	**24.7**	0.222
ω_5	3.491	4.4	11.5	0.827	**1.883**	**5.8**	**13.9**	**0.481**
ω_6	2.605	4.8	**13.5**	**0.291**	**2.471**	**4.9**	11.2	0.335

Table S-3: Diagnostics for identity-metric NUTS and diagonal PMMH, using four chains and iterations 501–1,500 of each chain. The columns give rank-normalized split $\widehat{R}$, bulk and tail ESS, and MCSE of the mean on the estimation scale. Bold entries indicate the more favorable value within each pair, as in Table S-2. The large between-chain differences limit interpretation of ESS and MCSE; bold entries do not indicate convergence.

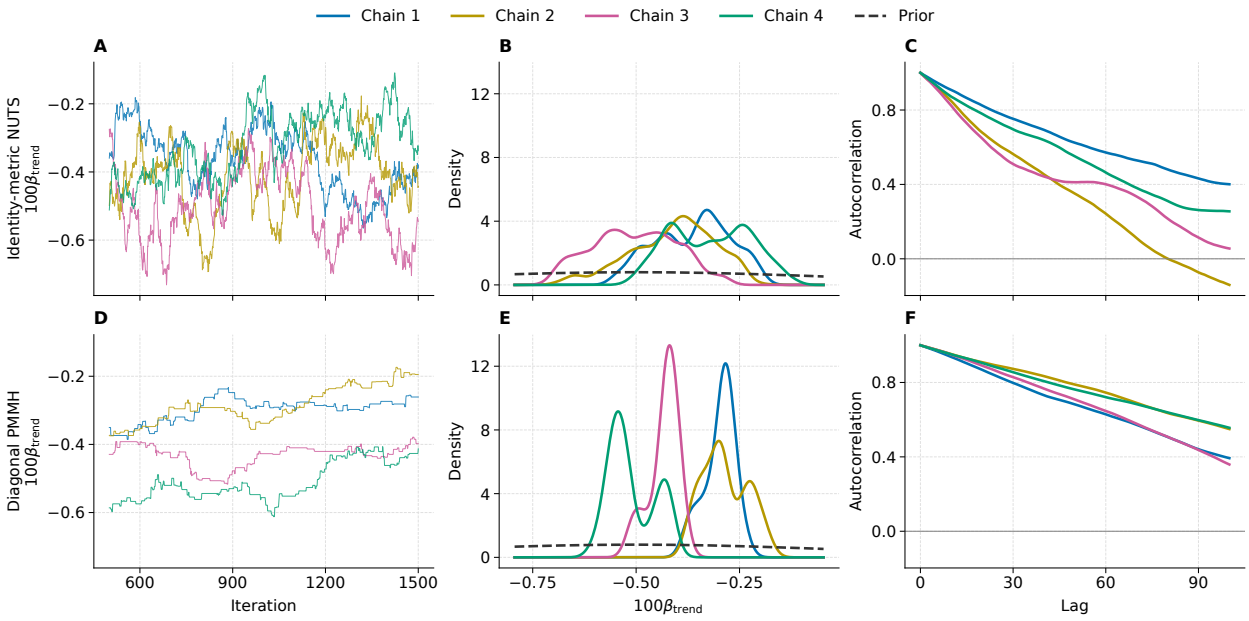


Figure S-4: Trend-coordinate diagnostics for identity-metric NUTS (A–C) and diagonal PMMH (D–F). The columns show traces, marginal sample densities and autocorrelations for iterations 501–1,500. Colors identify the four chains; dashed curves in the density panels show the common prior. Corresponding panels have the same scales. The density estimates use a common Gaussian-kernel bandwidth across methods and chains.

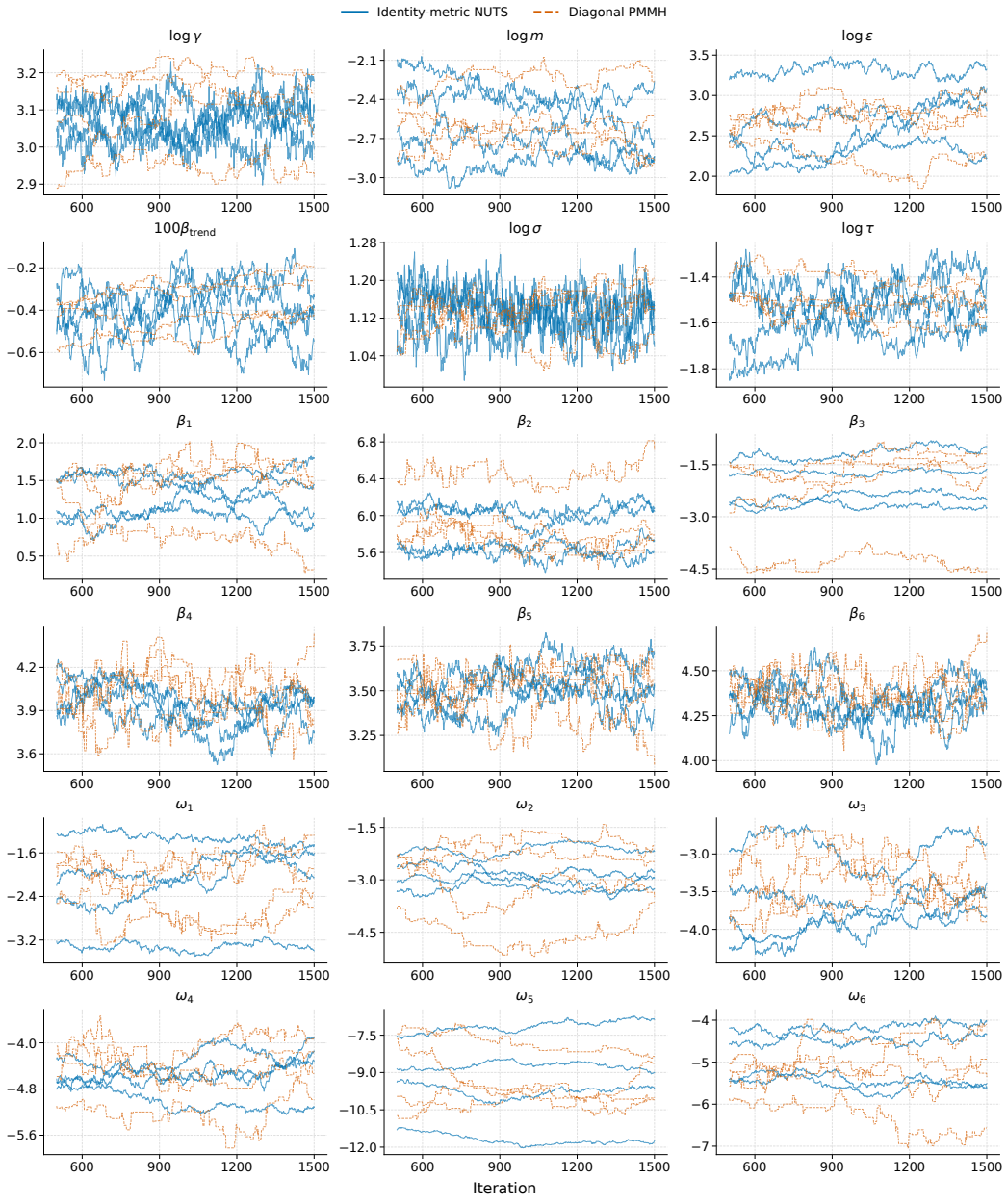


Figure S-5: Traces for all 18 estimation coordinates under identity-metric NUTS (blue solid) and diagonal PMMH (orange dashed). Each method has four chains, shown over iterations 501–1,500. Several coordinates retain substantial differences between chains under both methods.

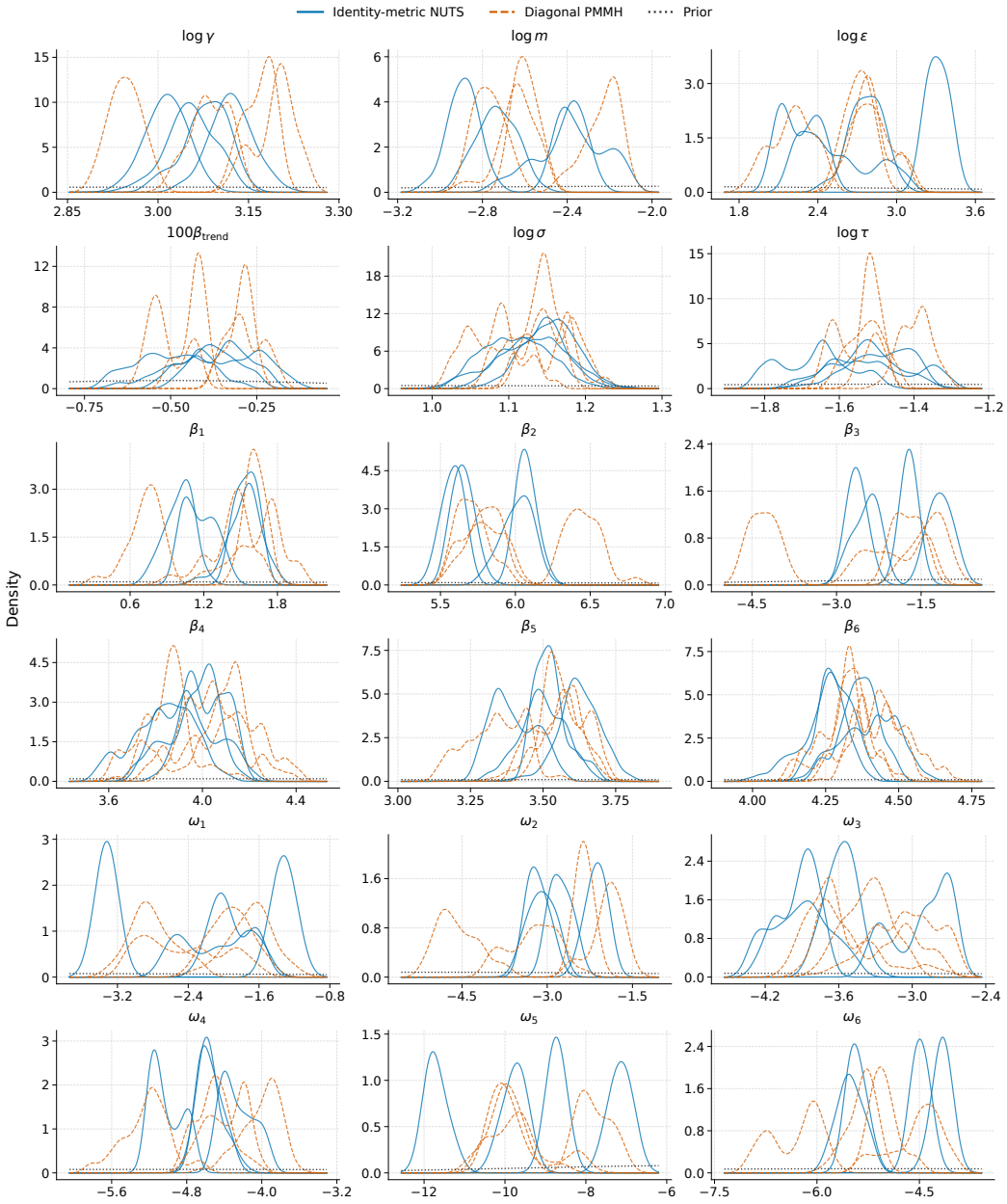


Figure S-6: Marginal sample distributions for the chains in Figure S-5. Blue solid curves show identity-metric NUTS, orange dashed curves show diagonal PMMH, and dotted curves show the common prior. Each curve uses all 1,000 retained draws from one chain. For each coordinate, all eight estimates use the same absolute Gaussian-kernel bandwidth, obtained by Scott's rule from the pooled draws. Differences between curves reflect incomplete mixing and cannot be interpreted as distinct posterior modes.

S9 Algorithmic Parameters

We include the full set of algorithmic parameters used for the Dhaka results in Table S-4. The warm start iterations are IF2 iterations, which are followed by gradient descent (GD) iterations when using IFAD. The IF2 cooling rate is the fractional reduction in the parameter perturbations over 50 iterations, so that a cooling rate of 0.5 corresponds to approximately 1% cooling per iteration. The random walk standard deviation (RWSD) scales the noise added at each filtering step to each regular parameter. Initial value parameters (IVPs), that affect the value of the latent process only at time t_0, are treated differently: they are perturbed only at time t_0, with a larger RWSD. Parameters are transformed to a unit scale, for example by taking logarithms or a logistic transformation, so that a single RWSD value works for all parameters.

Hyperparameter	**Standard IF2**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Number of Particles (J)	5,000	5,000	5,000	5,000
Warm-start Iterations	1,950	60	175	60
GD Iterations	—	410	400	425
IF2 Cooling Rate	0.93	0.5	0.5	0.5
Initial RWSD	0.02	0.02	0.02	0.02
Initial RWSD (IVPs)	0.16	0.16	0.16	0.16
Gradient Optimizer	—	Adam	Adam	Adam
Momentum (β_1)	—	0.0	0.9	0.9
LR Schedule	—	Cosine Decay	Cosine Decay	Cosine Decay
Initial LR (η_0)	—	0.1	0.1	0.1
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	5,000 / 36	5,000 / 36	5,000 / 36	5,000 / 36

Table S-4: Algorithmic parameters for the global search benchmarking on the Dhaka cholera model.

Following some experimentation, IFAD was implemnenteed by plugging the DMOP gradient estimate into the Adam optimizer (Kingma and Ba, 2015). The momentum parameter in Adam was turned off for IFAD-0, since this low-variance estimate did not reqquire it. Decreasing the learning rate (LR) was found to be helpful, and we selected the commonly used cosine decay. Warm-up steps were taken with Adam to stabilize its initial performance and avoid noisy

S10 Application to a Daphnia Population Model

S10.1 Model and data

The SIRJPF2 model developed by Yang et al. (2025) describes interactions between two species of *Daphnia*, their algal food, and a fungal parasite. We use the version with unit-specific infected-adult mortality to compare likelihood maximization by IFAD and the marginalized panel iterated filter (MPIF) of Wheeler et al. (2025).

The data come from the mesocosm experiment of Searle et al. (2016). We use the treatment containing the native species *D. dentifera*, the invasive species *D. lumholtzi*, and the parasite *Metschnikowia bicuspidata*, with eight mesocosms, each treated as one unit in the panel. Each mesocosm was observed at ten times, $t_n = 5n + 2$ days for $n = 1, \ldots, 10$. At each observation time, a one-liter sample was taken and susceptible and infected adult females of both species were counted. Juvenile counts are excluded from the likelihood.

We write $S_u^a(t)$, $I_u^a(t)$, and $J_u^a(t)$ for the densities of susceptible adult females, infected adult females, and juveniles, where $u = 1, \ldots, 8$ indexes the mesocosm and $a \in \{\mathrm{n}, \mathrm{i}\}$ denotes the native or invasive species. The densities of food and parasite spores are $F_u(t)$ and $P_u(t)$. These variables form an eight-dimensional latent state in each mesocosm. Removal by death, consumption, or sampling is denoted by R in the model name and does not require an additional state variable. The two host species interact through their shared food and parasite populations. The processes in different mesocosms are independent conditional on the parameters.

As in Yang et al. (2025), the transition dynamics follow an Itô stochastic differential equation (SDE):

$$
\begin{aligned}
dS_u^a &= \{\lambda_J J_u^a - (\theta_S^a + p^a f_S^a P_u + \delta) S_u^a\}\, dt + \sigma_S^a S_u^a\, dB_{S,u}^a, \\
dI_u^a &= \{p^a f_S^a S_u^a P_u - (\theta_{I,u}^a + \delta) I_u^a\}\, dt + \sigma_I^a I_u^a\, dB_{I,u}^a, \\
dJ_u^a &= \{r^a f_S^a F_u S_u^a - (\theta_J^a + \delta + \lambda_J) J_u^a\}\, dt + \sigma_J^a J_u^a\, dB_{J,u}^a, \\
dP_u &= \left\{ \sum_a \left[\beta \theta_{I,u}^a I_u^a - f_S^a (S_u^a + \xi I_u^a) P_u\right] - (\theta_P + \delta) P_u \right\} dt + \sigma_P P_u\, dB_{P,u}, \\
dF_u &= \left\{ \mu - \delta F_u - \sum_a f_S^a F_u (S_u^a + \xi I_u^a + J_u^a) \right\} dt + \sigma_F F_u\, dB_{F,u},
\end{aligned}
$$

with independent standard Brownian motions B. The time argument is suppressed in the equations. Except for infected-adult mortality, estimated parameters are shared across mesocosms. The parameters r^a and f_S^a are the reproductive rate and host filtration rate, p^a controls infection following spore ingestion, and ξ is the filtration rate of infected adults relative to susceptible adults. The mortality or loss rates of susceptible adults, infected adults, juveniles, and spores are θ_S^a, $\theta_{I,u}^a$, θ_J^a, and θ_P, respectively; the σ parameters are process-noise coefficients. The relative filtration rate of juveniles is fixed at one.

We use the units of the public Python implementation:[1] *Daphnia* densities are individuals per liter, F is in 10^6 algal cells per liter, and P is in 10^3 spores per liter, with parameters expressed in the corresponding units. The sampling loss rate is fixed at $\delta = 0.013$ per day, the food replenishment rate at $\mu = 0.37$, the juvenile maturation rate at $\lambda_J = 0.1$ per day, and the spore yield per

[1] `pypomp/Daphnia-tutorial`, advanced tutorial, revision `f318330`.

infected-host death at $\beta = 30$, corresponding to 3×10^4 spores. The initial states are $S_u^{\mathrm{n}} = 2.333$, $S_u^{\mathrm{i}} = 0.667$, and $F_u = 16.667$, with the remaining states set to zero. Simulation begins at $t_0 = 1$ day, and parasites are introduced on day 4 at a density of 25×10^3 spores per liter. We numerically solve the SDE using the Euler-Maruyama method with a time step of 0.25 days, following the initial conditions, inoculation, and numerical boundary rules of the public implementation.

The measurement model for the adult counts is

$$Y^a_{S,u,n} \sim \mathrm{NBinomial}(S^a_u(t_n), \tau^a_S), \qquad Y^a_{I,u,n} \sim \mathrm{NBinomial}(I^a_u(t_n), \tau^a_I),$$

where $\mathrm{NBinomial}(m, \tau)$ has mean m and variance $m + m^2/\tau$ (Yang et al., 2025). The four counts at each observation time are independent conditional on the latent state. Since the sampled volume is one liter, the state densities and observed counts are on the same scale. The numerical implementation sets a lower bound of 10^{-10} on the negative binomial mean and size parameters. As in the public Python code, a state variable that crosses a numerical boundary is reset to zero, and that particle's log weight for the mesocosm at the next observation is set to -150. The boundary thresholds and the treatment of the inoculation time are specified in the code.

We fix $\sigma_S^{\mathrm{n}} = \sigma_S^{\mathrm{i}} = 0$ and estimate $\theta_{I,u}^{\mathrm{n}}$ and $\theta_{I,u}^{\mathrm{i}}$ separately for each mesocosm. The remaining 22 estimated parameters are shared across mesocosms, giving $22 + 2 \times 8 = 38$ estimated parameters. This is the model with unit-specific infected-adult mortality in Table S3 of the archived revision (Yang et al., 2026).

S10.2 Experimental settings

We tested IFAD with discount factors $\alpha = 0$, $\alpha = 0.97$, and $\alpha = 1$ against MPIF using Pypomp 1.0.5 (Abkemeier et al., 2025). For each method, we performed 50 searches from the same starting parameter vectors, generated by independent lognormal perturbations of the initial parameter vector in the public tutorial, with a standard deviation of 0.45 on the log scale. All positive estimated parameters were optimized on the log scale. Each IFAD search used an MPIF warm start of 200 iterations followed by 200 iterations of the Adam optimizer (Kingma and Ba, 2015), using DMOP-α gradient estimates. We ran MPIF alone for 680 iterations, chosen to approximate the computing time of IFAD; measured times are reported in Table S-6.

We used 1,000 particles per mesocosm during optimization. The initial random walk standard deviation (RWSD) for MPIF was 0.05 on the log scale, with geometric cooling by a factor of 0.7 over 50 iterations. Adam used $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with all eight mesocosms contributing to each update. The learning-rate schedule used ten steps of linear warmup multiplied by cosine decay with a final factor of 0.1. A summary of the algorithmic parameters is provided in Table S-5.

The learning-rate scale was selected using four additional starting points, separate from the 50 used for the comparison. We compared the six combinations of learning-rate scales 0.001, 0.005, and 0.01 with $\alpha \in \{0.9, 0.97\}$. The combination with the highest median final log-likelihood, evaluated by independent particle filtering, was 0.01 with $\alpha = 0.97$. We used the same learning-rate schedule for $\alpha = 0$ and $\alpha = 1$.

All reported log-likelihoods were evaluated independently of optimization using ordinary particle filters, with 2,000 particles per mesocosm and ten replicates. Likelihood estimates were averaged within each mesocosm before taking logarithms, and the eight mesocosm log-likelihoods were then summed. Each method's best parameter vector was then held fixed and re-evaluated with ten

Hyperparameter	**MPIF**	**IFAD-0**	**IFAD-0.97**	**IFAD-1**
Particles per mesocosm (J)	1,000	1,000	1,000	1,000
MPIF Iterations	680	200	200	200
Adam Iterations	—	200	200	200
MPIF Cooling Factor	0.7	0.7	0.7	0.7
Initial RWSD (log scale)	0.05	0.05	0.05	0.05
Momentum (β_1)	—	0.9	0.9	0.9
Learning-rate Scale	—	0.01	0.01	0.01
Warm-up Steps	—	10	10	10
Eval. Particles / Reps	2,000 / 10	2,000 / 10	2,000 / 10	2,000 / 10

Table S-5: Algorithmic parameters for the Daphnia experiment. Particle numbers are per mesocosm; the cooling factor is the multiplier over 50 iterations.

fresh replicates at the same particle count, using the same model and Pypomp version on a CPU. The same evaluation method was used for the final results and for intermediate saved parameter vectors. The progress plot uses 25, 27, 26, and 22 saved parameter vectors per search for IFAD-0, IFAD-0.97, IFAD-1, and MPIF, respectively, including the final evaluations. Time is assigned in proportion to iteration number within each stage, using the measured total optimization time for that stage. Each set of 50 searches ran on one Tesla V100 16 GB GPU, with two searches per batch. Optimization times are summed over the 25 batches and include initial compilation but exclude the additional particle filter evaluations.

S10.3 Results

Our findings are summarized in Table S-6 and Figures S-7 and S-8. IFAD-0.97 and IFAD-1 attained maximum log-likelihoods of −858.48 and −858.27, respectively, each with a Monte Carlo standard error (MCSE) of approximately 0.17. The maximum for MPIF was −860.06, with MCSE 0.16. IFAD-0 attained −860.36 and had a median log-likelihood close to that of MPIF.

	Maximum	Re-evaluated	Median	MPIF time (s)	DMOP time (s)
IFAD-0.97	−858.48 (0.17)	−858.23 (0.16)	−861.42	535	1,309
IFAD-1	−858.27 (0.17)	−858.00 (0.14)	−861.46	520	1,313
IFAD-0	−860.36 (0.24)	−860.30 (0.21)	−864.12	503	1,314
MPIF	−860.06 (0.16)	−860.26 (0.14)	−864.27	1,591	-

Table S-6: Log-likelihoods from 50 searches per method, with MCSEs in parentheses. Re-evaluated uses fresh particle filters at each method's original maximizing parameter vector.

Comparing searches from the same starting points, IFAD-0.97 and IFAD-1 each attained a higher final log-likelihood than MPIF in 49 of the 50 searches. The median paired differences were 2.86 and 2.79 log-likelihood units, respectively. The distributions of the final log-likelihoods for IFAD-0.97 and IFAD-1 were similar, which may be due to the short time series in each mesocosm.

Yang et al. (2026) report a maximum log-likelihood of −863.03 in the revised Table S3. Our MPIF searches attained a higher value, and IFAD-0.97 and IFAD-1 improved on this further.

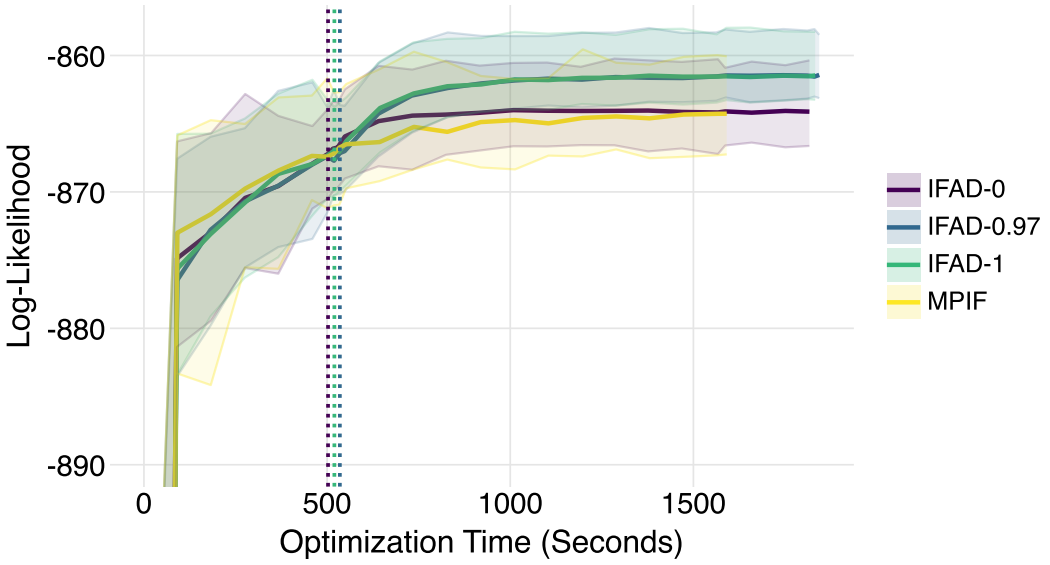


Figure S-7: Optimization progress of IFAD and MPIF for the Daphnia model. The solid lines show the median log-likelihood across 50 searches, and the shaded regions cover the area between the 10th and 100th percentiles. The vertical dotted lines indicate where the IFAD searches switch from MPIF to Adam optimization. The plot shows log-likelihoods above −890; all searches contribute to the percentiles.

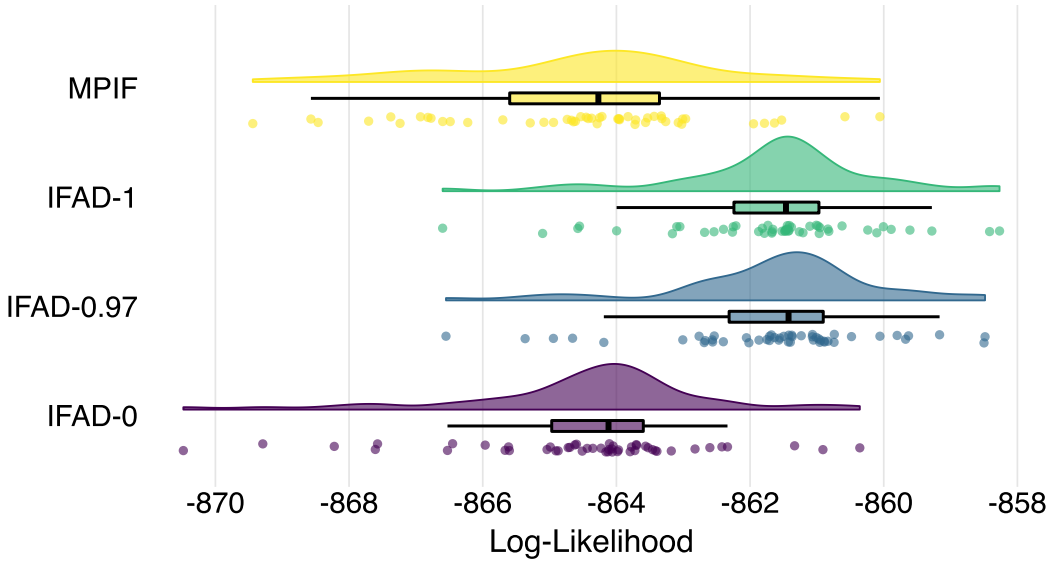


Figure S-8: Raincloud plot of the searches for the Daphnia model. The half-violin plots show the distributions of all 50 final log-likelihood estimates for each method, the boxplots show the medians and quartiles, and each point represents one search. IFAD-0.97 and IFAD-1 have similar distributions, with generally higher log-likelihoods than MPIF and IFAD-0.

S11 Comparison with Other Methods for the Dhaka Model

We compared IFAD with the transition-density method of Ditlevsen and Samson (2019) (DS19) and the differentiable particle filter of Corenflos et al. (2021) (CTDD21) on the Dhaka cholera model. We tested whether these methods could find high-likelihood parameter estimates from the wide range of starting points used in the main text, and whether they could improve estimates obtained by IF2. We describe our implementations below.

S11.1 Methods

DS19. The method of Ditlevsen and Samson (2019) combines an approximate transition density with particle smoothing and stochastic approximation expectation maximization (SAEM). Their construction assumes that one smooth state coordinate is observed directly and that noise acts directly on the remaining coordinates. Applying it to Dhaka requires some modifications: the six state coordinates (S, I, R^1, R^2, R^3, M) are driven by one Brownian motion, and the monthly death counts are observed with measurement error.

The SAEM objective includes the process log-density, which DS19 must therefore be able to evaluate. Replacing the SAEM maximization with a score update retains this requirement: the score includes derivatives of the process log-density, evaluated along the imputed path. Our backward smoother also evaluates the transition density to sample that path. For Dhaka, a single Euler step is driven by one Brownian increment and has no density with respect to Lebesgue measure on the six-dimensional state space. Composing 20 Euler steps gives a monthly transition that we can simulate, but whose density we cannot evaluate directly after integrating out the intermediate states. IF2 and IFAD use this simulator without evaluating its transition density; IFAD additionally differentiates through the simulation. They retain the Euler discretization error but do not require a separate approximation to the monthly transition density.

We approximate the transition over each month by a Gaussian distribution. The local strong-order-1.5 approximation has a covariance matrix of rank at most two, so it does not have a density in the six-dimensional state space. We instead calculate the mean and covariance over a full month by propagating the second-order mean and covariance through 20 time steps. We linearize the dynamics at each step and integrate out the intermediate states. The resulting monthly covariance matrix can have full rank, although it remains poorly conditioned. To stabilize the calculation, we add 10^{-12} times its largest diagonal entry to each diagonal entry. This gives a monthly transition density that differs from the Euler approximation used to evaluate the final parameter estimates.

The rank deficiency also affects parameter updates on an augmented path. At a fixed interior state, the range of the unregularized local covariance is

$$\operatorname{span}\left\{(-1,1,0,0,0,0)^{\mathsf{T}},\ (0,-\gamma-m,\gamma,0,0,m)^{\mathsf{T}}\right\}.$$

This follows by writing the diffusion vector as $g = q(-1,1,0,0,0,0)^{\mathsf{T}}$: its diffusion derivatives are parallel to g, while the drift derivative $(Db)g$ supplies the second direction. The plane depends on m/γ. Changing this ratio changes the support of the local Gaussian transition, so augmenting the states at every internal step does not provide an ordinary full-state density for parameter optimization. Changes in the transition mean can also change its affine support. An augmentation in terms of driving noise would require a different parameter-update scheme. Our implementation adds the covariance floor only after composing the monthly transition, not at each internal step.

There is a separate difficulty with finely augmented paths even when the transition densities are nonsingular. As the time step decreases, transitions become concentrated and observations at intermediate steps are uninformative, which can make particle resampling and backward sampling ineffective (Karppinen et al., 2024). This concerns mixing of the path sampler and does not by itself establish mutual singularity of path laws at different parameter values. Our smoother samples monthly endpoints, so it does not assess this fine-grid behavior. We restrict the comparison to this implementable adaptation of DS19; constructing and validating a different augmented-path method is beyond the present comparison.

We use each month's observation to guide the particle filter toward states that are consistent with the observed death count. For each particle, we calculate the measurement variance using the predicted monthly death count and hold this variance fixed when conditioning on the observation. This gives a Gaussian distribution from which we sample the next state. We weight each sample by the original measurement density multiplied by the ratio of the transition density to the proposal density.

At each optimization step, we run the particle filter through the data, then sample a trajectory of monthly states backward through the stored particles. We differentiate the complete-data log-likelihood of this trajectory and average the resulting scores using stochastic approximation. For the 23-parameter Dhaka model, these searches use averaged-score ascent. This differs from averaging and maximizing the complete-data objective in SAEM. Section S12.1 gives both updates and describes the separate numerical-SAEM comparison. We clip the scores and use backtracking, with periodic checks of the log-likelihood under the monthly Gaussian transition approximation. The 100- and 1,000-particle fits use 100 particles for these checks; the separately tuned 5,000-particle experiment uses 5,000. We also checked the particle filter on the harmonic oscillator example of Ditlevsen and Samson (2019), for which an analytic Gaussian filter is available. With 100 particles and 1,000 observations, the root mean squared difference between the particle and analytic filtered means was 0.00523.

CTDD21. The method of Corenflos et al. (2021) uses entropy-regularized optimal transport to make particle resampling differentiable. We implemented it using the same Euler simulator as in the main text, with 20 steps per month. The transport calculation uses all six state coordinates, scaled as in the published implementation. We solve the transport problem using Sinkhorn iterations, gradually decreasing the regularization parameter to 0.25 with an annealing scale of 0.75. We use a convergence tolerance of 10^{-3} and a maximum of 100 Sinkhorn iterations. To calculate derivatives, we hold the converged potentials fixed and differentiate through one further fixed-point update. As in the published implementation, we clip each component of the derivative propagated through the transport step to $[-1, 1]$. We resample when the effective sample size falls below half the number of particles.

We maximize the resulting log-likelihood estimate using Adam, with new random numbers for the particle filter at each iteration and a gradient norm limited to 100. If a search becomes invalid or repeatedly deteriorates, we return to its best valid parameter estimate, reset the Adam moments, and halve the learning rate. We allow at most six such restarts. Entropic resampling changes the particle filter's likelihood approximation, so the objective used for optimization can differ from the ordinary particle filter estimate used to evaluate the results.

S11.2 Experimental settings

For each method, we performed 100 global searches from starting parameter vectors drawn uniformly from the same wide bounding box as in the main text. We tested 100 and 1,000 particles, using learning rates of 0.02 for CTDD21 and 0.10 for DS19. We allowed each search 800 seconds and selected parameter estimates at approximately 700 seconds. For CTDD21, we selected the saved estimate with the highest fitting objective before 700 seconds. If none was valid, we used another valid saved estimate or the initial parameter vector. For DS19, we selected the first saved estimate at or after 700 seconds, or the last estimate if the search ended earlier. Thus a DS19 search could exceed the selection time by one iteration.

To assess the benefit of an IF2 warm start, we also initialized both methods at the 100 IF2 estimates used to initialize IFAD-0.97 in the main text's comparable computational effort experiment. We used these estimates directly, including those outside the initial bounding box. The IF2 stage used 175 iterations and 5,000 particles, taking 92.160 seconds under the timing procedure described in the main text. We then allowed each method 400 iterations or another 850.657 seconds, giving a total time limit of 942.817 seconds. The learning rates were 0.0002 for CTDD21 and 0.0005 for DS19, at both 100 and 1,000 particles. We selected the last valid CTDD21 estimate before the time limit and the first DS19 estimate at or after it, or the last estimate if the search ended earlier.

We kept the other optimization settings fixed when increasing the number of particles from 100 to 1,000. The 1,000-particle experiments ran sequentially on one RTX 3090 GPU, as did the 100-particle experiments initialized by IF2. The 100-particle DS19 global searches ran four at a time on that GPU. Consequently, both the particle count and GPU scheduling affect the number of iterations completed within the time limit. We also ran DS19 with 5,000 particles from the global starting points. For that experiment, we separately tuned the learning rate to 0.20 and began decreasing the stochastic approximation step size earlier.

We evaluated all selected parameter vectors independently using the Euler approximation with 20 steps per month, including estimates from searches that ended early. Each evaluation used 36 particle filter replicates with 5,000 particles. We averaged the likelihood estimates before taking the logarithm. For the optimization progress plots, we also evaluated each saved iterate using one replicate with 5,000 particles. These plots show the iterates visited during optimization, which can differ from the estimates selected for the final comparison. As in the main text, we excluded the additional evaluations from the reported computation times.

S11.3 Results

Our findings are summarized in Tables S-7 and S-8 and Figures S-9–S-12. The figures show CTDD21 with 100 particles and DS19 with 1,000, using the same particle count for each method with and without an IF2 warm start. IFAD-0.97 uses 5,000 particles. The tables include all particle counts tested.

Both methods attained lower median log-likelihoods than IFAD-0.97 in the global searches. At the particle counts shown in the figures, the medians were -4554.67 for CTDD21 and -3998.93 for DS19, compared with -3744.40 for IFAD-0.97. Increasing the number of particles did not consistently improve performance within the allotted time. For CTDD21, increasing from 100 to 1,000 particles reduced the median number of updates from 255 to 28, and the median log-likelihood fell to -6312.89. For DS19, the median improved from -4100.34 with 100 particles to -3998.93 with 1,000 particles, while the median number of updates rose from 55 to 147. This comparison

also includes the change in GPU scheduling described above. With 5,000 particles and separately tuned settings, DS19 attained a median of −4100.40 after a median of 47 updates.

Method	Start	J	Median	Maximum	Updates
CTDD21	Box	100	-4554.67	-3842.43	255
CTDD21	Box	1,000	-6312.89	-4366.30	28
DS19	Box	100	-4100.34	-3872.97	55
DS19	Box	1,000	-3998.93	-3798.03	147
DS19	Box	5,000	-4100.40	-3871.57	47
CTDD21	IF2	100	-3765.91	-3749.10	213
CTDD21	IF2	1,000	-3766.61	-3749.23	20
DS19	IF2	100	-3768.72	-3752.20	252
DS19	IF2	1,000	-3769.04	-3751.03	171
IF2 checkpoint	Box	5,000	-3766.64	-3749.30	175
IFAD-0.97	IF2	5,000	-3744.40	-3743.68	400

Table S-7: Log-likelihoods from 100 searches per method and particle count for the Dhaka model. J is the number of particles used for optimization. Box denotes initialization from the bounding box, and IF2 denotes an IF2 warm start. Median and maximum use independent particle filter evaluations with 20 Euler steps per month. Updates gives the median number of optimization updates. The IF2 checkpoint row reports the 175 iterations used for initialization; the IFAD-0.97 row reports the 400 subsequent gradient updates. The 5,000-particle DS19 experiment used separate tuning, as described in the text.

Starting from IF2 estimates substantially improved the final log-likelihoods of both methods, but neither consistently improved on the IF2 estimates themselves. With 100 particles, CTDD21 improved on 45 of the 100 starting estimates and DS19 improved on 18. The median changes were −0.10 and −1.78 log-likelihood units, respectively. With 1,000 particles, these counts were 48 and 24, and the median changes were −0.03 and −1.15. CTDD21 completed a median of 213 updates with 100 particles and 20 with 1,000 particles; DS19 completed 252 and 171, respectively. IFAD-0.97 continued from the same IF2 estimates to a median log-likelihood of −3744.40, compared with −3766.64 at the end of IF2.

Method	J	Fitting change	Euler-20 change	Improved
CTDD21	100	5.56	-0.10	45/100
DS19	100	17.24	-1.78	18/100
CTDD21	1,000	0.73	-0.03	48/100
DS19	1,000	15.21	-1.15	24/100

Table S-8: Changes during optimization from an IF2 warm start. For each search, we subtract the value at its IF2 starting point from the value at its selected parameter estimate. Fitting change and Euler-20 change give the medians of these differences for the fitting objective and the independently evaluated log-likelihood. Improved is the number of searches whose independently evaluated log-likelihood increased.

For both methods and both particle counts, the median change in the fitting objective was positive even though the median change in the independently evaluated log-likelihood was negative

(Table S-8). The two evaluations use different likelihood approximations: DS19 fits the monthly Gaussian transition model, while CTDD21 uses entropic resampling. The fitting objectives are also estimated using a single particle filter, and the updates depend on clipping and the numerical safeguards described above. These experiments do not determine how much each of these factors contributes to the disagreement.

S11.4 Reproducibility

Source code and algorithmic settings are available at https://github.com/pypomp/dmop, in the `ditlevsen/` and `corenflos/` directories. The file `imgs/competitors/README.md` gives instructions for reproducing the tables and figures, identifies the numerical results used, and links to plots showing the full range of log-likelihoods. The repository includes the tables and figures needed to compile the supplement. Instructions for retrieving archived parameter estimates and evaluations at individual iterations are in `artifacts/README.md`.

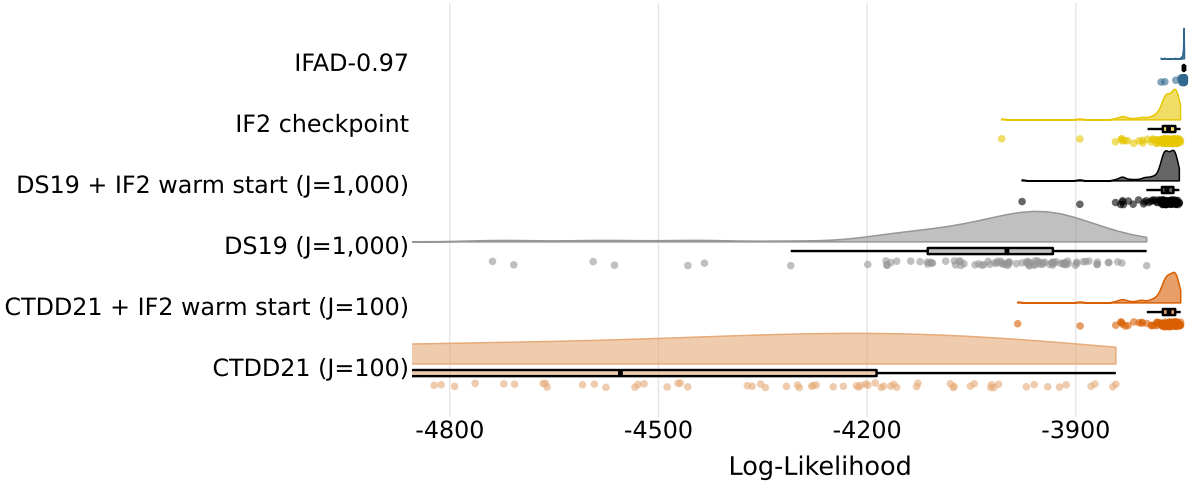


Figure S-9: Raincloud plots of the log-likelihoods found in 100 searches for the Dhaka model, with and without an IF2 warm start. CTDD21 uses 100 particles, DS19 1,000, and IFAD-0.97 5,000 during optimization. The IF2 checkpoint shows the starting estimates for the warm starts. The plot shows log-likelihoods between −4800 and −3735; all 100 searches contribute to the densities, medians, and quartiles. Each half-violin is scaled to the same maximum height.

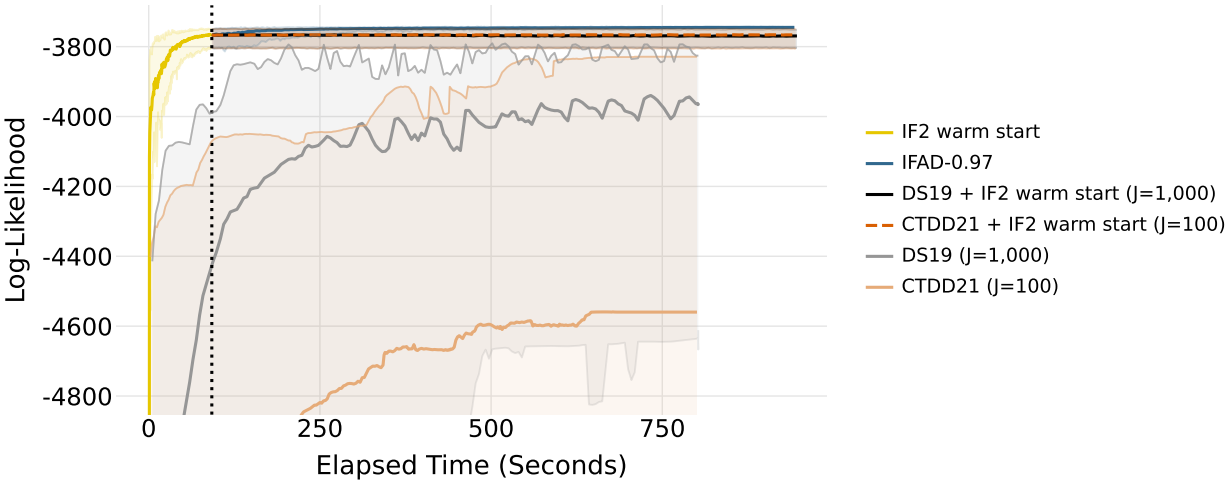


Figure S-10: Optimization progress for the Dhaka model, with and without an IF2 warm start. CTDD21 uses 100 particles, DS19 1,000, and IFAD-0.97 5,000 during optimization. Heavy lines show the median log-likelihood, and shaded regions cover the area between the 10th and 100th percentiles. The vertical dotted line indicates the end of IF2 at 92.160 seconds. The plot shows log-likelihoods between −4800 and −3735; all 100 searches contribute to the percentiles. Each saved iterate is evaluated with one particle filter replicate, compared with 36 replicates for the selected estimates in Table S-7, using 5,000 particles in both cases.

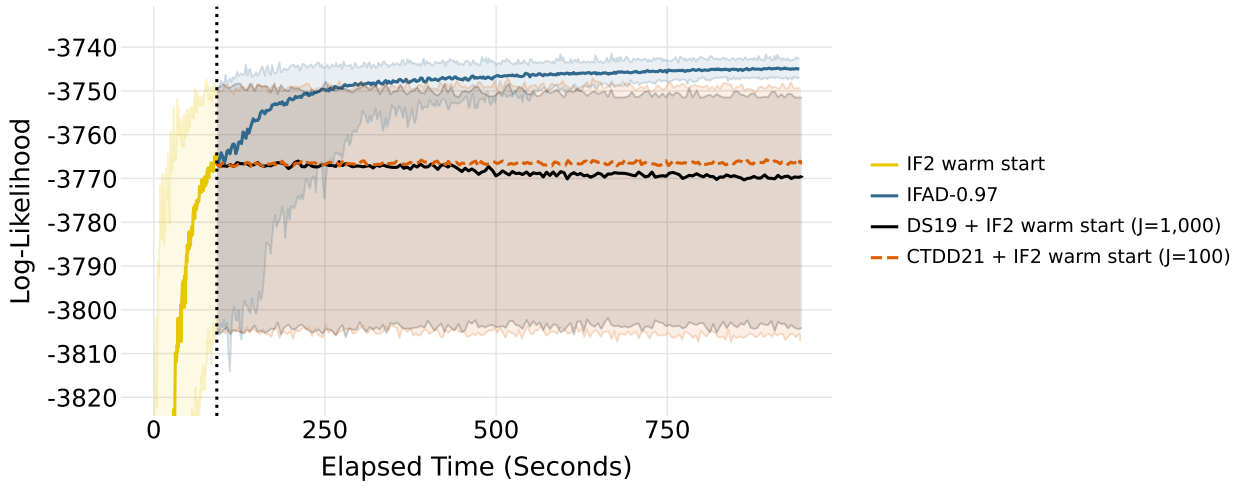


Figure S-11: Optimization progress with an IF2 warm start, showing log-likelihoods between −3820 and −3735. Particle counts, lines, and shading are as in Figure S-10. Elapsed time includes the 92.160-second IF2 stage.

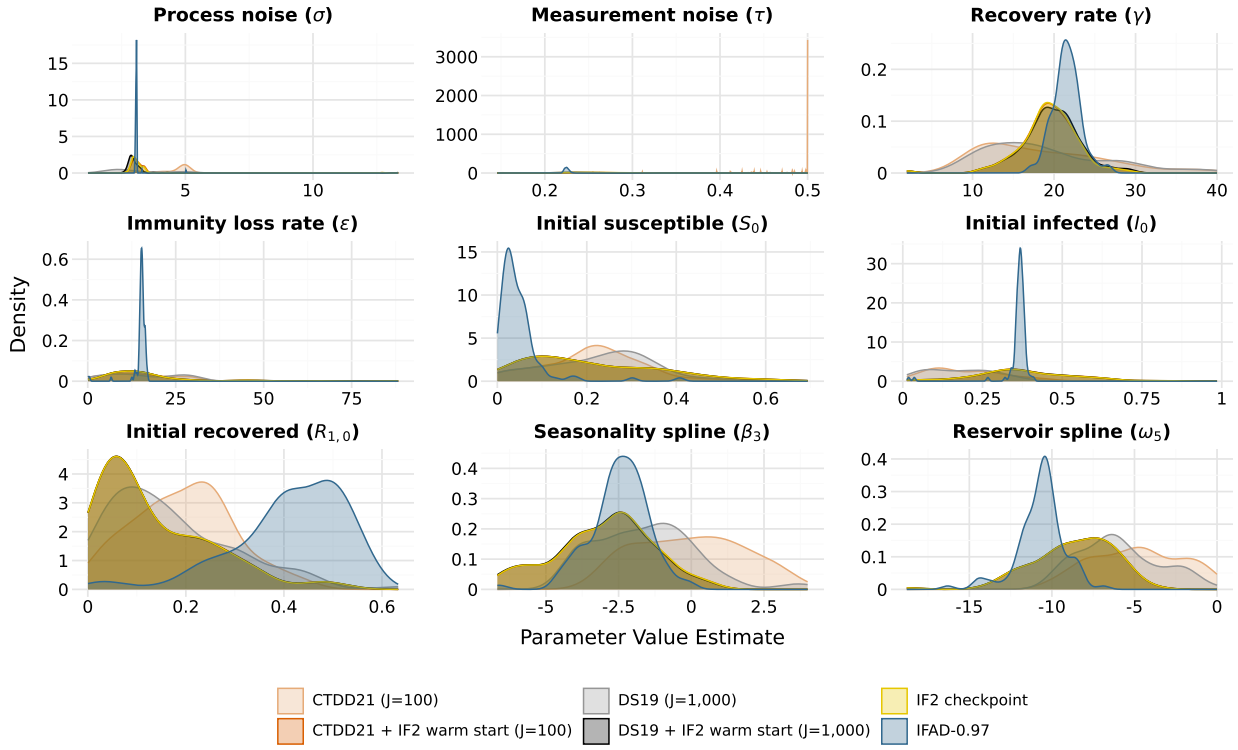


Figure S-12: Density plots of the final parameter estimates across 100 searches for nine selected parameters of the Dhaka model. Each panel has its own density scale. CTDD21 uses 100 particles and DS19 1,000, with and without an IF2 warm start; IFAD-0.97 uses 5,000 particles. The IF2 checkpoint shows the starting estimates for the warm starts.

S12 Comparisons Across Models

We compared IFAD, IF2, DS19, and CTDD21 on the harmonic oscillator of Ditlevsen and Samson (2019), the two-dimensional linear Gaussian model of Corenflos et al. (2021), and the Pypomp library implementation of a stochastic volatility model at `pypomp.models.spx`, subsequently called `spx`, in addition to the Daphnia and Dhaka models. For each example, we used the same starting parameter vectors for all methods and evaluated the resulting estimates under the same model. The two Gaussian examples have analytic likelihoods, allowing us to assess maximization error without Monte Carlo evaluation error.

Table S-9 summarizes the DS19 implementation for each panel. All five implementations evaluate process transition densities. For the linear Gaussian model and `spx`, these densities are available directly. For the oscillator, all methods fit the same order-1.5 Gaussian approximation, whose likelihood we evaluate by Kalman filtering. Thus these three comparisons use the same transition model for fitting and evaluation.

For Daphnia and Dhaka, we also use moments from the order-1.5 expansion, but compose them over smaller time steps and approximate the transition between observations by a Gaussian distribution. This composition adds a linearization of the nonlinear dynamics. It does not imply that the resulting Gaussian approximation has strong order 1.5 over the full observation interval. For Dhaka, a single local Gaussian approximation has covariance rank at most two in six dimensions, as explained in Section S11. Composing the local covariances allows noise to propagate into additional state directions. For Daphnia, we compose moments at the 0.25-day simulation step rather than apply a single Taylor expansion across the five-day observation interval. We have not tested the accuracy of such a single coarse step. Both models are evaluated using their original Euler simulators, so the Gaussian approximation introduces a difference between the fitting and evaluation likelihoods.

S12.1 SAEM and averaged-score updates

Stochastic approximation expectation maximization (SAEM) alternates imputation of the latent states with maximization of an averaged complete-data log-likelihood. Let x denote a latent trajectory and write $C(\vartheta; x) = \log \widetilde{p}_\vartheta(x, y)$ for the complete-data objective of the fitted model. The tilde allows for a transition-density approximation. Deterministic initial states are functions of the candidate parameter; only random states are imputed. At iteration m, a particle filter followed by backward simulation draws one trajectory $X^{(m)}$ from an approximation to the smoothing distribution at ϑ_{m-1}. SAEM updates a function of the candidate parameter ϑ:

$$\overline{Q}_m(\vartheta) = (1-\rho_m)\overline{Q}_{m-1}(\vartheta) + \rho_m C(\vartheta; X^{(m)}), \tag{S354}$$

$$\vartheta_m \in \operatorname*{argmax}_\vartheta \overline{Q}_m(\vartheta). \tag{S355}$$

In the maximization step, we hold the imputed paths and their averaging weights fixed. This maximization can be carried out numerically; a step that increases $\overline{Q}_m$ without maximizing it gives a generalized EM update.

The averaged-score implementation instead calculates

$$g_m = \nabla_\vartheta C(\vartheta; X^{(m)})\Big|_{\vartheta=\vartheta_{m-1}}, \tag{S356}$$

$$\overline{g}_m = (1-\rho_m)\overline{g}_{m-1} + \rho_m g_m, \tag{S357}$$

Model	DS19 transition density	Parameter update	Likelihood evaluation
A: Harmonic oscillator	Order-1.5 Gaussian approximation at the observation interval	SAEM; numerical maximization of averaged sufficient statistics	Kalman likelihood of the same approximation
B: Linear Gaussian	Exact Gaussian density of the specified discrete-time model	SAEM; explicit maximization using averaged sufficient statistics	Kalman likelihood of the same model
C: SPX	`spx` transition law: Gaussian above the variance floor, with a point mass at the floor	Averaged scores and Adam	Particle-filter likelihood of the same `spx` model
D: Daphnia	Gaussian approximation across each observation interval, composed from local order-1.5 moments	Averaged scores and Adam	Particle-filter likelihood of the original Euler model; transition and boundary rules differ
E: Dhaka	Monthly Gaussian approximation composed from local order-1.5 moments	Averaged scores and Adam; numerical SAEM in a separate comparison	Particle-filter likelihood of the original Euler model; transition differs

Table S-9: DS19 implementations for panels A–E of Figure S-13. Each uses transition-density evaluation for backward smoothing and either maximization or differentiation of a complete-data objective. For SPX, the density includes a discrete mass and is exact for the implemented model, not for a continuous-time diffusion. The Daphnia and Dhaka covariances are regularized as described below. The separate Dhaka SAEM comparison retains the same Gaussian transition.

then uses the clipped $\overline{g}_m$ in an Adam step. Derivatives hold the imputed path fixed; they do not pass through the particle filter or backward sampler. With exact smoothing, the conditional expectation of g_m is the score of the fitted model's observed-data likelihood. The particle smoother approximates this expectation. Before clipping or safeguard resets, the distinction between the two averages can be written as

$$\nabla\overline{Q}_m(\vartheta) = \sum_{r=1}^{m} w_{mr}\nabla_\vartheta C(\vartheta; X^{(r)}), \qquad \overline{g}_m = \sum_{r=1}^{m} w_{mr}\nabla_\vartheta C(\vartheta_{r-1}; X^{(r)}), \tag{S358}$$
$$w_{mr} = \rho_r \prod_{k=r+1}^{m}(1-\rho_k).$$

The SAEM gradient evaluates all path contributions at the same candidate parameter, whereas $\overline{g}_m$ averages gradients evaluated at the parameter values used in earlier iterations. Consequently, $\overline{g}_m$ generally differs from the gradient of $\overline{Q}_m$ at the current parameter. Our gain schedule is $\rho_m = 1$ for $m \le b$ and $\rho_m = (m-b)^{-0.9}$ thereafter. We use $b = 30$ for the Gaussian SAEM and averaged-score runs, and $b = 3$ for numerical SAEM on Dhaka. Clipping and Adam act on the existing estimation scales. In the Dhaka score implementation, the clipped average is also carried into the next iteration, and failed safeguard checks can reset it.

Linear Gaussian model: explicit SAEM. The latent states satisfy $X_t = \mathrm{diag}(a_1, a_2)X_{t-1} + \varepsilon_t$, with $\varepsilon_t \sim \mathcal{N}(0, 0.5I)$, $Y_t \mid X_t \sim \mathcal{N}(X_t, 0.1I)$, and $X_0 \sim \mathcal{N}(0, I)$. For a fixed path, the terms depending on $a = (a_1, a_2)$ are

$$C(a; x) = -\frac{1}{2(0.5)}\sum_{t=1}^{T}\sum_{d=1}^{2}(x_{t,d} - a_d x_{t-1,d})^2 + \text{constant}.$$

For each coordinate, the sufficient statistics are

$$S_{00,d}(x) = \sum_t x_{t-1,d}^2, \qquad S_{10,d}(x) = \sum_t x_{t,d}x_{t-1,d}.$$

We update their averages by $\overline{S}_m = (1-\rho_m)\overline{S}_{m-1} + \rho_m S(X^{(m)})$. The M-step is

$$a_{d,m} = \mathrm{clip}_{[-0.98,0.98]}\left(\frac{\overline{S}_{10,d,m}}{\overline{S}_{00,d,m}}\right). \tag{S359}$$

Each iteration therefore requires a particle filter and backward simulation of one path, after which the maximization reduces to two scalar ratios.

Harmonic oscillator: SAEM with a numerical M-step. Write $x_t = (v_t, u_t)^\mathsf{T}$, where the positions v_t are observed without measurement error and the velocities u_t are imputed. For $\theta = (D, \gamma, \sigma)$ and observation interval $\Delta = 0.02$, the fitted Gaussian transition has

$$A_\theta = \begin{pmatrix} 0 & 1 \\ -D & -\gamma \end{pmatrix}, \qquad F_\theta = I + \Delta A_\theta + \tfrac{1}{2}\Delta^2 A_\theta^2,$$
$$Q_\theta = \sigma^2 \begin{pmatrix} \Delta^3/3 & \Delta^2/2 - \gamma\Delta^3/3 \\ \Delta^2/2 - \gamma\Delta^3/3 & \Delta - \gamma\Delta^2 + \gamma^2\Delta^3/3 \end{pmatrix}.$$

Conditional on the first position, $U_0 \sim \mathcal{N}(0, v_\theta)$, where $v_\theta = \sigma^2/(2\gamma)$. Let $S_{ij} = \sum_{t=1}^{T} x_{t-1+i} x_{t-1+j}^\mathsf{T}$ for $(i,j) = (0,0), (1,0), (1,1)$, and let $h = u_0^2$. We average these quantities using the recursion above, giving the following objective up to constants independent of θ:

$$\overline{Q}_m(\theta) = -\tfrac{1}{2}\left\{\log v_\theta + \frac{\overline{h}_m}{v_\theta} + T \log\det Q_\theta + \operatorname{tr}\big(Q_\theta^{-1}\overline{R}_m(\theta)\big)\right\}, \tag{S360}$$

where

$$\overline{R}_m(\theta) = \overline{S}_{11,m} - F_\theta \overline{S}_{10,m}^\mathsf{T} - \overline{S}_{10,m} F_\theta^\mathsf{T} + F_\theta \overline{S}_{00,m} F_\theta^\mathsf{T}.$$

We use L-BFGS-B on $(\log D, \log\gamma, \log\sigma)$, retaining a finite step only if this objective does not decrease, up to numerical tolerance. To reduce numerical cancellation, we evaluate this quadratic form using accumulated state increments. The M-step thus uses averaged sufficient statistics of the imputed paths.

SPX: averaged complete-data scores. Let r_t be the observed return and v_t the latent variance. The `spx` conditional variance transition has Gaussian mean and standard deviation

$$m_t(v) = v + \kappa(\bar{v} - v) + \xi\rho(r_{t-1} - \mu + v/2), \qquad s(v) = \xi\sqrt{v(1-\rho^2)},$$

followed by a floor $\epsilon = 10^{-32}$. Writing $\phi(z; m, s^2)$ for a Gaussian density and Φ for the standard normal distribution function, its transition is

$$f_{\theta,t}(v' \mid v) = \begin{cases} \phi(v'; m_t(v), s(v)^2), & v' > \epsilon, \\ \Phi((\epsilon - m_t(v))/s(v)), & v' = \epsilon. \end{cases}$$

The second case is a probability mass; the reference measure is Lebesgue measure plus an atom at ϵ. The complete-path objective is

$$C(\theta; v) = \sum_{t=1}^{T} \left\{\log f_{\theta,t}(v_t \mid v_{t-1}) + \log\phi(r_t; \mu - v_t/2, v_t)\right\}. \tag{S361}$$

The initial variance $v_0 = V_0$ is a parameter, so its current candidate value enters the first transition when differentiating. For $v > 0$, $\xi > 0$, and $|\rho| < 1$, the conditional transition variance $\xi^2 v(1-\rho^2)$ is positive. This scalar latent process therefore has no missing covariance direction in the interior; its boundary atom is included explicitly in the density above. For this example, we update the parameters with Adam using the averaged scores in Equations (S356)–(S357), without a SAEM maximization step. We obtain these complete-data derivatives by AD with the imputed path fixed, without differentiating through the filtering and resampling calculations. The runtime difference from IFAD also reflects the particle counts and optimization settings below.

Daphnia: averaged scores for the Gaussian block approximation. For mesocosm u, let $m_{\theta,u,t}$ and $\Sigma_{\theta,u,t}$ be the mean and covariance obtained by composing the local Gaussian transitions over an observation interval, including the inoculation in the first interval. Let $g_{\theta,u}$ be the product of the four negative-binomial measurement probabilities in Section S10. For the eight imputed mesocosm paths, the complete-path objective is

$$C(\theta; x) = \sum_{u=1}^{8}\sum_{t=1}^{10} \left\{\log\phi_8(x_{u,t}; m_{\theta,u,t}(x_{u,t-1}), \Sigma_{\theta,u,t}(x_{u,t-1})) + \log g_{\theta,u}(y_{u,t} \mid x_{u,t})\right\}. \tag{S362}$$

Initial states are fixed. The covariance uses a relative diagonal floor of 10^{-8} on scaled states. The particle filter uses the observations to guide proposals, with the original negative-binomial measurement probabilities in the importance weights. We approximate each observation interval by one eight-dimensional Gaussian transition, composing local moments over 24 steps for the first interval and 20 thereafter. The step size is 0.25 days, so these transitions cover six days initially and five days between observations. This approximates the distribution produced by the nonlinear dynamics over the whole interval. Boundary handling also differs: we assign zero weight to Gaussian proposals outside the allowed state range, whereas the Euler simulator resets an offending state to zero and assigns a log weight of -150 to that particle at the next observation. Both differences can change the likelihood being maximized. The local covariance need not be rank deficient in this model. At a strictly positive state, the six independent noises span the six directly forced coordinates. The integrated juvenile noises contribute two further directions through maturation into susceptible adults, with nonzero coefficients $\lambda_J \sigma_J^{\mathrm{n}} J^{\mathrm{n}}$ and $\lambda_J \sigma_J^{\mathrm{i}} J^{\mathrm{i}}$. The implemented order-1.5 covariance is therefore full rank in this interior when the six fitted noise coefficients are positive, and its range is independent of the parameter values there. At the specified initial state several populations are zero, and the local covariance has rank three. Daphnia thus differs from Dhaka's rank-two interior covariance. The algebra and numerical checks, including parameter perturbations at fixed states, are provided in `code/benchmarks/covariance_support.py` and its accompanying note. After backward simulation of one path per mesocosm, we differentiate this sum with respect to the 38 estimated parameters and update them with Adam using the averaged scores. As for SPX, we do not carry out a SAEM maximization step. We evaluate the resulting parameter estimates using the original Euler model from Section S10.

Dhaka: score updates and numerical generalized SAEM. We compare score updates with numerical SAEM using the monthly Gaussian transition described in Section S11, with 20 internal time steps and a relative covariance floor of 10^{-12}. Let $\mathcal{R}x$ reset the accumulated monthly deaths to zero, and write $d_t = P_t M_t$ for the predicted deaths at a sampled endpoint. Here P_t is the population and M_t is the accumulated death count divided by population. Both methods use the complete-data objective

$$C(\theta; x) = \sum_{t=1}^{T} \Big\{ \log \phi_6(x_t; m_{\theta,t}(\mathcal{R}x_{t-1}), \Sigma_{\theta,t}(\mathcal{R}x_{t-1})) + \log[\phi(y_t; d_t, (\tau d_t + \delta)^2) + \delta] \Big\}, \qquad \delta = 10^{-18}. \tag{S363}$$

The initial state $x_0 = x_0(\theta)$ is recalculated at each candidate parameter, while the sampled monthly endpoints are held fixed. For the score updates, we use the gradient average in Equation (S357), with Adam and the safeguards described above. For numerical SAEM, we retain the paths and weights to calculate $\overline{Q}_m(\theta) = \sum_{r=1}^{m} w_{mr} C(\theta; X^{(r)})$. Each L-BFGS-B evaluation recalculates every retained path's objective and gradient at the same candidate θ. Within each M-step, we divide the objective and its gradient by the fixed scale $\max\{1, \|\nabla \overline{Q}_m(\theta_{m-1})\|_\infty\}$. This reduces numerical line-search failures without changing the maximizer. We allow at most 25 L-BFGS-B iterations per M-step and retain a candidate only if it is finite and increases or leaves unchanged this fixed $\overline{Q}_m$. An M-step interrupted by the time limit is discarded. This gives a generalized-SAEM update when an explicit maximizer in terms of sufficient statistics is unavailable. We evaluate both methods with the Euler likelihood, so the evaluation model differs from the monthly Gaussian approximation used for fitting.

Computation times for the Gaussian examples. We used 500 particles for DS19, IFAD, and IF2 in both Gaussian examples. DS19 ran for 80 SAEM iterations, IFAD for 100 IF2 iterations followed by 300 gradient iterations, and IF2 for 600 iterations. CTDD21 used 25 particles and 300 gradient iterations. The common particle count for DS19, IFAD, and IF2 therefore does not imply equal computational effort.

DS19 runs a particle filter and samples one trajectory by backward simulation at every iteration, and both are included in its computation time. Its maximization step is inexpensive for these models: the linear Gaussian model has an explicit update, while the oscillator requires optimization over three parameters using averaged sufficient statistics. Neither maximization requires differentiation through the particle filter. The analytic likelihoods are used only to evaluate the fitted parameters. For all methods, we wait for the numerical calculations to finish before recording the time, excluding compilation and likelihood evaluation. IFAD, IF2, and DS19 used four efficiency cores of an Intel Core i9-13900K; CTDD21 used two performance cores with four hardware threads. Concurrent batches used separate core groups. These hardware differences also affect the comparison of computation times.

S12.2 Search and evaluation settings

We ran 20 searches per method for the oscillator, linear Gaussian, SPX, and Daphnia models, using the same starting parameter vectors for each method. For Dhaka, we used the 100 searches from Section S11.

For SPX, we used 500 particles for IFAD and IF2, 100 for DS19, and 25 for CTDD21. IFAD used $\alpha = 0.97$ and ran for 100 IF2 iterations followed by 300 gradient iterations, IF2 for 600 iterations, and DS19 and CTDD21 for 300 updates each. The Adam learning rates started at 0.001 for IFAD and 0.01 for DS19 and CTDD21, decreasing to one tenth of these values by cosine decay.

For Daphnia, we used 1,000 particles per mesocosm for MPIF and IFAD, and 100 for DS19 and CTDD21. We ran MPIF for 680 iterations. IFAD used a separate 200-iteration MPIF warm start followed by 200 gradient steps. To compare refinement of the same starting estimate, we initialized DS19 and CTDD21 at this MPIF warm start and allowed each the time taken by the IFAD gradient stage, with a maximum of 2,000 updates. Their learning rates decreased from 0.001 to 0.0001 by cosine decay over the allotted time. We discarded updates completed after the time limit and included the common MPIF warm start in the computation times for all three methods.

We evaluated the final SPX estimates using 24 independent particle-filter replicates with 5,000 particles. For Daphnia, we used 10 replicates with 2,000 particles per mesocosm under the original Euler model. We averaged the likelihood estimates for each mesocosm before taking logarithms, then summed the log-likelihoods over mesocosms, combining their independent Monte Carlo errors. For the optimization trajectories, we used four replicates with 1,000 particles for SPX and two replicates with 500 particles per mesocosm for Daphnia. All these evaluations were independent of fitting; we did not use them to select the reported parameter estimates.

SPX fits used two performance cores with four hardware threads, except for one recovered starting point for which all four methods used four efficiency cores. Daphnia fits used four efficiency cores. These runs used the same processor as the Gaussian comparisons. The archived Dhaka comparison used GPU computations as described in Section S11.

S12.3 Repeated-search results

IFAD and DS19 both approached the maximum likelihood for the two Gaussian models (Table S-10 and Figure S-13). DS19 did so more quickly, benefiting from the inexpensive maximization steps described above. IF2 gave more variable results, while CTDD21 attained lower likelihoods, particularly for the oscillator. DS19 also had the highest median log-likelihood and shortest median computation time for SPX. We therefore found no consistent improvement from IFAD on these examples.

For Daphnia, IFAD attained the highest maximum and median log-likelihoods. The median was -861.30, compared with -864.98 for MPIF, -867.02 for DS19, and -865.03 for CTDD21. Figure S-14 shows IFAD continuing to improve after the common MPIF warm start, while DS19 and CTDD21 made less progress with the same time allowance. The full trajectories are shown in Figure S-15; the shaded regions in both figures cover the 10th through 100th percentiles, as in Section S11. We retained the last finite estimate from one DS19 search that stopped prematurely. For Dhaka, the differences were larger: IFAD attained a median log-likelihood of -3744.40, compared with -3755.41 for IF2, -3769.04 for DS19, and -3765.91 for CTDD21.

The Daphnia comparison does not determine how much of the DS19 shortfall comes from the Gaussian transition approximation. Smoothing error and optimization may also contribute. DS19 used 100 particles per mesocosm, compared with 1,000 for IFAD, and estimated each gradient from one backward-sampled path per mesocosm for the 38-parameter optimization. Moreover, the Daphnia implementation uses averaged-score Adam updates, whereas the Gaussian examples use SAEM maximization with sufficient statistics. We have not compared these updates with numerical SAEM on Daphnia. Evaluating both the Gaussian fitting objective and the Euler likelihood along the same searches would help assess the effect of the transition approximation: improvement in the former accompanied by deterioration in the latter would indicate that improving the approximate objective need not improve the likelihood of interest. We have not carried out this diagnostic.

The Gaussian examples provide useful numerical checks because their likelihoods are available analytically. Once methods repeatedly approach these maxima, the examples offer limited evidence about further improvements in likelihood maximization. The SPX comparison still shows differences between methods and is not saturated in this sense. Daphnia and Dhaka retain difficulties encountered in scientific applications, including many estimated parameters, nonlinear dynamics, partially observed states, and, for Daphnia, multiple experimental units. Assessing methods for such applications requires benchmarks that retain these difficulties alongside tractable examples used to verify implementations.

S12.4 Score updates and numerical SAEM on Dhaka

To assess whether the score updates accounted for the poor performance of DS19 on Dhaka, we repeated the comparison using numerical SAEM with the same monthly Gaussian transition approximation and particle smoother. We used 1,000 particles and allowed 850 seconds per search, starting from 20 global parameter vectors and 20 IF2 estimates. Numerical SAEM gave higher Euler log-likelihoods than the score updates for only two of the 20 global starts (Figure S-16 and Table S-11). The median paired difference was -608.83 log-likelihood units, taking numerical SAEM minus score updates. The numerical M-steps were expensive: SAEM completed a median of 15.5 outer iterations in the allotted time, compared with 86 for the score updates.

Starting from the IF2 estimates reduced this difference substantially. The median paired dif-

ference was -0.62 log-likelihood units, although one SAEM search finished 39.59 units below its score counterpart. SAEM gave higher likelihoods in four of the 20 pairs. Neither method improved on the IF2 estimates at the median, with changes of -0.27 for score updates and -1.37 for numerical SAEM. Thus numerical maximization of the SAEM objective did not improve the Dhaka results. Both methods retained the same monthly Gaussian transition and particle smoother, so this comparison does not measure the effect of the transition approximation. Increasing the number of particles can improve the particle approximation to this Gaussian model, but does not remove its difference from the Euler model. The results therefore leave open the relative contributions of transition approximation, smoothing error, numerical conditioning, and optimization cost to the Dhaka shortfall.

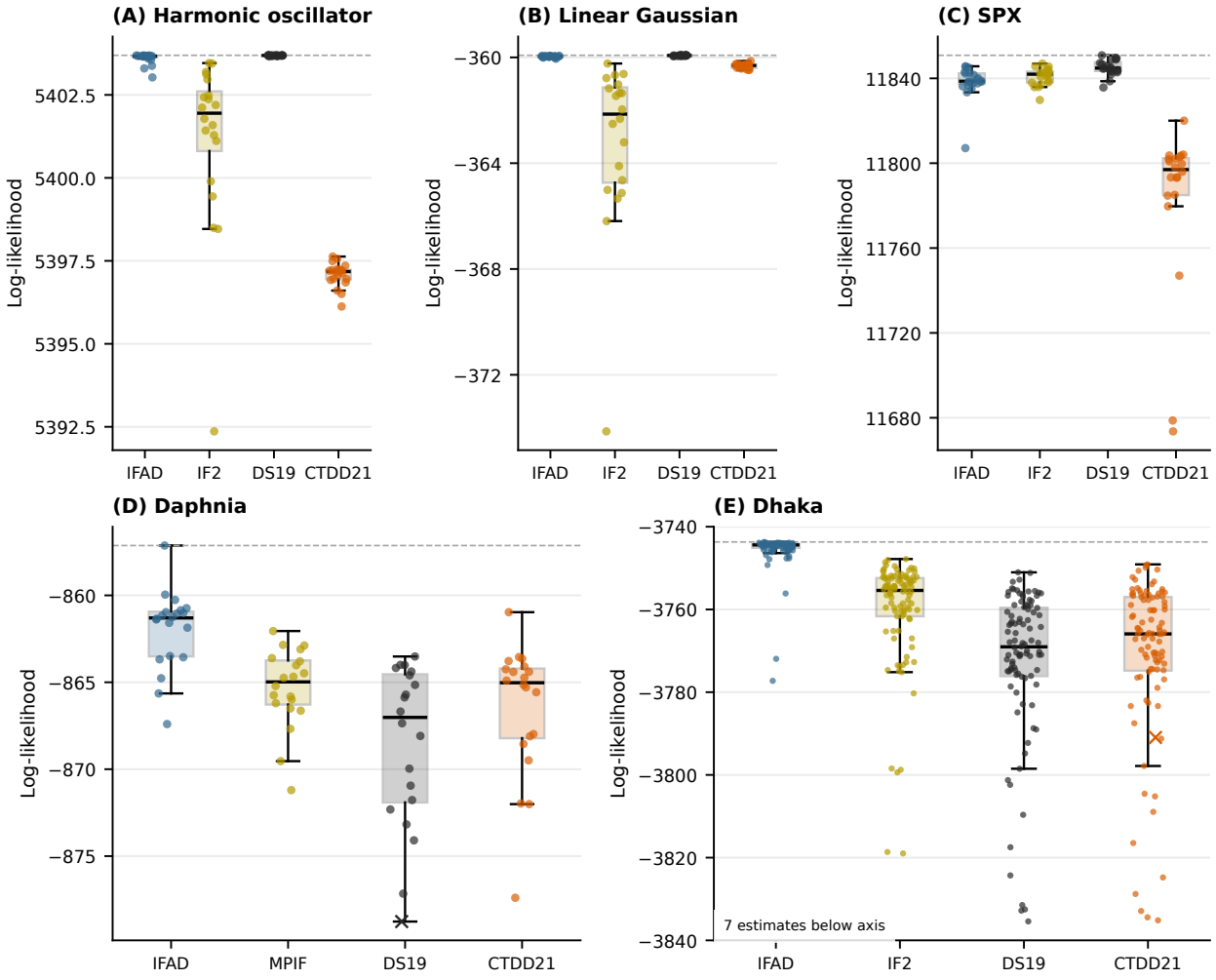


Figure S-13: Final log-likelihoods from repeated parameter searches. Each point represents one of 20 searches per method in panels A–D or 100 searches in panel E. Boxes show the median and interquartile range; whiskers extend to the most extreme values within 1.5 interquartile ranges of the box. All estimates are shown in panels A–D. Panel E uses the range $[-3840, -3740]$; the number of estimates below the axis is indicated. The uncropped figure is available with the source code, and all estimates contribute to the boxes and numerical summaries. Crosses mark searches that stopped prematurely, for which we evaluated the last estimate obtained before failure or the time limit. Dashed lines mark the analytic maximum in panels A and B and the best displayed estimate in panels C–E. Likelihoods are evaluated independently of fitting, using the same model within each panel. DS19 and CTDD21 use the common MPIF warm start in D and the common IF2 warm start in E. Panel E uses the same estimates as the warm-start comparison in Figure S-9. All axes are linear.

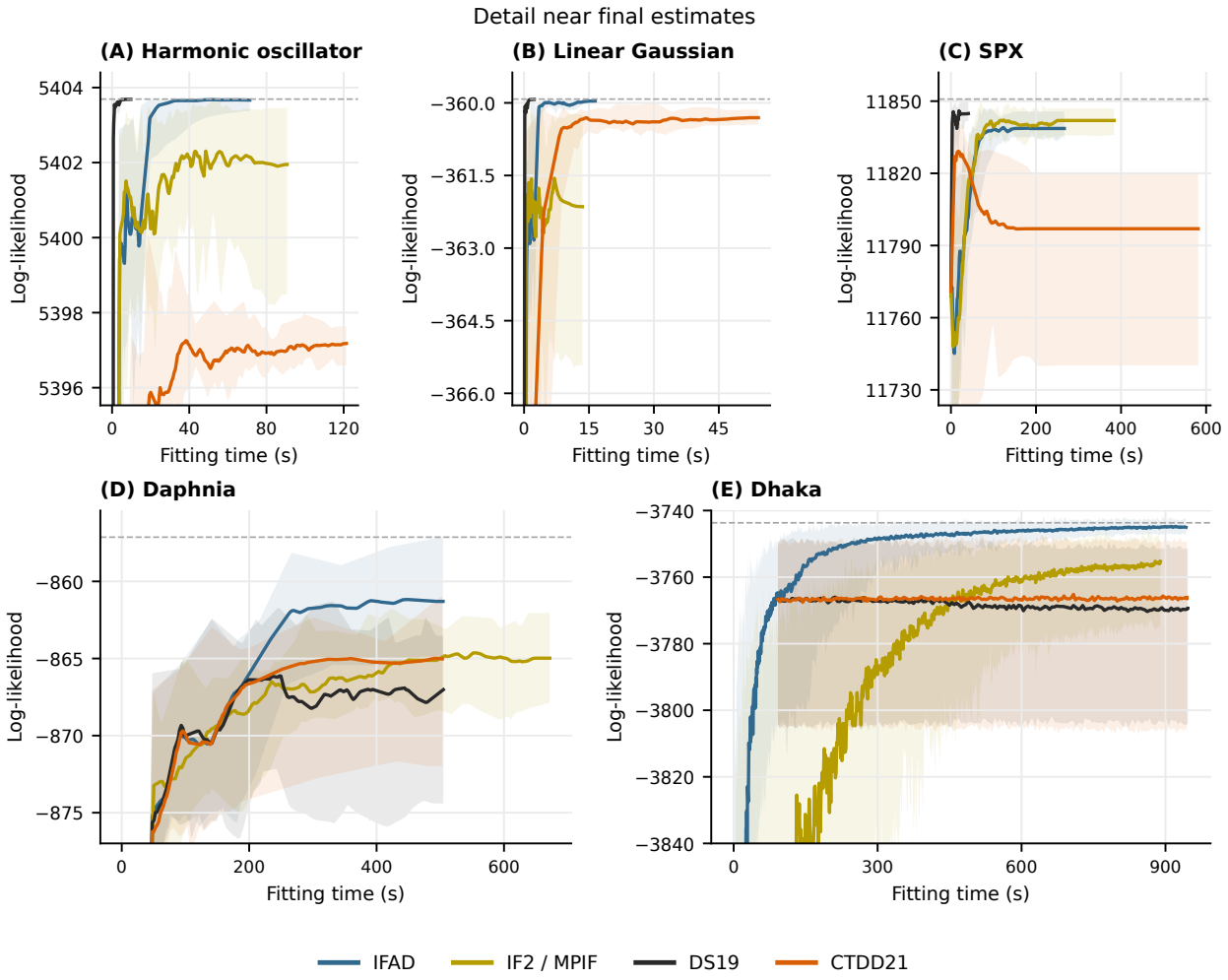


Figure S-14: Optimization progress near the final estimates. Solid lines show the median log-likelihoods, and the shaded regions cover the area between the 10th and 100th percentiles across searches. We evaluated likelihoods at saved iterations and interpolated linearly between them, retaining each search's final value after it stopped. These evaluations were not used to select the fitted parameters. The vertical axes are linear and restricted to the region near the final estimates; Figure S-15 shows the full range, with the same shading opacity. Panel E uses the range $[-3840, -3740]$. The starting points, reference lines and final evaluations are those of Figure S-13.

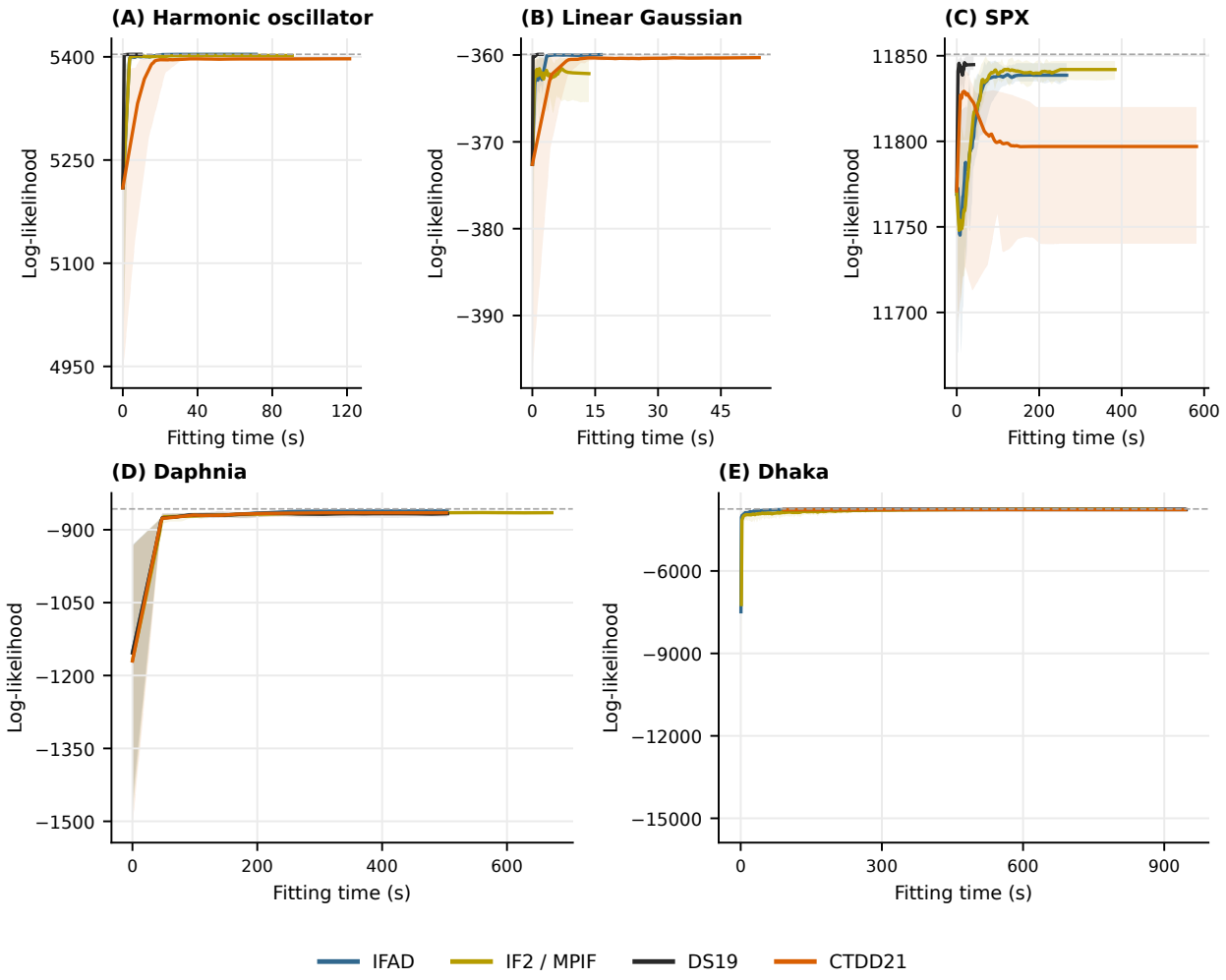


Figure S-15: Optimization progress over the full range of the median trajectories and percentile bands. Lines, shading and reference lines are defined in Figure S-14. All axes are linear, with separate ranges for each model and data set. Computation times exclude compilation and independent likelihood evaluation. The hardware and algorithmic settings are given in Section S12.2 and the preceding paragraph.

Method	Median	Maximum	IQR	MCSE	Time (s)	Failed
Harmonic oscillator (20 starts)						
IFAD	5403.67	5403.69	0.04	0.00	70.0	0
IF2	5401.95	5403.46	1.80	0.00	89.4	0
DS19	5403.69	5403.69	0.01	0.00	9.4	0
CTDD21	5397.18	5397.63	0.28	0.00	117.3	0
Linear Gaussian (20 starts)						
IFAD	-359.96	-359.94	0.03	0.00	16.1	0
IF2	-362.14	-360.23	3.60	0.00	13.3	0
DS19	-359.93	-359.92	0.01	0.00	2.1	0
CTDD21	-360.31	-360.14	0.13	0.00	51.6	0
SPX (20 starts)						
IFAD	11838.63	11845.69	4.95	0.31	144.5	0
IF2	11841.96	11846.95	5.06	0.31	249.8	0
DS19	11844.83	11850.85	4.11	0.47	21.5	0
CTDD21	11796.97	11820.04	17.37	0.19	165.9	0
Daphnia (20 starts)						
IFAD	-861.30	-857.13	2.57	0.19	487.8	0
MPIF	-864.98	-862.05	2.53	0.19	647.3	0
DS19	-867.02	-863.52	7.37	0.20	488.1	1
CTDD21	-865.03	-860.97	4.00	0.17	488.7	0
Dhaka (100 starts)						
IFAD	-3744.40	-3743.68	0.95	0.14	942.8	0
IF2	-3755.41	-3747.82	9.21	0.15	889.3	0
DS19	-3769.04	-3751.03	16.53	0.17	942.8	0
CTDD21	-3765.91	-3749.10	17.78	0.15	942.8	2

Table S-10: Repeated-search results for the five models. Median, maximum and interquartile range (IQR) summarize final log-likelihoods across searches. MCSE is the median Monte Carlo standard error of an individual final log-likelihood estimate; it is zero for the analytic Gaussian evaluations. Time is the median fitting time, including any IF2 or MPIF warm start and excluding compilation and independent evaluation. Failed is the number of searches that stopped prematurely. We retained their last estimates before failure or the time limit in all summaries. Hardware and fitting settings differ between experiments, as described in the text.

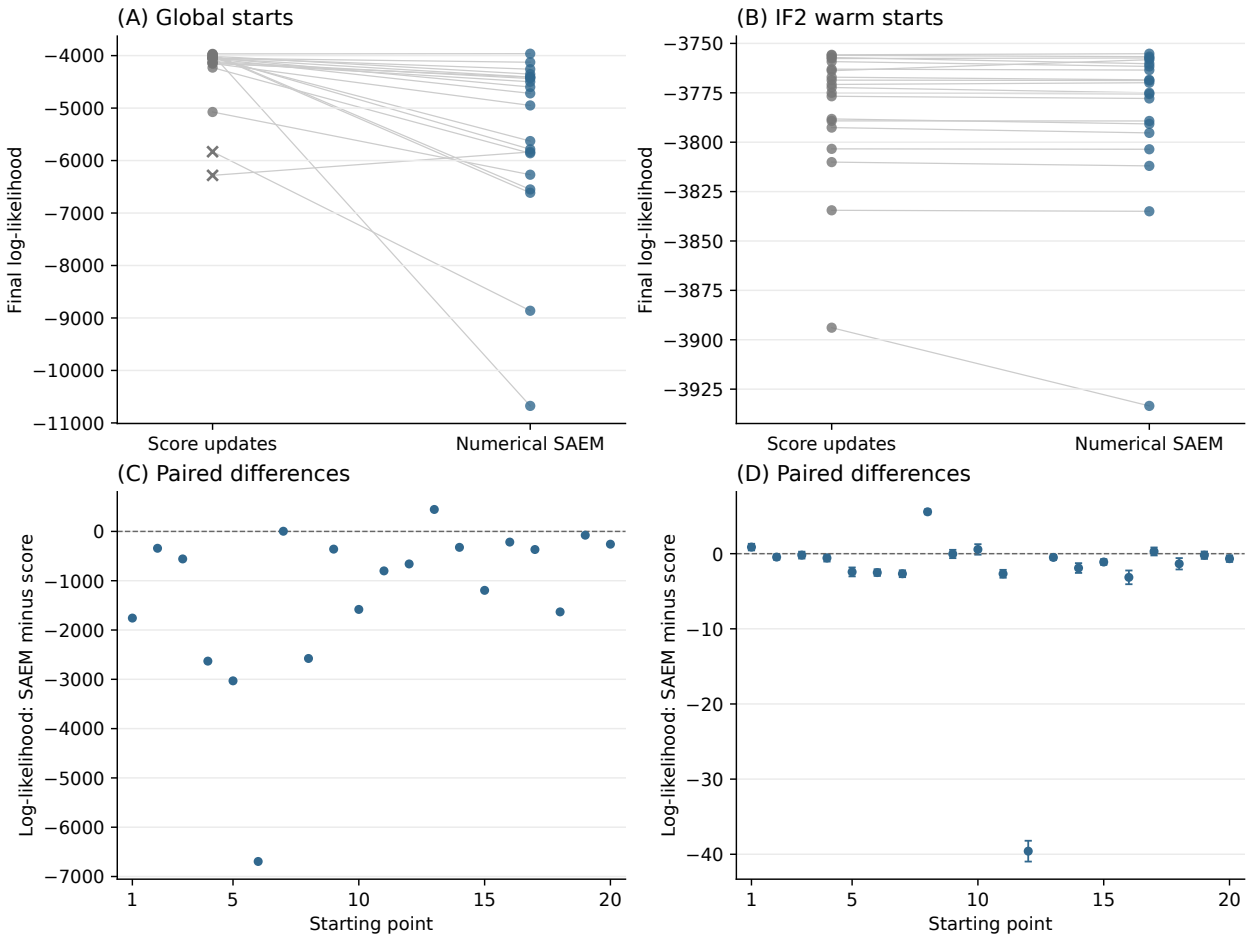


Figure S-16: Averaged-score updates and numerical SAEM on Dhaka, using the same monthly Gaussian transition approximation and particle smoother. Both methods use 1,000 particles and are allowed 850 seconds per search, from 20 global starts (A, C) and 20 saved IF2 estimates (B, D). Panels A and B join final estimates from the same starting point. Crosses mark searches that stopped prematurely, for which we retained the last estimate before failure or the time limit. Panels C and D show paired log-likelihood differences, numerical SAEM minus score updates; bars show twice the paired Monte Carlo standard error. Final likelihoods use the original Euler model, with 5,000 particles and 36 evaluation replicates. Evaluation random numbers are shared within a pair and independent of fitting. All axes are linear.

Initialization	Optimizer	Median ℓ	Median change	Iterations	Time (s)	Failed
Global	Score updates	-4050.49	3827.36	86	854.5	2
Global	Numerical SAEM	-4834.43	3109.60	15.5	851.3	0
IF2	Score updates	-3771.54	-0.27	89	855.7	0
IF2	Numerical SAEM	-3772.44	-1.37	13	851.1	0

Table S-11: Dhaka score and numerical-SAEM comparison in Figure S-16. Median ℓ is the median final Euler log-likelihood. Median change is the median difference between each search's final and initial log-likelihoods. Iterations is the median number of outer iterations completed at the reported estimate, including iterations without a parameter change. Time is the median computation time, including computation past the time limit; we discarded updates that did not finish within the allotted time. Compilation, independent evaluation and the cost of producing the saved IF2 starting estimates are excluded. For failed searches, we retained the last estimate before failure or the time limit in all summaries.

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